Egyptian Baccalaureate Mathematics

Student Edition • Open Workbook / Guided Practice • Part2

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Concept 1 | Defining Angles and Basic Trigonometric Functions

English Student Edition • Focused formulas and guided practice

Formula opener

Radians

$$\pi \text{ rad} = 180^\circ$$

Arc length

$$s = r\theta$$

Trig

$$\sin\theta = \frac{y}{r}, \quad \cos\theta = \frac{x}{r}$$

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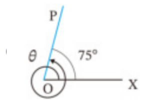
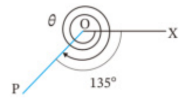
Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q001 **Written**

Find the measures of θ in the two figures: (a) one full counter-clockwise revolution plus $75°$; (b) two full clockwise revolutions and a further $135°$.



(b)



Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q002 Written

Draw the terminal side for $300°$ and $-450°$.

Q003 **Written**

Express $420°$ **and** $-210°$ **as** $\alpha + 360° \times n$, **where** n **is an integer and** $0° \leq \alpha < 360°$.

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q004 **Written**

Find the measures of θ in the two figures: (a) a clockwise turn of $30°$ from the positive x-axis; (b) a terminal side $40°$ above the negative x-axis, measured counter-clockwise from the positive x-axis).

2 Draw the terminal side for the following angles.

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q005 **Written**

Draw the terminal side for $500°$ and $-120°$.

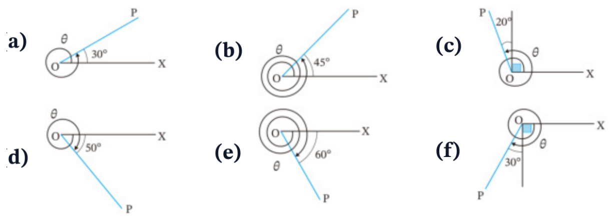
Q006 **Written**

Express $520°$ **and** $-140°$ **as** $\alpha + 360° \times n$.

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q007 **Written**

Find the six diagram measures θ shown in the source figure (a)–(f).



2 Draw the terminal side for the following angles.

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q008 **Written**

Draw the terminal sides for $470°, 780°, 225°, -380°, -990°, -200°$.

Q009 **Written**

Express $610°, 920°, 840°, -430°, -960°, -20°$ **as** $\alpha + 360° \times n$.

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q010 **Written**

Express $129720°$ **and** $-2592100°$ **as** $\alpha + 360° \times n$, **with** $0° \leq \alpha < 360°$.

Q011 *MCQ*

The principal angle coterminal with $765°$ is:

Choose the correct option.

A $45°$

B $135°$

C $225°$

D $315°$

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q012 MCQ

The principal angle coterminal with $-410°$ is:

Choose the correct option.

A $-50°$
B $50°$
C $310°$
D $410°$

Q013 **MCQ**

The terminal side of $540°$ lies on the:

Choose the correct option.

A positive x-axis
B negative x-axis
C positive y-axis
D negative y-axis

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q014 MCQ

A positive angle of $315°$ terminates in:

Choose the correct option.

A QI

B QII

C QIII

D QIV

Q015 *MCQ*

Which is the general-angle form for $-75°$?

Choose the correct option.

A $75° + 360°$ n

B $285° + 360°$ n

C $-285° + 360°$ n

D $360°\, n - 75°$

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q016 MCQ

The direction of a negative angle is:

Choose the correct option.

A counter-clockwise
B clockwise
C radial only
D undefined

Q017 **MCQ**

The terminal side of $1080°$ is the:

Choose the correct option.

A positive x-axis

B negative x-axis

C positive y-axis

D negative y-axis

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q018 MCQ

For $\alpha = 0°$, the general-angle family is:

Choose the correct option.

A $360°n$

B $0°n$

C $180° + 360°n$

D $90° + 360°n$

Q019 **MCQ**

The principal angle for $-990°$ is:

Choose the correct option.

A $90°$

B $180°$

C $270°$

D $-270°$

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q020 MCQ

A ray (P) is $25°$ above the negative x-axis. Its standard-position angle is:

Choose the correct option.

A $25°$

B $155°$

C $205°$

D $335°$

Q021 **Written**

(1) Convert $630°$ to radians. (2) Convert $-\frac{7\pi}{4}$ to degrees. (3) For a sector with $r = 3$ and $\theta = \frac{4\pi}{3}$, find arc length L and area S.

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q022 Written

Complete the standard-angle degree/radian table, then: convert $288°, -400°$ to radians; convert $\frac{11\pi}{6}, -\frac{3\pi}{5}$ to degrees; and find L, S for $r = 6, \theta = \frac{3\pi}{4}$.

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q023 **Written**

Convert $108°, 36°, -240°, -225°, 420°, 400°, -570°, -2700°$ **to radians.**

Q024 **Written**

Convert $\frac{7\pi}{6}, \frac{4\pi}{5}, \frac{2\pi}{3}, \frac{4\pi}{3}, -\frac{5\pi}{4}, -\frac{\pi}{2}, -\frac{11\pi}{2}, -\frac{7\pi}{18}$ to degrees.

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q025 **Written**

Find L and S for: (a) $r = 6, \theta = \frac{7\pi}{6}$; (b) $r = 4, \theta = \frac{2\pi}{5}$; (c) $r = 6, \theta = \frac{5\pi}{4}$.

Q026 **Written**

Draw terminal sides for $\frac{\pi}{4}, \frac{\pi}{3}, \frac{5\pi}{6}, \frac{3\pi}{4}, \frac{4\pi}{3}, \frac{7\pi}{6}, -\frac{\pi}{3}, -\frac{\pi}{4}$.

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q027 Written

A sector of radius $9\,\mathrm{cm}$ and central angle $4\pi/3$ is joined to form a cone. Determine the exact vertical height and total volume.

Q028 MCQ

Convert $150°$ to radians:

Choose the correct option.

A $\frac{5\pi}{6}$

B $\frac{3\pi}{5}$

C $\frac{5\pi}{3}$

D $\frac{\pi}{6}$

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q029 MCQ

Convert $-\frac{5\pi}{6}$ to degrees:

Choose the correct option.

A $-120°$

B $-150°$

C $150°$

D $-300°$

Q030 MCQ

For $r = 5, \theta = \frac{2\pi}{3}$, the arc length is:

Choose the correct option.

A $\frac{5\pi}{3}$

B $\frac{10\pi}{3}$

C $\frac{25\pi}{3}$

D 10π

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q031 MCQ

For $r = 4, \theta = \frac{\pi}{2}$, the sector area is:

Choose the correct option.

A π

B 2π

C 4π

D 8π

Q032 MCQ

The radian measure of a full turn is:

Choose the correct option.

A π

B 2π

C 180

D 360

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q033 MCQ

The degree measure of $\frac{7\pi}{4}$ is:

Choose the correct option.

A 210°

B 270°

C 315°

D 360°

Q034 MCQ

The principal angle for $-\frac{11\pi}{6}$ is:

Choose the correct option.

A $\frac{\pi}{6}$

B $-\frac{\pi}{6}$

C $\frac{5\pi}{6}$

D $\frac{11\pi}{6}$

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q035 MCQ

For $r = 3, \theta = \pi$, the sector area is:

Choose the correct option.

A $\frac{3\pi}{2}$

B 3π

C 9π

D $\frac{9\pi}{2}$

Q036 MCQ

The angle $\frac{5\pi}{3}$ lies in:

Choose the correct option.

A QI

B QII

C QIII

D QIV

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q037 **Written**

Convert $252°$ to radians.

Q038 **Written**

A sector has $r = 8$ and $\theta = \frac{3\pi}{4}$. Find L and S.

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q039 **Written**

If $\theta = \frac{4\pi}{3}$**, find** $\sin\theta, \cos\theta, \tan\theta$**.**

Q040 **Written**

Find the exact values (write undefined where appropriate): $\sin\frac{7\pi}{4}, \cos\frac{\pi}{3}, \tan\frac{\pi}{6}, \sin(-\frac{2\pi}{3}), \cos(-\pi), \tan(-\frac{\pi}{2})$.

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q041 **Written**

Find the exact values of the eighteen trigonometric functions listed in the source exercise (a)–(r), including negative and undefined cases.

Q042 **Written**

The panel height is $h(t) = 3\sin(\frac{2\pi}{3}t - \frac{\pi}{4}) + 1.5$. Find the exact height at $t = 0$.

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q043 MCQ

$\sin\frac{\pi}{6}$ **equals:**

Choose the correct option.

A 0

B $\frac{1}{2}$

C $\frac{\sqrt{3}}{2}$

D 1

Q044 MCQ

$\cos \frac{2\pi}{3}$ **equals:**

Choose the correct option.

A -1

B $-\frac{1}{2}$

C $\frac{1}{2}$

D $\frac{\sqrt{3}}{2}$

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q045 MCQ

$\tan\frac{3\pi}{4}$ **equals:**

Choose the correct option.

A -1

B 0

C 1

D undefined

Q046 MCQ

$\sin(-\frac{\pi}{2})$ **equals:**

Choose the correct option.

A -1

B 0

C 1

D undefined

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q047 MCQ

$\tan \frac{\pi}{2}$ **is:**

Choose the correct option.

A 0

B 1

C -1

D undefined

Q048 MCQ

$\cos \frac{5\pi}{3}$ **equals:**

Choose the correct option.

A $-\frac{1}{2}$

B $\frac{1}{2}$

C $-\frac{\sqrt{3}}{2}$

D $\frac{\sqrt{3}}{2}$

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q049 MCQ

If $P(x, y)$ lies on the unit circle, then $\sin\theta$ is:

Choose the correct option.

A x
B y
C x/y
D 1/x

Q050 MCQ

$\tan(-\frac{3\pi}{4})$ **equals:**

Choose the correct option.

A -1

B 0

C 1

D undefined

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q051 *MCQ*

$\sin \frac{11\pi}{6}$ **equals:**

Choose the correct option.

A $-\frac{1}{2}$

B $\frac{1}{2}$

C $-\frac{\sqrt{3}}{2}$

D $\frac{\sqrt{3}}{2}$

Q052 **Written**

Find $\sin\frac{5\pi}{6}, \cos\frac{4\pi}{3}, \tan\frac{7\pi}{4}$.

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q053 **Written**

Find $\sin(-\frac{5\pi}{6}), \cos(-\frac{7\pi}{6}), \tan(-\frac{\pi}{3})$.

Q054 **Written**

For $P = (-\frac{1}{2}, \frac{\sqrt{3}}{2})$ on the unit circle, find the corresponding $\sin\theta, \cos\theta, \tan\theta$.

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q055 **Written**

Find the exact value of $3\sin\frac{3\pi}{2} - 2\cos\pi$**.**

Q056 **Written**

State the two standard angles where $\tan\theta$ **is undefined in** $[0, 2\pi)$**.**

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q057 Written

Solve for θ, $0 \leq \theta < 2\pi$: **(a)** $\sin\theta = -\frac{1}{2}$; **(b)** $\cos\theta = \frac{1}{\sqrt{2}}$; **(c)** $\tan\theta = -\sqrt{3}$.

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q058 **Written**

Solve the six trigonometric equations in the source Try set for $0 \le \theta < 2\pi$: $\sin\theta = 1/\sqrt{2}$, $\cos\theta = -\sqrt{3}/2$, $\tan\theta = 1/\sqrt{3}$, $\sin\theta = 0$, $\cos\theta = 1/2$, $\tan\theta = -1$.

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q059 **Written**

Solve the sixteen equations in the source Exercise set (a)–(p) for $0 \le \theta < 2\pi$.

Q060 **Written**

Solve $\tan^2\theta + (1-\sqrt{3})\tan\theta = \sqrt{3}$ **for** $0 \leq \theta < 360°$.

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q061 MCQ

For $0 \leq \theta < 2\pi$, $\sin\theta = \frac{1}{2}$ gives:

Choose the correct option.

A $\frac{\pi}{6}, \frac{5\pi}{6}$

B $\frac{\pi}{3}, \frac{2\pi}{3}$

C $\frac{\pi}{6}, \frac{7\pi}{6}$

D $\frac{5\pi}{6}, \frac{11\pi}{6}$

Q062 MCQ

For $0 \leq \theta < 2\pi$, $\cos\theta = -\frac{1}{2}$ gives:

Choose the correct option.

A $\frac{\pi}{3}, \frac{5\pi}{3}$

B $\frac{2\pi}{3}, \frac{4\pi}{3}$

C $\frac{\pi}{6}, \frac{5\pi}{6}$

D $\frac{4\pi}{3}, \frac{5\pi}{3}$

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q063 MCQ

For $0 \leq \theta < 2\pi$, $\tan\theta = 1$ gives:

Choose the correct option.

A $\frac{\pi}{4}, \frac{5\pi}{4}$

B $\frac{3\pi}{4}, \frac{7\pi}{4}$

C $\frac{\pi}{4}, \frac{7\pi}{4}$

D $\frac{3\pi}{4}, \frac{5\pi}{4}$

Q064 MCQ

The solutions of $\sin\theta = 0$ in $[0, 2\pi)$ are:

Choose the correct option.

A $0, \pi$
B $\frac{\pi}{2}, \frac{3\pi}{2}$
C $0, 2\pi$
D π

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q065 MCQ

If $\cos\theta = 1$, then in $[0, 2\pi)$, θ is:

Choose the correct option.

A 0

B $\frac{\pi}{2}$

C π

D $\frac{3\pi}{2}$

Q066 **MCQ**

For $0 \leq \theta < 2\pi$, $\tan\theta = -\sqrt{3}$ gives:

Choose the correct option.

A $\frac{\pi}{3}, \frac{4\pi}{3}$

B $\frac{2\pi}{3}, \frac{5\pi}{3}$

C $\frac{\pi}{6}, \frac{7\pi}{6}$

D $\frac{5\pi}{6}, \frac{11\pi}{6}$

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q067 **MCQ**

For $0 \leq \theta < 2\pi$, $\sin\theta = -1$ gives:

Choose the correct option.

A $\frac{\pi}{2}$

B $\frac{3\pi}{2}$

C π

D 0

Q068 MCQ

For $0 \leq \theta < 2\pi$, $\cos\theta = 0$ gives:

Choose the correct option.

A $0, \pi$
B $\frac{\pi}{2}, \frac{3\pi}{2}$
C π
D $\frac{\pi}{4}, \frac{5\pi}{4}$

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q069 MCQ

The equation $\tan\theta = 0$ has solutions in $[0, 2\pi)$:

Choose the correct option.

A $0, \pi$
B $\frac{\pi}{2}, \frac{3\pi}{2}$
C $\frac{\pi}{4}, \frac{5\pi}{4}$
D $\pi, 2\pi$

Q070 **Written**

Solve $\cos\theta = \frac{\sqrt{3}}{2}$ **for** $0 \leq \theta < 2\pi$**.**

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q071 **Written**

Solve $\sin\theta = -\frac{\sqrt{3}}{2}$ **for** $0 \le \theta < 2\pi$**.**

Q072 **Written**

Solve $\tan\theta = \frac{1}{\sqrt{3}}$ **for** $0 \le \theta < 2\pi$.

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Q073 **Written**

Solve $2\cos\theta - 1 = 0$ **for** $0 \leq \theta < 2\pi$.

Q074 **Written**

Solve $2\sin\theta + \sqrt{2} = 0$ **for** $0 \leq \theta < 2\pi$.

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Quiz MCQ 1 **Concept Quiz · MCQ**

The radian measure of $240°$ is:

Choose the correct option.

A $\frac{2\pi}{3}$

B $\frac{4\pi}{3}$

C $\frac{3\pi}{4}$

D $\frac{5\pi}{3}$

Quiz MCQ 2 **Concept Quiz · MCQ**

If $\sin\theta = -\frac{\sqrt{3}}{2}, 0 \le \theta < 2\pi$, then θ is:

Choose the correct option.

A $\frac{\pi}{3}, \frac{2\pi}{3}$

B $\frac{4\pi}{3}, \frac{5\pi}{3}$

C $\frac{2\pi}{3}, \frac{4\pi}{3}$

D $\frac{5\pi}{6}, \frac{11\pi}{6}$

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Quiz MCQ 3 **Concept Quiz · MCQ**

For a sector with $r = 6$ and $\theta = \frac{\pi}{3}$, the area is:

Choose the correct option.

A 3π

B 6π

C 12π

D 18π

Quiz MCQ 4 **Concept Quiz · MCQ**

The general-angle family for $-300°$ is:

Choose the correct option.

A $60° + 360°\,n$

B $300° + 360°\,n$

C $-60° + 360°\,n$

D $120° + 360°\,n$

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Quiz WRITTEN 5 **Concept Quiz · Written**

Solve $\cos\theta = -\frac{\sqrt{2}}{2}$ **for** $0 \le \theta < 2\pi$**.**

Quiz WRITTEN 6 **Concept Quiz · Written**

Convert $-\frac{13\pi}{6}$ to degrees and state its principal angle.

Classified Practice **Defining Angles and Basic Trigonometric Functions**

Quiz WRITTEN 7 **Concept Quiz · Written**

Find L and S for a sector with $r = 10$ and $\theta = \frac{2\pi}{5}$.

Quiz WRITTEN 8 **Concept Quiz · Written**

Evaluate $3\sin\frac{7\pi}{6} - 2\cos\frac{4\pi}{3}$.

Answer key

Use the question number shown on each page.

Q001 $(a)\ 435°\quad (b)\ -855°$

Q002 $300°\ :\ QIV$

$-450°\ :\ negative\ y-axis$

Q003 $60° + 360° \times n\quad ;\quad 150° + 360° \times n$

Q004 $(a)\ -30°\quad (b)\ 140°$

Q005 $500°\ :\ QI\qquad -120°\ :\ QIII$

Q006 $160° + 360° \times n\quad ;\quad 220° + 360° \times n$

Q007 $30°;\ 405°;\ 110°;\ 310°;\ 300°;\ 240°$

Q008 QII; QII; QIII; QI; QII; QII

Q009 $250° + 360°n;\ 200° + 360°n;\ 120° + 360°n;\ 290° + 360°n;\ 120° + 360°n;\ 340° + 360°n$

Q010 $120° + 360°n\quad ;\quad 260° + 360°n$

Q011 A

Q012 C

Q013 B

Q014 D

Q015 B

Q016 B

Q017 A

Q018 A

Q019 C

Q020 B

Q021 $\frac{7\pi}{2};\ -315°;\ L = 4\pi,\ S = 6\pi$

Q022 $\frac{8\pi}{5}, -\frac{20\pi}{9};\ 330°, -108°;\ L = \frac{9\pi}{2}, S = \frac{27\pi}{2}$

Q023 $\frac{3\pi}{5}, \frac{\pi}{5}, -\frac{4\pi}{3}, -\frac{5\pi}{4}, \frac{7\pi}{3}, \frac{20\pi}{9}, -\frac{19\pi}{6}, -15\pi$

Q024 $210°, 144°, 120°, 240°, -225°, -90°, -990°, -70$

Q025 $(a)\ (7\pi, 21\pi);\ (b)\ (\frac{8\pi}{5}, \frac{16\pi}{5});\ (c)\ (\frac{15\pi}{2}, \frac{45\pi}{2})$

Q026 QI; QI; QII; QII; QIII; QIII; QIV; QIV

Q027 $h = 3\sqrt{5}\,\text{cm};\quad V = 36\pi\sqrt{5}\,\text{cm}^3$

Q028 A

Q029 B

Q030 B

Q031 B

Q032 B

Q033 C

Q034 A

Q035 D

Q036 D

Q037 $\frac{7\pi}{5}$

Q038 $L = 6\pi,\quad S = 24\pi$

Q039 $-\frac{\sqrt{3}}{2},\ -\frac{1}{2},\ \sqrt{3}$

Q040 $-\frac{\sqrt{2}}{2};\ \frac{1}{2};\ \frac{1}{\sqrt{3}};\ -\frac{\sqrt{3}}{2};\ -1;\ undefined$

Q041 $\frac{\sqrt{3}}{2}; -\frac{1}{2}; 0; -\frac{\sqrt{2}}{2}; \frac{1}{2}; -\frac{\sqrt{3}}{2}; -\sqrt{3}; 1; 0; -\frac{1}{2}, \frac{\sqrt{2}}{2}; \frac{\sqrt{3}}{2}; \frac{\sqrt{2}}{2}; -\frac{\sqrt{3}}{2}; -\frac{1}{2}; -\sqrt{3}; \frac{1}{\sqrt{3}}; undefined$

Q042 $\frac{3(1-\sqrt{2})}{2}\,\text{m}$

Q043 B

Q044 B

Q045 A

Q046 A

Q047 D

Q048 D

Q049 B

Q050 C

Q051 A

Q052 $\frac{1}{2}, -\frac{1}{2}, -1$

Q053 $-\frac{1}{2}, -\frac{\sqrt{3}}{2}, -\sqrt{3}$

Q054 $\frac{\sqrt{3}}{2}, -\frac{1}{2}, -\sqrt{3}$

Q055 -1

Q056 $\frac{\pi}{2}, \frac{3\pi}{2}$

Q057 $(a)\ \frac{7\pi}{6}, \frac{11\pi}{6};\ (b)\ \frac{\pi}{4}, \frac{7\pi}{4};\ (c)\ \frac{2\pi}{3}, \frac{5\pi}{3}$

Q058 $\frac{\pi}{4}, \frac{3\pi}{4};\ \frac{5\pi}{6}, \frac{7\pi}{6};\ \frac{\pi}{6}, \frac{7\pi}{6};\ 0, \pi;\ \frac{\pi}{3}, \frac{5\pi}{3};\ \frac{3\pi}{4}, \frac{7\pi}{4}$

Q059 $\frac{\pi}{6}, \frac{5\pi}{6};\ \frac{\pi}{3}, \frac{2\pi}{3};\ \frac{\pi}{2}, \frac{3\pi}{2};\ 0;\ 0, \pi;\ \frac{\pi}{3}, \frac{4\pi}{3};\ \frac{\pi}{2};\ \frac{5\pi}{4}, \frac{7\pi}{4};\ \frac{3\pi}{4}, \frac{5\pi}{4};\ \frac{2\pi}{3}, \frac{4\pi}{3};\ \frac{5\pi}{6}, \frac{11\pi}{6};\ \frac{\pi}{4}, \frac{5\pi}{4};\ \frac{4\pi}{3}, \frac{5\pi}{3};\ \frac{3\pi}{2};\ \pi;\ \frac{\pi}{6}, \frac{11\pi}{6}$

Q060 $60°, 135°, 240°, 315°$

Q061 A

Q062 B

Q063 A

Q064 A

Q065 A

Q066 B

Q067 B

Q068 B

Q069 A

Q070 $\frac{\pi}{6}, \frac{11\pi}{6}$

Q071 $\frac{4\pi}{3}, \frac{5\pi}{3}$

Q072 $\frac{\pi}{6}, \frac{7\pi}{6}$

Q073 $\frac{\pi}{3}, \frac{5\pi}{3}$

Q074 $\frac{5\pi}{4}, \frac{7\pi}{4}$

Quiz MCQ 1 B

Quiz MCQ 2 B

Quiz MCQ 3 C

Quiz MCQ 4 A

Quiz WRITTEN 5 $\frac{3\pi}{4}, \frac{5\pi}{4}$

Quiz WRITTEN 6 $-390°;\ 330°$

Quiz WRITTEN 7 $L = 4\pi,\quad S = 20\pi$

Quiz WRITTEN 8 $-\frac{1}{2}$

Concept 2 | Trigonometric Identities and Expression Evaluation

English Student Edition • Focused formulas and guided practice

Formula opener

Identity

$$\sin^2\theta + \cos^2\theta = 1$$

Tangent

$$\tan\theta = \frac{\sin\theta}{\cos\theta}$$

Secant

$$1 + \tan^2\theta = \sec^2\theta$$

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Classified Practice Trigonometric Identities and Expression Evaluation

Q001 Written

Given (a) $\sin\theta = -\frac{1}{5}$ **in QIII and (b)** $\tan\theta = -\sqrt{6}$ **with** $\frac{3\pi}{2} < \theta < 2\pi$**, find the other two trigonometric values in each case.**

Classified Practice **Trigonometric Identities and Expression Evaluation**

Q002 **Written**

Given one value and the quadrant, find the other two: (a) $\cos\theta = -1/\sqrt{3}$**, QIII; (b)** $\tan\theta = -2$**, QIV; (c)** $\sin\theta = 3/5$**, QII; (d)** $\tan\theta = \sqrt{3}/2$, $\pi < \theta < 3\pi/2$**.**

Classified Practice **Trigonometric Identities and Expression Evaluation**

Q003 **Written**

For each of the twelve numbered cases in the source Exercise, one of $\sin\theta, \cos\theta, \tan\theta$ and a quadrant/interval is given. Find the other two values.

Q004 **Written**

Using trigonometric identities, find the other two values given $\tan\theta + \cot\theta = 12$ **and** $\pi < \theta < 2\pi$**.**

Classified Practice **Trigonometric Identities and Expression Evaluation**

Q005 **Written**

Given $\sin\theta = \frac{5}{13}$ **in QII, find** $\cos\theta$ **and** $\tan\theta$**.**

Q006 **Written**

Given $\cos\theta = -\frac{3}{5}$ **in QIII, find** $\sin\theta$ **and** $\tan\theta$**.**

Classified Practice **Trigonometric Identities and Expression Evaluation**

Q007 **Written**

Given $\tan\theta = \frac{3}{4}$ **in QI, find** $\sin\theta$ **and** $\cos\theta$**.**

Q008 **Written**

Given $\tan\theta = -\frac{5}{12}$ **in QIV, find** $\sin\theta$ **and** $\cos\theta$**.**

Classified Practice **Trigonometric Identities and Expression Evaluation**

Q009 **Written**

Given $\sin\theta = -\frac{\sqrt{2}}{2}$ **in QIII, find** $\cos\theta$ **and** $\tan\theta$**.**

Q010 MCQ

If $\sin\theta = -\frac{3}{5}$ in QIII, $\cos\theta$ is:

Choose the correct option.

A $\frac{4}{5}$

B $-\frac{4}{5}$

C $\frac{3}{5}$

D $-\frac{3}{5}$

Classified Practice **Trigonometric Identities and Expression Evaluation**

Q011 MCQ

If $\tan\theta = 2$ in QI, $\sin\theta$ is:

Choose the correct option.

A $\frac{2\sqrt{5}}{5}$

B $\frac{\sqrt{5}}{5}$

C $-\frac{2\sqrt{5}}{5}$

D $\frac{2}{5}$

Q012 MCQ

If $\cos\theta = \frac{1}{\sqrt{2}}$ **in QIV,** $\tan\theta$ **is:**

Choose the correct option.

A 1

B -1

C $\sqrt{2}$

D $-\sqrt{2}$

Classified Practice **Trigonometric Identities and Expression Evaluation**

Q013 MCQ

For $\tan\theta = -\sqrt{3}$ **in QII,** $\sin\theta$ **is:**

Choose the correct option.

A $\frac{1}{2}$

B $-\frac{1}{2}$

C $\frac{\sqrt{3}}{2}$

D $-\frac{\sqrt{3}}{2}$

Q014 MCQ

If $\sin\theta = \frac{7}{25}$ **in QI,** $\tan\theta$ **is:**

Choose the correct option.

A $\frac{7}{24}$

B $\frac{24}{7}$

C $-\frac{7}{24}$

D $\frac{25}{7}$

Classified Practice Trigonometric Identities and Expression Evaluation

Q015 Written

Given $\sin\theta + \cos\theta = \frac{1}{2}$**, with** θ **in QII, evaluate (a)** $\sin\theta\cos\theta$**, (b)** $\sin^3\theta + \cos^3\theta$**, (c)** $\sin\theta - \cos\theta$**.**

Q016 **Written**

Given $\sin\theta + \cos\theta = \frac{1}{3}$**, with** θ **in QII, evaluate (a)** $\sin\theta\cos\theta$**, (b)** $\sin^3\theta + \cos^3\theta$**, (c)** $\sin\theta - \cos\theta$**.**

Classified Practice **Trigonometric Identities and Expression Evaluation**

Q017 **Written**

Solve the three numbered Exercise problems: (1) given $\sin\theta - \cos\theta = \sqrt{3}/2$ in QI; (2) given $\sin\theta + \cos\theta = \sqrt{3}/2$ in QIV; (3) given $\sin\theta - \cos\theta = 1/2$ in QIII. Evaluate the three expressions requested in each case.

Classified Practice **Trigonometric Identities and Expression Evaluation**

Q018 **Written**

Given $\sin\theta - \cos\theta = 0.5$ **and** θ **in QIII, evaluate the exact value of** $\sin^6\theta - \cos^6\theta$**.**

Q019 **Written**

Given $\sin\theta + \cos\theta = \frac{2}{3}$ **in QII, find** $\sin\theta\cos\theta$ **and** $\sin\theta - \cos\theta$**.**

Classified Practice **Trigonometric Identities and Expression Evaluation**

Q020 **Written**

Given $\sin\theta - \cos\theta = \frac{\sqrt{5}}{2}$ **in QI, find** $\sin\theta\cos\theta$ **and** $\sin^3\theta - \cos^3\theta$**.**

Q021 **Written**

Given $\sin\theta + \cos\theta = -\frac{1}{2}$ **in QIII, find** $\sin^3\theta + \cos^3\theta$**.**

Classified Practice **Trigonometric Identities and Expression Evaluation**

Q022 **Written**

Given $\sin\theta - \cos\theta = \frac{1}{3}$ **in QIV, find** $\sin\theta + \cos\theta$**.**

Q023 **Written**

Given $\sin\theta - \cos\theta = 1$ **in QIII, find** $\sin^3\theta - \cos^3\theta$.

Classified Practice **Trigonometric Identities and Expression Evaluation**

Q024 MCQ

If $s + c = \frac{1}{2}$, then sc equals:

Choose the correct option.

A $-\frac{3}{8}$

B $\frac{3}{8}$

C $-\frac{1}{4}$

D $\frac{1}{4}$

Q025 MCQ

If $s + c = \frac{1}{2}$, then $s^3 + c^3$ equals:

Choose the correct option.

A $\frac{5}{16}$

B $\frac{11}{16}$

C $-\frac{11}{16}$

D $\frac{3}{16}$

Classified Practice **Trigonometric Identities and Expression Evaluation**

Q026 MCQ

If $s - c = \frac{1}{2}$ in QIII, then $s + c$ equals:

Choose the correct option.

A $\frac{\sqrt{7}}{2}$

B $-\frac{\sqrt{7}}{2}$

C $\frac{1}{2}$

D $-\frac{1}{2}$

Q027 MCQ

If $s - c = \frac{\sqrt{3}}{2}$ in QI, then sc equals:

Choose the correct option.

A $\frac{1}{8}$

B $-\frac{1}{8}$

C $\frac{3}{8}$

D $-\frac{3}{8}$

Classified Practice **Trigonometric Identities and Expression Evaluation**

Q028 MCQ

If $s - c = 0.5$ in QIII, then $s^6 - c^6$ equals:

Choose the correct option.

A $-\frac{55\sqrt{7}}{256}$

B $\frac{55\sqrt{7}}{256}$

C $-\frac{11\sqrt{7}}{16}$

D $\frac{11}{16}$

Q029 **Written**

Prove $\frac{\sin\theta}{1+\cos\theta} + \frac{1}{\tan\theta} = \frac{1}{\sin\theta}$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q030 Written

Prove (a) $\frac{1-\tan\theta}{1+\tan\theta}=\frac{\cos\theta-\sin\theta}{\cos\theta+\sin\theta}$**; (b)** $\frac{\cos\theta}{1-\sin\theta}-\tan\theta=\frac{1}{\cos\theta}$**.**

Classified Practice **Trigonometric Identities and Expression Evaluation**

Q031 **Written**

Prove the six identities in the source Exercise: (a) $\tan^2\theta - \sin^2\theta = \tan^2\theta\sin^2\theta$; **(b)** $\tan\theta + 1/\tan\theta = 1/(\sin\theta\cos\theta)$; **(c)** $\cos\theta/(1 - \sin\theta) - 1/\cos\theta = \tan\theta$; **(d)** $\tan\theta/\sin\theta - \sin\theta/\tan\theta = \sin\theta\tan\theta$; **(e)** $(1 + \sin\theta)/\cos\theta + \cos\theta/(1 + \sin\theta) = 2/\cos\theta$; **(f)** $(\tan^2\theta - \sin^2\theta)/\sin^2\theta = \tan^2\theta$.

Classified Practice **Trigonometric Identities and Expression Evaluation**

Q032 **Written**

Prove that, for all defined values of x, $\frac{1+\sin x-\cos x}{1+\sin x+\cos x}+\frac{1+\sin x+\cos x}{1+\sin x-\cos x}=2\csc x$.

Q033 **Written**

Prove $\frac{1+\sin\theta}{\cos\theta} = \frac{\cos\theta}{1-\sin\theta}$.

Classified Practice **Trigonometric Identities and Expression Evaluation**

Q034 **Written**

Prove $\frac{1}{1+\cos\theta} + \frac{1}{1-\cos\theta} = 2\sec^2\theta$.

Q035 **Written**

Prove $\frac{\tan\theta}{1+\tan\theta} = \frac{\sin\theta}{\sin\theta+\cos\theta}$.

Classified Practice **Trigonometric Identities and Expression Evaluation**

Q036 **Written**

Prove $\frac{\sin\theta}{1-\cos\theta} = \frac{1+\cos\theta}{\sin\theta}$.

Q037 **Written**

Prove $\frac{\cos\theta}{1+\sin\theta} = \frac{1-\sin\theta}{\cos\theta}$.

Classified Practice **Trigonometric Identities and Expression Evaluation**

Q038 MCQ

The first move in proving a trig identity is usually to:

Choose the correct option.

A Change both sides at once

B Simplify one side

C Substitute random values

D Take a square root

Q039 MCQ

The identity used to replace $1 - \sin^2\theta$ is:

Choose the correct option.

A $\cos^2\theta$

B $\tan^2\theta$

C $\sec^2\theta$

D $\sin^2\theta$

Classified Practice **Trigonometric Identities and Expression Evaluation**

Q040 **MCQ**

$\tan\theta$ **should be rewritten as:**

Choose the correct option.

A $\sin\theta\cos\theta$

B $\sin\theta/\cos\theta$

C $\cos\theta/\sin\theta$

D $1/\sin\theta$

Q041 **MCQ**

To add two trig fractions, first find a:

Choose the correct option.

A common numerator

B common denominator

C new quadrant

D reference angle

Classified Practice **Trigonometric Identities and Expression Evaluation**

Q042 MCQ

The result of a valid identity proof is:

Choose the correct option.

A LHS=0

B LHS=RHS

C RHS=1

D undefined

Q043 **MCQ**

If $\cos\theta = -\frac{12}{13}$ in QII, $\sin\theta$ is:

Choose the correct option.

A $\frac{5}{13}$

B $-\frac{5}{13}$

C $\frac{12}{13}$

D $-\frac{12}{5}$

Classified Practice **Trigonometric Identities and Expression Evaluation**

Q044 MCQ

If $\tan\theta = -\frac{1}{2}$ **in QII,** $\cos\theta$ **is:**

Choose the correct option.

A $\frac{2\sqrt{5}}{5}$

B $-\frac{2\sqrt{5}}{5}$

C $\frac{\sqrt{5}}{5}$

D $-\frac{\sqrt{5}}{5}$

Q045 MCQ

If $\sin\theta = \frac{8}{17}$ **in QIV,** $\tan\theta$ **is:**

Choose the correct option.

A $\frac{8}{15}$

B $-\frac{8}{15}$

C $\frac{15}{8}$

D $-\frac{15}{8}$

Classified Practice **Trigonometric Identities and Expression Evaluation**

Q046 MCQ

If $\tan\theta = \sqrt{3}$ **and** θ **is in QIII**, $\sin\theta$ **is:**

Choose the correct option.

A $\frac{1}{2}$

B $-\frac{1}{2}$

C $\frac{\sqrt{3}}{2}$

D $-\frac{\sqrt{3}}{2}$

Q047 MCQ

If $s + c = 1$, then sc equals:

Choose the correct option.

A 0
B $\frac{1}{2}$
C $-\frac{1}{2}$
D 1

Classified Practice **Trigonometric Identities and Expression Evaluation**

Q048 MCQ

If $s - c = \sqrt{2}$, then sc equals:

Choose the correct option.

A $-\frac{1}{2}$

B $\frac{1}{2}$

C -1

D 1

Q049 MCQ

If $s + c = \frac{2}{3}$, then $s^3 + c^3$ equals:

Choose the correct option.

A $\frac{19}{27}$

B $\frac{5}{27}$

C $-\frac{19}{27}$

D $\frac{11}{27}$

Classified Practice **Trigonometric Identities and Expression Evaluation**

Q050 MCQ

If $s - c = \frac{1}{2}$ in QIII, then $s^3 - c^3$ equals:

Choose the correct option.

A $\frac{11}{16}$

B $-\frac{11}{16}$

C $\frac{7}{16}$

D $-\frac{7}{16}$

Q051 **MCQ**

Which identity converts $\frac{1}{\tan\theta}$ into sine and cosine?

Choose the correct option.

A $\frac{\sin\theta}{\cos\theta}$

B $\frac{\cos\theta}{\sin\theta}$

C $\sin\theta\cos\theta$

D $\sec\theta$

Classified Practice **Trigonometric Identities and Expression Evaluation**

Q052 MCQ

The identity $1 - \cos^2\theta$ simplifies to:

Choose the correct option.

A $\sin^2\theta$

B $\tan^2\theta$

C $\cos^2\theta$

D 1

Q053 **MCQ**

When proving an identity, which side should normally be transformed?

Choose the correct option.

A Both sides simultaneously

B One side only

C Neither side

D Only the denominator

Classified Practice **Trigonometric Identities and Expression Evaluation**

Q054 **MCQ**

After combining trig fractions, the next step is usually to use:

Choose the correct option.

A A random angle
B The Pythagorean identity
C A logarithm
D A derivative

Quiz MCQ 1 **Concept Quiz · MCQ**

If $\sin\theta + \cos\theta = \frac{1}{2}$, **then** $\sin\theta\cos\theta$ **is:**

Choose the correct option.

A $-\frac{3}{8}$

B $\frac{3}{8}$

C $-\frac{1}{4}$

D $\frac{1}{4}$

Classified Practice **Trigonometric Identities and Expression Evaluation**

Quiz MCQ 2 **Concept Quiz · MCQ**

If $\tan\theta = \frac{3}{4}$ in QI, then $\cos\theta$ is:

Choose the correct option.

A $\frac{3}{5}$

B $\frac{4}{5}$

C $-\frac{4}{5}$

D $\frac{3}{4}$

Quiz MCQ 3 **Concept Quiz · MCQ**

Which identity is always valid where defined?

Choose the correct option.

A $\tan\theta = \frac{\cos\theta}{\sin\theta}$

B $1 + \tan^2\theta = \frac{1}{\cos^2\theta}$

C $\sin\theta + \cos\theta = 1$

D $\sin\theta\cos\theta = 1$

Classified Practice

Trigonometric Identities and Expression Evaluation

Quiz MCQ 4 **Concept Quiz · MCQ**

If $s - c = \frac{1}{2}$ in QIII, then $s + c$ is:

Choose the correct option.

A $\frac{\sqrt{7}}{2}$

B $-\frac{\sqrt{7}}{2}$

C $\frac{1}{2}$

D $-\frac{1}{2}$

Quiz WRITTEN 5 **Concept Quiz · Written**

Given $\cos\theta = -\frac{5}{13}$ **in QIII, find** $\sin\theta$ **and** $\tan\theta$**.**

Classified Practice **Trigonometric Identities and Expression Evaluation**

Quiz WRITTEN 6 Concept Quiz · Written

Given $s + c = \frac{2}{3}$, find sc and $s^3 + c^3$.

Quiz WRITTEN 7 **Concept Quiz · Written**

Prove $\frac{1-\sin\theta}{\cos\theta} = \frac{\cos\theta}{1+\sin\theta}$.

Classified Practice **Trigonometric Identities and Expression Evaluation**

Quiz WRITTEN 8 **Concept Quiz · Written**

Given $\tan\theta = -2$ in QIV, find $\sin\theta$ and $\cos\theta$, then verify $1 + \tan^2\theta = 1/\cos^2\theta$.

Answer key

Use the question number shown on each page.

Q001 $(a)\ \cos\theta = -\frac{2\sqrt{6}}{5},\ \tan\theta = \frac{\sqrt{6}}{12};\quad (b)\ \cos\theta = \frac{\sqrt{7}}{7},\ \sin\theta = -\frac{\sqrt{42}}{7}$

Q002 $(a)\ \sin\theta = -\frac{\sqrt{6}}{3},\ \tan\theta = \sqrt{2};\ (b)\ \cos\theta = \frac{\sqrt{5}}{5},\ \sin\theta = -\frac{2\sqrt{5}}{5};\ (c)\ \cos\theta = -\frac{4}{5},\ \tan\theta = -\frac{3}{4};\ (d)\ \sin\theta = -\frac{\sqrt{21}}{7},\ \cos\theta = -\frac{2\sqrt{7}}{7}$

Q003 $(a)\ (-\frac{\sqrt{7}}{4}, \frac{3\sqrt{7}}{7})\ (b)\ (-\frac{\sqrt{5}}{5}, -\frac{1}{2})\ (c)\ (\frac{\sqrt{2}}{4}, -\frac{\sqrt{14}}{4})\ (d)\ (-\frac{\sqrt{5}}{3}, -\frac{2\sqrt{5}}{5})\ (e)\ (-\frac{\sqrt{15}}{4}, -\sqrt{15})\ (f)\ (\frac{1}{3}, -\frac{2\sqrt{2}}{3})\ (g)\ (-\frac{3\sqrt{10}}{10}, \frac{\sqrt{10}}{10})\ (h)\ (-\frac{\sqrt{5}}{5}, -2)\ (i)\ (-\frac{12}{13}, -\frac{12}{5})\ (j)\ (\frac{12}{13}, -\frac{5}{12})\ (k)\ (-\frac{3}{5}, \frac{3}{4})\ (l)\ (\frac{2\sqrt{5}}{5}, -\frac{\sqrt{5}}{5})$

Q004 $\sin\theta = -\frac{1}{2},\quad \cos\theta = -\frac{\sqrt{3}}{2}$ or the swapped pair

Q005 $-\frac{12}{13}, -\frac{5}{12}$

Q006 $-\frac{4}{5}, \frac{4}{3}$

Q007 $\frac{3}{5}, \frac{4}{5}$

Q008 $-\frac{5}{13}, \frac{12}{13}$

Q009 $-\frac{\sqrt{2}}{2}, 1$

Q010 B

Q011 A

Q012 B

Q013 C

Q014 A

Q015 $(a)\ -\frac{3}{8}\quad (b)\ \frac{11}{16}\quad (c)\ \frac{\sqrt{7}}{2}$

Q016 $(a)\ -\frac{4}{9}\quad (b)\ \frac{37}{27}\quad (c)\ \frac{\sqrt{17}}{3}$

Q017 $(1)\ \sin\theta\cos\theta = \frac{1}{8},\ \sin^3\theta - \cos^3\theta = \frac{7\sqrt{3}}{16},\ \sin\theta + \cos\theta = \frac{\sqrt{5}}{2};\ (2)\ -\frac{1}{8},\ \frac{9\sqrt{3}}{16},\ -\frac{\sqrt{5}}{2};\ (3)\ \frac{3}{8},\ \frac{11}{16},\ -\frac{\sqrt{7}}{2}$

Q018 $-\frac{55\sqrt{7}}{256}$

Q019 $-\frac{5}{18}, \frac{\sqrt{14}}{3}$

Q020 $-\frac{1}{8}, \frac{7\sqrt{5}}{16}$

Q021 $-\frac{11}{16}$

Q022 $\frac{\sqrt{11}}{3}$

Q023 $\frac{3}{2}$

Q024 A

Q025 B

Q026 B

Q027 A

Q028 A

Q029 LHS = RHS

Q030 Both identities verified

Q031 LHS = RHS for (a)+(f)

Q032 LHS = $2\csc x$

Q033 Identity proved

Q034 Identity proved

Q035 Identity proved

Q036 Identity proved

Q037 Identity proved

Q038 B

Q039 A

Q040 B

Q041 B

Q042 B

Q043 A

Q044 A

Q045 B

Q046 D

Q047 A

Q048 A

Q049 A

Q050 A

Q051 B

Q052 A

Q053 B

Q054 B

Quiz MCQ 1 A

Quiz MCQ 2 B

Quiz MCQ 3 B

Quiz MCQ 4 B

Quiz WRITTEN 5 $-\frac{12}{13}, \frac{12}{5}$

Quiz WRITTEN 6 $-\frac{5}{18}, \frac{19}{27}$

Quiz WRITTEN 7 Identity proved

Quiz WRITTEN 8 $\sin\theta = -\frac{2\sqrt{5}}{5},\ \cos\theta = \frac{\sqrt{5}}{5};\ 6 = 6$

Concept 3 | Properties and Graphs of Trigonometric Functions

English Student Edition • Focused formulas and guided practice

Formula opener

Amplitude

$$y = a\sin(bx + c) + d \Rightarrow |a|$$

Period

$$T = \frac{2\pi}{|b|}$$

Range

$$d - |a| \le y \le d + |a|$$

Eng Eslam Ahmed | 01120009622

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q001 **Written**

(1) Express $\sin\frac{9\pi}{5}$ **as a trigonometric function of an acute angle. (2) Simplify** $\sin\theta + \sin(\theta + \frac{\pi}{2}) + \sin(\theta + \pi) + \sin(\theta + \frac{3\pi}{2})$**.**

Classified Practice   **Properties and Graphs of Trigonometric Functions**

Q002 **Written**

(1) Express $\cos(-\frac{\pi}{6})$ **as a trigonometric function of an acute angle. (2) Express** $\sin\frac{8\pi}{7}$ **as a trigonometric function of an acute angle. (3) Simplify** $\cos(\frac{\pi}{2}-\theta)\sin(\pi-\theta)-\sin(\frac{3\pi}{2}+\theta)\cos(\pi-\theta)$**.**

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q003 **Written**

Solve all three source demands: (1) express $\cos\frac{17\pi}{5}, \cos\frac{5\pi}{7}, \sin(-\frac{7\pi}{8}), \sin(-\frac{19\pi}{8})$ **using acute angles; (2) find** $\tan(-\frac{\pi}{6})$ **and** $\tan(-\frac{\pi}{4})$**; (3) simplify (a)** $\sin\theta + \cos(\pi+\theta) - \cos(\pi-\theta) + \sin(\pi+\theta)$**, (b)** $\sin(\theta+\frac{\pi}{2})\cos(\theta+\pi) + \sin(-\theta)\cos(\frac{\pi}{2}-\theta)$**.**

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q004 **Written**

Simplify $\sin(\frac{3\pi}{2} - \theta)\sec(\pi + \theta) - \tan(\theta - \frac{\pi}{2})\cot(\frac{5\pi}{2} + \theta)$.

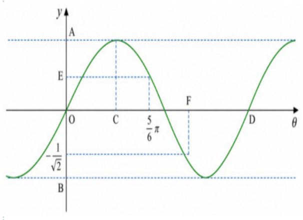
Q005 **Written**

Draw the parent graphs $y = \sin\theta$, $y = \cos\theta$, and $y = \tan\theta$, and state each period and range.

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q006 **Written**

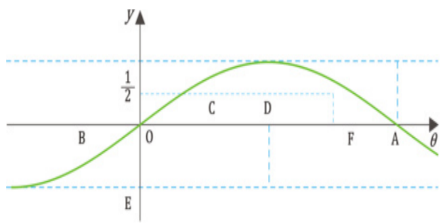
For the source graph of $y = \sin\theta$, determine the labelled scales A to F.



Classified Practice Properties and Graphs of Trigonometric Functions

Q007 Written

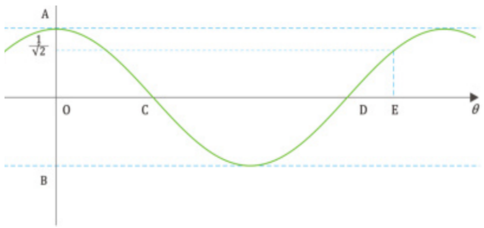
(1) Draw the parent graphs $y = \sin\theta$, $y = \cos\theta$, $y = \tan\theta$, and state their periods. (2) For the source graph of $y = \cos\theta$, determine the labelled scales A to F.

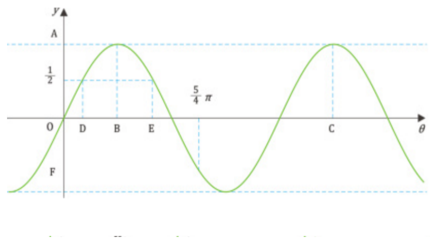


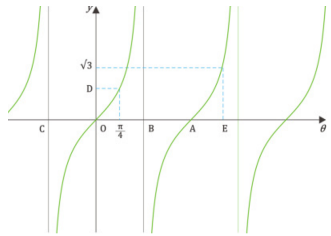
Classified Practice — **Properties and Graphs of Trigonometric Functions**

Q008 Written

Complete all four source graph tasks: draw the three parent graphs and find their periods; read the labelled scales on the source cosine graph; read the labelled scales on the source sine graph; and read the labelled scales on the source tangent graph.





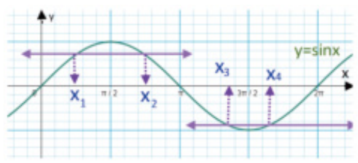


Classified Practice **Properties and Graphs of Trigonometric Functions**

Q009 Written

The source graph represents $y = \sin x$ for $0 < x < 2\pi$. Find $x_1 + x_2 + x_3 + x_4$ from the four marked x-coordinates.

$y = sinx, 0 < x < 2\pi$



Then find: $x_1 + x_2 + x_3 + x_4$

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q010 **Written**

Draw the graphs of $y = 2\cos\theta$ and $y = \tan\frac{\theta}{2}$, and find their periods.

Q011 **Written**

Draw the graphs of $y = 2\sin\theta$, $y = \tan 2\theta$, and $y = \cos\frac{\theta}{2}$, and find their periods.

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q012 **Written**

Draw the nine source graphs and find each period: **(a)** $3\sin\theta$, **(b)** $3\cos\theta$, **(c)** $\frac{1}{2}\cos\theta$, **(d)** $\sin 3\theta$, **(e)** $\cos 2\theta$, **(f)** $\tan 3\theta$, **(g)** $\sin\frac{\theta}{2}$, **(h)** $\cos\frac{\theta}{3}$, **(i)** $\tan\frac{\theta}{3}$.

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q013 **Written**

Draw $y = -2|\sin(3x)|$ and find its period.

Q014 **Written**

Draw $y = \sin(\theta - \frac{\pi}{6})$ and find its period.

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q015 **Written**

Draw the graphs and find their periods: (1) $y = \sin(\theta - \frac{\pi}{3})$**, (2)** $y = \cos(\theta + \frac{\pi}{4})$**, (3)** $y = \tan(\theta - \frac{\pi}{2})$**.**

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q016 Written

Draw all source graphs and find their periods: sine group $y = \sin(\theta + \frac{\pi}{4}), \sin(\theta - \frac{\pi}{2}), \sin(\theta + \frac{\pi}{3})$; **cosine group** $y = \cos(\theta - \frac{\pi}{3}), \cos(\theta - \frac{\pi}{4}), \cos(\theta + \frac{\pi}{6})$; **tangent group** $y = \tan(\theta + \frac{\pi}{6}), \tan(\theta - \frac{\pi}{3}), \tan(\theta - \frac{\pi}{4})$**.**

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q017 **Written**

Draw $y = 3\cos\left|x + \frac{\pi}{3}\right|$ and find its period.

Q018 **Written**

Draw the graphs and find their periods: (1) $y = 2\cos(2\theta - \frac{\pi}{2})$**, (2)** $y = \tan(\frac{\theta}{2} - \frac{\pi}{6})$**.**

Classified Practice Properties and Graphs of Trigonometric Functions

Q019 Written

Draw the graphs and find their periods: (a) $y = 3\sin\frac{\theta}{2}$, **(b)** $y = \tan(2\theta - \frac{\pi}{2})$, **(c)** $y = 2\cos(2\theta - \frac{\pi}{3})$.

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q020 Written

Draw all nine source graphs and find their periods: (a) $2\cos\frac{\theta}{2}$, **(b)** $3\sin 2\theta$, **(c)** $2\cos 2\theta$, **(d)** $2\cos(\theta-\frac{\pi}{4})$, **(e)** $2\sin(\theta-\frac{\pi}{3})$, **(f)** $\tan(2\theta-\frac{\pi}{3})$, **(g)** $3\cos(2\theta-\frac{\pi}{2})$, **(h)** $3\sin(2\theta-\frac{2\pi}{3})$, **(i)** $2\cos(\frac{\theta}{2}-\frac{\pi}{4})$.

Classified Practice Properties and Graphs of Trigonometric Functions

Q021 Written

Draw $y = 3\sin|2x - \pi|$ and find its period.

Q022 **MCQ**

$\sin(\theta + \pi)$ **equals:**

Choose the correct option.

A $\sin \theta$

B $-\sin \theta$

C $\cos \theta$

D $-\cos \theta$

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q023 **MCQ**

$\cos(\theta + \pi)$ **equals:**

Choose the correct option.

A $\cos\theta$

B $-\cos\theta$

C $\sin\theta$

D $-\sin\theta$

Q024 **MCQ**

$\sin(\frac{\pi}{2} - \theta)$ **equals:**

Choose the correct option.

A $\sin\theta$

B $-\sin\theta$

C $\cos\theta$

D $-\cos\theta$

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q025 **MCQ**

$\cos(-\theta)$ **equals:**

Choose the correct option.

A $\cos\theta$

B $-\cos\theta$

C $\sin\theta$

D $-\sin\theta$

Q026 **MCQ**

$\sin(-\theta)$ **equals:**

Choose the correct option.

A $\sin\theta$

B $-\sin\theta$

C $\cos\theta$

D $-\cos\theta$

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q027 **MCQ**

$\cos(\pi - \theta)$ **equals:**

Choose the correct option.

A $\cos\theta$

B $-\cos\theta$

C $\sin\theta$

D $-\sin\theta$

Q028 MCQ

$\sin(\frac{3\pi}{2} + \theta)$ **equals:**

Choose the correct option.

A $\sin\theta$

B $-\sin\theta$

C $\cos\theta$

D $-\cos\theta$

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q029 **MCQ**

$\tan(\theta - \frac{\pi}{2})$ **equals:**

Choose the correct option.

A $\tan\theta$

B $-\tan\theta$

C $\cot\theta$

D $-\cot\theta$

Q030 MCQ

The period of $y = \sin\theta$ is:

Choose the correct option.

A $\pi/2$

B π

C 2π

D 4π

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q031 **MCQ**

The range of $y = \cos\theta$ is:

Choose the correct option.

A $y \geq 0$

B $-1 \leq y \leq 1$

C $y \in \mathbb{R}$

D y>1

Q032 **MCQ**

The period of $y = \tan\theta$ is:

Choose the correct option.

A $\pi/2$

B π

C 2π

D 4π

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q033 MCQ

A vertical asymptote of $y = \tan\theta$ occurs at:

Choose the correct option.

A $\theta = k\pi$

B $\theta = \frac{\pi}{2} + k\pi$

C $\theta = 2k\pi$

D $\theta = \frac{\pi}{4} + k\pi$

Q034 MCQ

The maximum value of $\sin\theta$ is:

Choose the correct option.

A -1

B 0

C 1

D π

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q035 MCQ

The x-coordinate of the first positive sine maximum is:

Choose the correct option.

A $\pi/4$
B $\pi/2$
C π
D $3\pi/2$

Q036 MCQ

At $\theta = 0$, $\cos\theta$ equals:

Choose the correct option.

A -1

B 0

C 1

D undefined

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q037 MCQ

At $\theta = 0$, $\tan\theta$ equals:

Choose the correct option.

A -1

B 0

C 1

D undefined

Q038 **MCQ**

The range of $y = 3\sin\theta$ is:

Choose the correct option.

A [-1,1]

B [-3,3]

C [0,3]

D $\mathbb{R}$

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q039 MCQ

The period of $y = \sin 4\theta$ is:

Choose the correct option.

A $\pi/4$

B $\pi/2$

C π

D 2π

Q040 MCQ

The period of $y = \cos(\theta/3)$ is:

Choose the correct option.

A $2\pi/3$

B 2π

C 3π

D 6π

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q041 **MCQ**

The period of $y = \tan 5\theta$ is:

Choose the correct option.

A $\pi/5$

B π

C 5π

D $2\pi/5$

Q042 **MCQ**

The range of $y = \frac{1}{2}\cos\theta$ is:

Choose the correct option.

A [-2,2]

B [-1,1]

C [-1/2,1/2]

D [0,1/2]

Classified Practice Properties and Graphs of Trigonometric Functions

Q043 MCQ

The graph of $y = -2\sin\theta$ is the parent sine graph:

Choose the correct option.

A shifted only

B reflected and stretched vertically

C compressed horizontally only

D undefined

Q044 MCQ

The period of $y = \tan(\theta/2)$ is:

Choose the correct option.

A $\pi/2$

B π

C 2π

D 4π

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q045 MCQ

The period of $y = \sin(3\theta)$ is:

Choose the correct option.

A $\pi/3$

B $2\pi/3$

C 3π

D 6π

Q046 **MCQ**

The graph of $y = \sin(\theta - \pi/4)$ is shifted:

Choose the correct option.

A left by $\pi/4$
B right by $\pi/4$
C up by $\pi/4$
D down by $\pi/4$

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q047 **MCQ**

The graph of $y = \cos(\theta + \pi/3)$ is shifted:

Choose the correct option.

A right by $\pi/3$

B left by $\pi/3$

C up by $\pi/3$

D down by $\pi/3$

Q048 MCQ

A horizontal shift changes the sine period from 2π to:

Choose the correct option.

A π

B 2π

C 4π

D 0

Classified Practice Properties and Graphs of Trigonometric Functions

Q049 MCQ

A horizontal shift changes the tangent period from π to:

Choose the correct option.

A $\pi/2$

B π

C 2π

D 4π

Q050 **MCQ**

The graph of $y = \sin(\theta + \pi/2)$ equals:

Choose the correct option.

A $\sin\theta$
B $-\sin\theta$
C $\cos\theta$
D $-\cos\theta$

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q051 MCQ

The graph of $y = \cos(\theta - \pi)$ equals:

Choose the correct option.

A $\cos\theta$

B $-\cos\theta$

C $\sin\theta$

D $-\sin\theta$

Q052 **MCQ**

The graph of $y = \tan(\theta - \pi/2)$ has its central zero at:

Choose the correct option.

A $\theta = 0$

B $\theta = \pi/2$

C $\theta = \pi$

D $\theta = 2\pi$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q053 MCQ

Since cosine is even, $\cos|x|$ **equals:**

Choose the correct option.

A $\cos(x)$

B $-\cos(x)$

C $\sin(x)$

D $|\cos(x)|$

Q054 MCQ

In $y = a\sin(b\theta - q)$, the horizontal shift is:

Choose the correct option.

A q
B q/b
C b/q
D a/b

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q055 **MCQ**

The period of $y = 2\cos(3\theta - \pi)$ is:

Choose the correct option.

A $\pi/3$

B $2\pi/3$

C π

D 2π

Q056 MCQ

The range of $y = 4\sin(2\theta - 1)$ is:

Choose the correct option.

A [-1,1]

B [-2,2]

C [-4,4]

D $\mathbb{R}$

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q057 **MCQ**

For $y = \tan(2\theta - \pi/3)$, the shift is:

Choose the correct option.

A left by $\pi/3$

B right by $\pi/6$

C right by $\pi/3$

D left by $\pi/6$

Q058 MCQ

The period of $y = 3\sin(\theta/2)$ is:

Choose the correct option.

A π
B 2π
C 3π
D 4π

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q059 MCQ

The correct transformation order for $y = af(b\theta - q)$ starts with:

Choose the correct option.

A vertical scale

B horizontal shift

C horizontal scale

D reflection only

Q060 MCQ

The period of $y = \tan(4\theta + q)$ is:

Choose the correct option.

A $\pi/4$

B $\pi/2$

C π

D 2π

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q061 MCQ

The range of $y = 3\sin|2x - \pi|$ is:

Choose the correct option.

A [-3,3]
B [0,3]
C [-1,1]
D $\mathbb{R}$

Q062 **Written**

Express $\cos \frac{11\pi}{6}$ **as a function of an acute angle.**

Classified Practice Properties and Graphs of Trigonometric Functions

Q063 **Written**

Simplify $\sin(\theta + \pi) - \cos(\frac{\pi}{2} - \theta)$.

Q064 **Written**

Simplify $\cos(\theta + \pi) + \cos(\frac{\pi}{2} - \theta)$.

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q065 **Written**

Evaluate $\tan(-\frac{3\pi}{4})$ using an acute reference angle.

Q066 **Written**

State the period and range of $y = \cos\theta$, then list its key points on one cycle.

Classified Practice Properties and Graphs of Trigonometric Functions

Q067 Written

State the period and asymptotes of $y = \tan\theta$ on $[-\pi, \pi]$.

Q068 **Written**

For the parent sine graph, find the x-values where $y = 0$ in $[0, 2\pi]$.

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q069 **Written**

For the parent cosine graph, find the x-values where $y = 0$ in $[0, 2\pi]$.

Q070 **Written**

Find the period and range of $y = 4\cos\theta$.

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q071 **Written**

Find the period and range of $y = \sin(5\theta)$.

Q072 **Written**

Find the period and range of $y = \tan(\theta/4)$.

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q073 **Written**

Describe the transformations from $y = \cos\theta$ to $y = -\frac{1}{2}\cos(2\theta)$.

Q074 **Written**

Find the shift and period of $y = \sin(\theta + \frac{\pi}{6})$**.**

Classified Practice Properties and Graphs of Trigonometric Functions

Q075 Written

Find the shift and period of $y = \cos(\theta - \frac{\pi}{2})$.

Q076 **Written**

Find the central zero and period of $y = \tan(\theta + \frac{\pi}{4})$**.**

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q077 **Written**

For $y = 2\sin(4\theta - \pi)$, state the shift, period and range.

Q078 **Written**

For $y = \cos(\theta/3 + \pi/6)$**, state the shift and period.**

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q079 **Written**

For $y = 3\tan(2\theta + \pi)$, state the shift and period.

Q080 **Written**

For $y = 2\cos(\theta/2 - \pi/8)$, state the shift, period and range.

Classified Practice **Properties and Graphs of Trigonometric Functions**

Quiz MCQ 1 **Concept Quiz · MCQ**

The period of $y = \sin(2\theta - \frac{\pi}{3})$ is:

Choose the correct option.

A $\pi/2$

B π

C 2π

D 4π

Quiz MCQ 2 **Concept Quiz · MCQ**

$\sin(\theta + \pi)$ **equals:**

Choose the correct option.

A $\sin\theta$

B $-\sin\theta$

C $\cos\theta$

D $-\cos\theta$

Classified Practice **Properties and Graphs of Trigonometric Functions**

Quiz MCQ 3 **Concept Quiz · MCQ**

The range of $y = 2\cos\theta$ is:

Choose the correct option.

A [-1,1]

B [-2,2]

C [0,2]

D $\mathbb{R}$

Quiz MCQ 4 **Concept Quiz · MCQ**

The graph of $y = \cos(\theta - \frac{\pi}{4})$ is shifted:

Choose the correct option.

A left by $\frac{\pi}{4}$

B right by $\frac{\pi}{4}$

C up by $\frac{\pi}{4}$

D down by $\frac{\pi}{4}$

Classified Practice Properties and Graphs of Trigonometric Functions

Quiz WRITTEN 5 Concept Quiz · Written

For $y = 3\sin(2\theta - \pi)$, state the horizontal shift, period and range.

Quiz WRITTEN 6 **Concept Quiz · Written**

Simplify $\sin(\theta + \frac{\pi}{2}) + \sin(\theta + \pi)$.

Classified Practice **Properties and Graphs of Trigonometric Functions**

Quiz WRITTEN 7 **Concept Quiz · Written**

Draw one cycle of $y = \tan(\theta - \frac{\pi}{3})$ and state its period and asymptotes.

Quiz WRITTEN 8 **Concept Quiz · Written**

For $y = -2|\sin(3x)|$, state its period and range and describe the reflection.

Answer key

Use the question number shown on each page.

Q001 (1) $-\sin\frac{\pi}{5}$ (2) 0

Q002 (1) $\cos\frac{\pi}{6}$ (2) $-\sin\frac{\pi}{7}$ (3) $\sin^2\theta-\cos^2\theta$

Q003 (1) $-\cos\frac{2\pi}{5}, -\cos\frac{2\pi}{7}, -\sin\frac{\pi}{8}, -\sin\frac{3\pi}{8}$ (2) $-\frac{1}{\sqrt{3}}, -1$ (3a) 0, (3b) -1

Q004 0

Q005 $\sin,\cos:\ T=2\pi,\ -1\le y\le 1;\quad \tan:\ T=\pi,\ y\in\mathbb{R}$

Q006 $A=1,\ B=-1,\ C=\frac{\pi}{2},\ D=2\pi,\ E=\frac{1}{2},\ F=\frac{5\pi}{4}$

Q007 (1) $T_{\sin}=2\pi,\ T_{\cos}=2\pi,\ T_{\tan}=\pi$; (2) read the marked extrema, intercepts and quarter-period points from the graph

Q008 Use the standard cycles: sine/cosine have period 2π and range $[-1,1]$; tangent has period π, range $\mathbb{R}$, and asymptotes at odd multiples of $\frac{\pi}{2}$. The requested A+F labels are read from the marked ticks and extrema in each supplied graph.

Q009 Read x_1, x_2, x_3, x_4 from the supplied graph and add the four marked coordinates; the result is the graph-reading total.

Q010 $y=2\cos\theta:\ T=2\pi,\ -2\le y\le 2;\quad y=\tan\frac{\theta}{2}:\ T=2\pi$

Q011 $T=2\pi,\ \frac{\pi}{2},\ 4\pi$ respectively; ranges $[-2,2],\ \mathbb{R},\ [-1,1]$

Q012 (a) 2π (b) 2π (c) 2π (d) $\frac{2\pi}{3}$ (e) π (f) $\frac{\pi}{3}$ (g) 4π (h) 6π (i) 3π; ranges follow the outside coefficient

Q013 $T=\frac{\pi}{3},\quad 0\ge y\ge -2$

Q014 shift right by $\frac{\pi}{6},\quad T=2\pi$

Q015 Shifts: right $\frac{\pi}{3}$, left $\frac{\pi}{4}$, right $\frac{\pi}{2}$; periods $2\pi, 2\pi, \pi$.

Q016 All sine/cosine periods are 2π; all tangent periods are π. The shifts are read directly from the brackets.

Q017 $y=3\cos(x+\frac{\pi}{3})$ since cosine is even; $T=2\pi$

Q018 (1) $y=2\cos 2(\theta-\frac{\pi}{4}),\ T=\pi,\ -2\le y\le 2$; (2) $y=\tan\frac{1}{2}(\theta-\frac{\pi}{3}),\ T=2\pi$

Q019 (a) $T=4\pi,\ -3\le y\le 3$; (b) $T=\frac{\pi}{2}$; (c) $T=\pi,\ -2\le y\le 2$

Q020 Periods: (a) 4π; (b) π; (c) π; (d) 2π; (e) 2π; (f) $\frac{\pi}{2}$; (g) π; (h) π; (i) 4π. Ranges are set by the outside factors.

Q021 $y=3|\sin(2x-\pi)|,\quad T=\frac{\pi}{2},\quad 0\le y\le 3$

Q022 B

Q023 B

Q024 C

Q025 A

Q026 B

Q027 B

Q028 D

Q029 D

Q030 C

Q031 B

Q032 B

Q033 B

Q034 C

Q035 B

Q036 C

Q037 B

Q038 B

Q039 B

Q040 D

Q041 A

Q042 C

Q043 B

Q044 C

Q045 B

Q046 B

Q047 B

Q048 B

Q049 B

Q050 C

Q051 B

Q052 B

Q053 A

Q054 B

Q055 B

Q056 C

Q057 B

Q058 D

Q059 B

Q060 A

Q061 B

Q062 $\cos\frac{\pi}{6}$

Q063 $-\sin\theta-\sin\theta=-2\sin\theta$

Q064 $-\cos\theta+\sin\theta$

Q065 1

Q066 $T=2\pi,\ -1\le y\le 1;\ (0,1),(\frac{\pi}{2},0),(\pi,-1),(\frac{3\pi}{2},0),(2\pi,1)$

Q067 $T=\pi;\ \theta=\pm\frac{\pi}{2}$

Q068 $0,\pi,2\pi$

Q069 $\frac{\pi}{2},\frac{3\pi}{2}$

Q070 $T=2\pi,\ -4\le y\le 4$

Q071 $T=\frac{2\pi}{5},\ -1\le y\le 1$

Q072 $T=4\pi,\ y\in\mathbb{R}$

Q073 horizontal compression by 1/2, reflection in x-axis and vertical scale 1/2; $T=\pi$

Q074 left by $\frac{\pi}{6},\ T=2\pi$

Q075 right by $\frac{\pi}{2},\ T=2\pi$

Q076 $\theta=-\frac{\pi}{4};\ T=\pi$

Q077 right by $\frac{\pi}{4},\ T=\frac{\pi}{2},\ -2\le y\le 2$

Q078 left by $\frac{\pi}{2},\ T=6\pi$

Q079 left by $\frac{\pi}{2},\ T=\frac{\pi}{2}$

Q080 right by $\frac{\pi}{4},\ T=4\pi,\ -2\le y\le 2$

Quiz MCQ 1 B

Quiz MCQ 2 B

Quiz MCQ 3 B

Quiz MCQ 4 B

Quiz WRITTEN 6 $\cos\theta - \sin\theta$

Quiz WRITTEN 7 $T = \pi;\ \theta = \dfrac{5\pi}{6} + k\pi$

Quiz WRITTEN 8 $T = \dfrac{\pi}{3},\ -2 \le y \le 0;$ reflected below the

Quiz WRITTEN 5 right by $\dfrac{\pi}{2},\ T = \pi,\ -3 \le y \le 3$

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

English Student Edition • Focused formulas and guided practice

Formula opener

Sine

$$\sin x = \sin \alpha$$

Cosine

$$\cos x = \cos \alpha$$

Tangent

$$\tan x = \tan \alpha$$

Eng Eslam Ahmed | 01120009622

Classified Practice **Solving Equations and Inequalities Involving Trigonometric Functions**

Q001 **Written**

Solve (a) $-2\cos\theta = \sqrt{2}$ and (b) $\tan\theta + \sqrt{3} = 0$, giving the general solution for (b).

Q002 **Written**

Solve (a) $2\cos\theta - 1 = 0$ **and (b)** $\tan\theta - 1 = 0$ **for** $0 \le \theta \le 2\pi$**.**

Classified Practice Solving Equations and Inequalities Involving Trigonometric Functions

Q003 Written

Solve all source equations on $0 \leq \theta \leq 2\pi$: **(a)** $2\sin\theta + 1 = 0$, **(b)** $2\sin\theta = \sqrt{3}$, **(c)** $3\sin\theta + 3 = 0$, **(d)** $\sqrt{2}\cos\theta + 1 = 0$, **(e)** $2\cos\theta = \sqrt{3}$, **(f)** $2\cos\theta - \sqrt{2} = 0$, **(g)** $\tan\theta + 1 = 0$, **(h)** $\sqrt{3}\tan\theta = 1$, **(i)** $\sqrt{3}\tan\theta = 3$, **and give general solutions for (j)** $2\sin\theta + \sqrt{3} = 0$, **(k)** $\cos\theta + 1 = 0$, **(l)** $\sqrt{3}\tan\theta + 3 = 0$.

Classified Practice **Solving Equations and Inequalities Involving Trigonometric Functions**

Q004 **Written**

Solve $\tan^2\theta - (2+\sqrt{3})\tan\theta + 2\sqrt{3} = 0$ **and give the general solution.**

Q005 **Written**

Solve for $0 \le \theta < 2\pi$**: (a)** $2\cos\theta > -\sqrt{3}$**, (b)** $\tan\theta \le \frac{1}{\sqrt{3}}$**.**

Classified Practice **Solving Equations and Inequalities Involving Trigonometric Functions**

Q006 Written

Solve for $0 \le \theta < 2\pi$**: (a)** $2\sin\theta \le -1$**, (b)** $\cos\theta > \frac{1}{2}$**, (c)** $\tan\theta > 1$**.**

Classified Practice **Solving Equations and Inequalities Involving Trigonometric Functions**

Q007 **Written**

Solve all source inequalities for $0 \le \theta < 2\pi$**:** **(a)** $\sin\theta > \frac{\sqrt{3}}{2}$, **(b)** $\sin\theta - \frac{\sqrt{2}}{2} > 0$, **(c)** $2\sin\theta + \sqrt{3} \le 0$, **(d)** $\cos\theta > 0$, **(e)** $\cos\theta - \frac{1}{2} \le 0$, **(f)** $2\cos\theta + \sqrt{2} \le 0$, **(g)** $\tan\theta \le -\frac{1}{\sqrt{3}}$, **(h)** $\tan\theta + \sqrt{3} > 0$, **(i)** $\tan\theta + 1 \ge 0$.

Classified Practice **Solving Equations and Inequalities Involving Trigonometric Functions**

Q008 **Written**

Solve $2\sin^2\theta - \cos\theta - 1 > 0$ **for** $0 \le \theta < 2\pi$.

Q009 **Written**

Solve for $0 \leq \theta < 2\pi$: **(a)** $2\cos^2\theta - 3\sin\theta = 0$, **(b)** $2\sin^2\theta - 3\cos\theta - 3 \leq 0$.

Classified Practice **Solving Equations and Inequalities Involving Trigonometric Functions**

Q010 **Written**

Solve for $0 \leq \theta < 2\pi$**: (a)** $2\sin^2\theta - \sin\theta = 0$**, (b)** $2\cos^2\theta - \sin\theta = 2$**, (c)** $2\sin^2\theta - 4 < 5\cos\theta$**.**

Classified Practice **Solving Equations and Inequalities Involving Trigonometric Functions**

Q011 Written

Solve all source equations and inequalities for $0 \le \theta < 2\pi$: **(a)** $2\sin^2\theta + \sin\theta - 1 = 0$, **(b)** $2\sin^2\theta + 3\cos\theta = 0$, **(c)** $2\sin^2\theta - \cos\theta = 2$, **(d)** $2\cos^2\theta - 5\cos\theta - 3 = 0$, **(e)** $2\cos^2\theta + \sin\theta - 1 = 0$, **(f)** $2\cos^2\theta + 3\sin\theta = 3$, **(g)** $2\cos^2\theta + 3\sin\theta \le 0$, **(h)** $2\sin^2\theta + \cos\theta > 1$, **(i)** $2\cos^2\theta - 4 \le 5\sin\theta$.

Classified Practice **Solving Equations and Inequalities Involving Trigonometric Functions**

Q012 **Written**

Solve $\frac{1-2\sin^2\theta+\sin\theta}{\cos^2\theta} \geq 1$ **for** $0 \leq \theta < 2\pi$.

Q013 **Written**

Solve for $0 \le \theta < 2\pi$**: (a)** $\sin(\theta + \frac{\pi}{3}) = \frac{1}{2}$, **(b)** $\sin(\theta + \frac{\pi}{6}) > -\frac{1}{2}$.

Classified Practice **Solving Equations and Inequalities Involving Trigonometric Functions**

Q014 **Written**

Solve for $0 \leq \theta < 2\pi$**: (a)** $\sin(\theta + \frac{\pi}{6}) = \frac{1}{2}$, **(b)** $\tan(\theta + \frac{\pi}{4}) = \sqrt{3}$, **(c)** $\cos(\theta + \frac{\pi}{4}) < -\frac{\sqrt{2}}{2}$**.**

Classified Practice Solving Equations and Inequalities Involving Trigonometric Functions

Q015 Written

Solve for $0 \leq \theta < 2\pi$: **(a)** $\cos(\theta + \frac{\pi}{3}) = -\frac{\sqrt{3}}{2}$, **(b)** $\cos(\theta + \frac{\pi}{6}) = -\frac{1}{2}$, **(c)** $\sin(\theta + \frac{\pi}{4}) = \frac{\sqrt{3}}{2}$, **(d)** $\tan(\theta + \frac{\pi}{4}) = -1$, **(e)** $\tan(\theta + \frac{\pi}{3}) = \frac{1}{\sqrt{3}}$, **(f)** $\tan(\theta + \frac{\pi}{6}) = 1$, **(g)** $\sin(\theta + \frac{\pi}{6}) \leq \frac{\sqrt{3}}{2}$, **(h)** $\sin(\theta + \frac{\pi}{4}) \geq \frac{\sqrt{2}}{2}$, **(i)** $\cos(\theta + \frac{\pi}{3}) > \frac{1}{2}$.

Classified Practice **Solving Equations and Inequalities Involving Trigonometric Functions**

Q016 Written

Solve $4\sin^2(\theta + \frac{\pi}{6}) \geq 1$ **for** $0 \leq \theta < 2\pi$.

Classified Practice **Solving Equations and Inequalities Involving Trigonometric Functions**

Q017 **Written**

A student divides $\tan x \times \cos x = \frac{1}{2}$ **by** $\cos x$**, obtaining** $\tan x = \frac{1}{2\cos x}$**. Evaluate this strategy and propose a better substitution on** $[0°, 360°[$**.**

Q018 MCQ

The period used for a general tangent solution is:

Choose the correct option.

A 2π
B π
C $\pi/2$
D 4π

Q019 **MCQ**

The solutions of $2\cos\theta = 1$ on $[0, 2\pi]$ are:

Choose the correct option.

A $\pi/6, 11\pi/6$

B $\pi/3, 5\pi/3$

C $2\pi/3, 4\pi/3$

D $\pi/4, 7\pi/4$

Classified Practice **Solving Equations and Inequalities Involving Trigonometric Functions**

Q020 MCQ

Which root is impossible when solving $2\sin^2\theta + 3\sin\theta - 2 = 0$?

Choose the correct option.

A $\sin\theta = 1/2$
B $\sin\theta = -2$
C $\sin\theta = 0$
D $\sin\theta = -1/2$

Q021 MCQ

The quadratic challenge factors into roots:

Choose the correct option.

A 1 and $\sqrt{3}$

B 2 and $\sqrt{3}$

C 2 and 3

D $\pi/3$ and 2

Classified Practice **Solving Equations and Inequalities Involving Trigonometric Functions**

Q022 MCQ

For $2\cos\theta > -\sqrt{3}$, the excluded arc is:

Choose the correct option.

A $[0, 5\pi/6]$

B $[5\pi/6, 7\pi/6]$

C $[7\pi/6, 2\pi)$

D $(\pi/6, 5\pi/6)$

Q023 **MCQ**

The inequality $\tan\theta > 1$ **on** $[0, 2\pi)$ **holds on:**

Choose the correct option.

A $(\pi/4, \pi/2) \cup (5\pi/4, 3\pi/2)$

B $[0, \pi/4]$

C $(\pi/2, \pi)$

D $[\pi, 2\pi)$

Classified Practice **Solving Equations and Inequalities Involving Trigonometric Functions**

Q024 **MCQ**

A closed endpoint in a trig inequality is used when the sign is:

Choose the correct option.

A strictly greater
B strictly less
C greater than or equal to
D never

Q025 MCQ

The 5-14 challenge reduces to which cosine condition?

Choose the correct option.

A $\cos\theta > 1/2$

B $-1 < \cos\theta < 1/2$

C $\cos\theta \leq -1$

D $\cos\theta \geq 1$

Q026 MCQ

When a quadratic gives $\sin\theta = -2$, we:

Choose the correct option.

A keep it
B reject it
C square it
D divide by 2

Q027 **MCQ**

The first step for $2\cos^2\theta - 3\sin\theta = 0$ is to use:

Choose the correct option.

A $\sin^2\theta + \cos^2\theta = 1$

B $\tan\theta = \sin\theta$

C $\cos\theta = 1$

D the tangent period

Classified Practice **Solving Equations and Inequalities Involving Trigonometric Functions**

Q028 MCQ

The solution of $2\sin^2\theta + \sin\theta - 1 = 0$ includes:

Choose the correct option.

A $\theta = \pi/6$
B $\theta = \pi/4$
C $\theta = \pi/3$
D $\theta = 2\pi/3$

Q029 *MCQ*

In the rational challenge, which angle must be excluded?

Choose the correct option.

A 0

B $\pi/4$

C $\pi/2$

D π

Classified Practice **Solving Equations and Inequalities Involving Trigonometric Functions**

Q030 MCQ

After setting $t = \theta + \pi/3$ with $0 \le \theta < 2\pi$, the t-range is:

Choose the correct option.

A $[0, 2\pi)$
B $[\pi/3, 7\pi/3)$
C $[-\pi/3, 5\pi/3)$
D $[\pi, 3\pi)$

Q031 **MCQ**

The solutions of $\tan(\theta + \pi/4) = \sqrt{3}$ **include:**

Choose the correct option.

A $\pi/12$

B $\pi/6$

C $\pi/3$

D $\pi/2$

Classified Practice **Solving Equations and Inequalities Involving Trigonometric Functions**

Q032 MCQ

For $\sin(\theta + \pi/4) \geq \sqrt{2}/2$, the θ-solution is:

Choose the correct option.

A $[0, \pi/2]$
B $(0, \pi/2)$
C $[\pi/2, \pi]$
D $[0, 2\pi)$

Q033 *MCQ*

The 5-16 challenge is equivalent to:

Choose the correct option.

A $|\sin(t)| \geq 1/2$

B $\sin(t) = 0$

C $\cos(t) > 1/2$

D $\tan(t) = 1$

Classified Practice **Solving Equations and Inequalities Involving Trigonometric Functions**

Q034 MCQ

In the Reflect problem, $\tan x \cos x$ **simplifies to:**

Choose the correct option.

A $\cos(x)$
B $\sin(x)$
C $\tan^2 x$
D 1

Quiz MCQ 1 **Concept Quiz · MCQ**

The general solution of $\tan\theta = -1$ is:

Choose the correct option.

A $-\pi/4 + 2n\pi$

B $-\pi/4 + n\pi$

C $\pi/4 + 2n\pi$

D $\pi/2 + n\pi$

Classified Practice **Solving Equations and Inequalities Involving Trigonometric Functions**

Quiz MCQ 2 **Concept Quiz · MCQ**

The solution of $\cos\theta > 0$ on $[0, 2\pi)$ is:

Choose the correct option.

A $(\pi/2, 3\pi/2)$
B $[0, \pi/2) \cup (3\pi/2, 2\pi)$
C $[0, \pi)$
D $(\pi, 2\pi)$

Quiz MCQ 3 **Concept Quiz · MCQ**

The valid sine root of $2\sin^2\theta + 3\sin\theta - 2 = 0$ is:

Choose the correct option.

A $\sin\theta = -2$

B $\sin\theta = 1/2$

C $\sin\theta = 2$

D $\sin\theta = -1/2$

Classified Practice **Solving Equations and Inequalities Involving Trigonometric Functions**

Quiz MCQ 4 **Concept Quiz · MCQ**

The first operation in a shifted-argument problem is to:

Choose the correct option.

A divide by $\cos\theta$

B set t equal to the whole argument and find its range

C change radians to degrees

D square both sides

Quiz WRITTEN 5 **Concept Quiz · Written**

Solve $2\cos\theta = \sqrt{3}$ **for** $0 \le \theta < 2\pi$**.**

Classified Practice **Solving Equations and Inequalities Involving Trigonometric Functions**

Quiz WRITTEN 6 **Concept Quiz · Written**

Solve $\tan\theta \leq -1$ **for** $0 \leq \theta < 2\pi$.

Quiz WRITTEN 7 **Concept Quiz · Written**

Solve $2\sin^2\theta + \sin\theta - 1 = 0$ **for** $0 \le \theta < 2\pi$.

Classified Practice **Solving Equations and Inequalities Involving Trigonometric Functions**

Quiz WRITTEN 8 **Concept Quiz · Written**

Solve $\cos(\theta + \frac{\pi}{3}) > \frac{1}{2}$ **for** $0 \leq \theta < 2\pi$.

Answer key

Use the question number shown on each page.

Q001 $(a)\ \theta = \frac{3\pi}{4}, \frac{5\pi}{4} \quad (b)\ \theta = \frac{2\pi}{3} + n\pi,\ n \in \mathbb{Z}$

Q002 $(a)\ \theta = \frac{\pi}{3}, \frac{5\pi}{3} \quad (b)\ \theta = \frac{\pi}{4}, \frac{5\pi}{4}$

Q003 $(a)\ \frac{7\pi}{6}, \frac{11\pi}{6}\ (b)\ \frac{\pi}{3}, \frac{2\pi}{3}\ (c)\ \frac{3\pi}{2}\ (d)\ \frac{3\pi}{4}, \frac{5\pi}{4}\ (e)\ \frac{\pi}{6}, \frac{11\pi}{6}\ (f)\ \frac{\pi}{4}, \frac{7\pi}{4}\ (g)\ \frac{3\pi}{4}, \frac{7\pi}{4}\ (h)\ \frac{\pi}{6}, \frac{7\pi}{6}\ (i)\ \frac{\pi}{3}, \frac{4\pi}{3}\ (j)\ \frac{4\pi}{3} + 2n\pi, \frac{5\pi}{3} + 2n\pi\ (k)\ \pi + 2n\pi\ (l)\ \frac{2\pi}{3}$ ...

Q004 $\tan\theta = 2$ or $\tan\theta = \sqrt{3}; \quad \theta = \arctan(2) + n\pi$ or $\theta = \frac{\pi}{3} + n\pi$

Q005 $(a)\ 0 \le \theta < \frac{5\pi}{6}$ or $\frac{7\pi}{6} < \theta < 2\pi \quad (b)\ 0 \le \theta \le \frac{\pi}{6}$ or $\frac{\pi}{2} < \theta \le \frac{7\pi}{6}$ or $\frac{3\pi}{2} < \theta < 2\pi$

Q006 $(a)\ \frac{7\pi}{6} \le \theta \le \frac{11\pi}{6} \quad (b)\ 0 \le \theta < \frac{\pi}{3}$ or $\frac{5\pi}{3} < \theta < 2\pi \quad (c)\ \frac{\pi}{4} < \theta < \frac{\pi}{2}$ or $\frac{5\pi}{4} < \theta < \frac{3\pi}{2}$

Q007 $(a)\ (\frac{\pi}{3}, \frac{2\pi}{3})\ (b)\ (\frac{\pi}{4}, \frac{3\pi}{4})\ (c)\ [\frac{4\pi}{3}, \frac{5\pi}{3}]\ (d)\ [0, \frac{\pi}{2}) \cup (\frac{3\pi}{2}, 2\pi)\ (e)\ [\frac{\pi}{3}, \frac{5\pi}{3}]\ (f)\ [\frac{3\pi}{4}, \frac{5\pi}{4}]\ (g)\ (\frac{\pi}{2}, \frac{5\pi}{6}] \cup (\frac{3\pi}{2}, \frac{11\pi}{6}]\ (h)\ [0, \frac{\pi}{2}) \cup (\frac{2\pi}{3}, \frac{3\pi}{2}) \cup (\frac{5\pi}{3}, 2\pi)\ (i)\ [0, \frac{\pi}{2}) \cup [\frac{3\pi}{4}, \frac{3\pi}{2}) \cup [\frac{7\pi}{4}, 2\pi)$

Q008 $\frac{\pi}{3} < \theta < \pi$ or $\pi < \theta < \frac{5\pi}{3}$

Q009 $(a)\ \theta = \frac{\pi}{6}, \frac{5\pi}{6} \quad (b)\ 0 \le \theta \le \frac{2\pi}{3}$ or $\theta = \pi$ or $\frac{4\pi}{3} \le \theta < 2\pi$

Q010 $(a)\ \theta = 0, \frac{\pi}{6}, \frac{5\pi}{6}, \pi \quad (b)\ \theta = 0, \pi, \frac{7\pi}{6}, \frac{11\pi}{6} \quad (c)\ 0 \le \theta < \frac{2\pi}{3}$ or $\frac{4\pi}{3} < \theta < 2\pi$

Q011 $(a)\ \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}\ (b)\ \frac{2\pi}{3}, \frac{4\pi}{3}\ (c)\ \frac{2\pi}{3}, \frac{\pi}{2}, \frac{4\pi}{3}, \frac{3\pi}{2}\ (d)\ \frac{2\pi}{3}, \frac{4\pi}{3}\ (e)\ \frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}\ (f)\ \frac{\pi}{6}, \frac{5\pi}{6}, \frac{\pi}{2}\ (g)\ \frac{7\pi}{6} \le \theta \le \frac{11\pi}{6}\ (h)\ 0 < \theta < \frac{2\pi}{3}$ or $\frac{4\pi}{3} < \theta < 2\pi\ (i)\ 0 \le \theta \le \frac{7\pi}{6}$ or $\frac{11\pi}{6} \le \theta < 2\pi$

Q012 $0 \le \theta < \frac{\pi}{2}$ or $\frac{\pi}{2} < \theta \le \pi$

Q013 $(a)\ \theta = \frac{\pi}{2}, \frac{11\pi}{6} \quad (b)\ 0 \le \theta < \pi$ or $\frac{5\pi}{3} < \theta < 2\pi$

Q014 $(a)\ \theta = 0, \frac{2\pi}{3} \quad (b)\ \theta = \pi 12, \frac{13\pi}{12} \quad (c)\ \frac{\pi}{2} < \theta < \pi$

Q015 $(a)\ \frac{\pi}{2}, \frac{5\pi}{6}\ (b)\ \frac{\pi}{2}, \frac{7\pi}{6}\ (c)\ \pi 12, \frac{5\pi}{12}\ (d)\ \frac{\pi}{2}, \frac{3\pi}{2}\ (e)\ \frac{5\pi}{6}, \frac{11\pi}{6}\ (f)\ \pi 12, \frac{13\pi}{12}\ (g)\ [0, \frac{\pi}{6}] \cup [\frac{\pi}{2}, 2\pi)\ (h)\ [0, \frac{\pi}{2}]\ (i)\ \frac{4\pi}{3} < \theta < 2\pi$

Q016 $0 \le \theta \le \frac{2\pi}{3}$ or $\pi \le \theta \le \frac{5\pi}{3}$

Q017 Dividing by $\cos(x)$ separates the interacting factors and is not helpful; use $\tan(x)\cos(x) = \sin(x)$ (with $\cos(x) \neq 0$), so $\sin(x) = \frac{1}{2}$ and $x = 30°, 150°$.

Q018 B

Q019 B

Q020 B

Q021 B

Q022 B

Q023 A

Q024 C

Q025 B

Q026 B

Q027 A

Q028 A

Q029 C

Q030 B

Q031 A

Q032 A

Q033 A

Q034 B

Quiz MCQ 1 B

Quiz MCQ 2 B

Quiz MCQ 3 B

Quiz MCQ 4 B

Quiz WRITTEN 5 $\theta = \frac{\pi}{6}, \frac{11\pi}{6}$

Quiz WRITTEN 6 $\frac{\pi}{2} < \theta \le \frac{3\pi}{4}$ or $\frac{3\pi}{2} < \theta \le \frac{7\pi}{4}$

Quiz WRITTEN 7 $\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$

Quiz WRITTEN 8 $\frac{4\pi}{3} < \theta < 2\pi$

Concept 5 | Advanced Theorems and Applications

English Student Edition • Focused formulas and guided practice

Formula opener

Addition

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

Double angle

$$\sin 2A = 2 \sin A \cos A$$

Cosine

$$\cos 2A = \cos^2 A - \sin^2 A$$

Eng Eslam Ahmed | 01120009622

Classified Practice **Advanced Theorems and Applications**

Q001 **Written**

Find the maximum and minimum of $y = sin^2\theta + cos\theta + 1, 0le\theta < 2pi$, **and the corresponding** θ.

Q002 **Written**

Find the maximum and minimum of $y = cos^2\theta - sin\theta, 0le\theta < 2pi$, and the corresponding θ.

Classified Practice Advanced Theorems and Applications

Q003 **Written**

Solve the four source maximum/minimum problems on $0le\theta < 2pi$**, including** $y = \sin^2\theta + \sin\theta - 1$, $y = \cos^2\theta - \cos\theta + 1$, $y = 2\sin\theta + \cos^2\theta - 1$**, and** $y = -2\sin^2\theta - 2\cos\theta + 1$. **Include the source challenge** $y = \cos^2\theta - \sin\theta$.

Classified Practice **Advanced Theorems and Applications**

Q004 **Written**

Use addition theorems to find $\sin 75°$ **and** $\cos(\pi/12)$**.**

Q005 **Written**

Given $\sin\alpha = \frac{\sqrt{3}}{2}$ **in QII and** $\cos\beta = -\frac{3}{5}$ **in QIII, find** $\sin(\alpha + \beta)$ **and** $\cos(\alpha - \beta)$**.**

Classified Practice **Advanced Theorems and Applications**

Q006 **Written**

Complete the source addition-theorem exercises $exactvalues, quadrant - controlled\alpha, \beta, andthethree - anglechallenge(\sin\alpha = 0.6, \cos\beta = 0.8, \tan\gamma = 0.75)$.

Q007 **Written**

Find $\tan 105°$**, then find the angle between** $y = 2x + 1$ **and** $y = -3x + 2$**.**

Classified Practice

Advanced Theorems and Applications

Q008 **Written**

Find the angle between each source pair of straight lines, including $y = \frac{1}{3}x$ with $y = 2x$, and $8x+2y-3 = 0$ with $-5x+3y+1 = 0$.

Classified Practice **Advanced Theorems and Applications**

Q009 Written

Find the exact angles for all source line pairs and solve the challenge: a line through $P(2,3)$ makes acute angle $\tan\theta = 3/4$ with $x + 2y - 4 = 0$; give both line equations and the areas of the triangles with the y-axis.

Classified Practice **Advanced Theorems and Applications**

Q010 **Written**

Given $\sin\alpha = 1/3$ **with α in QII, find** $\sin 2\alpha, \cos 2\alpha, \tan 2\alpha$**.**

Q011 **Written**

Given $\sin\alpha = -2/3$ **with** $2\pi < \alpha < 3\pi$, **find** $\sin 2\alpha, \cos 2\alpha, \tan 2\alpha$.

Classified Practice **Advanced Theorems and Applications**

Q012 **Written**

Solve the source double-angle exercises for given sine/cosine values and the challenge $\cos\theta = 0.6, \pi < \theta < 2\pi$, evaluating $\cos 3\theta \tan 2\theta$.

Q013 **Written**

Given $\cos\alpha = 3/5$, $0 < \alpha < \pi$, **find** $\sin(\alpha/2), \cos(\alpha/2), \tan(\alpha/2)$, **and evaluate** $\cos(5\pi/8)$.

Classified Practice **Advanced Theorems and Applications**

Q014 **Written**

Given $\cos\alpha = -1/3$, $0 < \alpha < \pi$, **find** $\sin(\alpha/2)$, $\cos(\alpha/2)$, $\tan(\alpha/2)$, **and evaluate** $\sin(3\pi/8)$.

Q015 **Written**

Solve the source half-angle exercises for the three quadrant intervals and the challenge with $\tan A = \frac{\sqrt{5}}{2}, 4\pi < A < 5\pi$, $\cos B = -\frac{\sqrt{5}}{3}, 5\pi < B < 6\pi$.

Classified Practice **Advanced Theorems and Applications**

Q016 **Written**

Solve on $0 \leq \theta < 2\pi$: $a \sin 2\theta + \sin\theta = 0$, $b \cos 2\theta - 7\cos\theta + 4 \geq 0$.

Q017 **Written**

Solve the source equation $\sin 2\theta + \cos\theta = 0$ **on** $0 \le \theta < 2\pi$**.**

Classified Practice **Advanced Theorems and Applications**

Q018 **Written**

Solve all source double-angle equations and inequalities on $0 \le \theta < 2\pi$, including the challenge $\cos 2\theta - 7\cos\theta + 4 \ge 0$.

Q019 **Written**

Rewrite $\sin\theta - \sqrt{3}\cos\theta$ **as** $r\sin(\theta+\alpha)$, $r > 0, -\pi < \alpha < \pi$.

Classified Practice **Advanced Theorems and Applications**

Q020 **Written**

Rewrite the source expressions $\sin\theta - \cos\theta$, $\sin\theta + \cos\theta$, $3\sin\theta - \sqrt{3}\cos\theta$, **and** $3\sin\theta - \cos\theta$ **in harmonic-addition form.**

Q021 **Written**

Rewrite all source harmonic-addition exercises in the form $r\sin(\theta+\alpha)$**, including** $2\sin\theta+2\cos\theta$ **and the challenge** $\sin\theta-\sqrt{3}\cos\theta$**.**

Classified Practice **Advanced Theorems and Applications**

Q022 **Written**

Solve on $0 \leq \theta < 2\pi$: $a \sin\theta - \cos\theta = 1$, $b \sin\theta + \cos\theta < 3/2$.

Q023 **Written**

Solve $\sin\theta \le \cos\theta$ **on** $0 \le \theta < 2\pi$.

Classified Practice **Advanced Theorems and Applications**

Q024 **Written**

Solve all source harmonic-addition equations and inequalities on $0 \le \theta < 2\pi$, including $\sin\theta + \cos\theta + 1 = 0$ and $\sin\theta - \cos\theta + 3/2 > 0$.

Q025 **Written**

Find the maximum and minimum of $y = 2\sin(\theta - \pi/3)$**, and of** $y = 2\cos 2\theta + 4\cos\theta + 1,\ 0 \leq \theta < 2\pi$**.**

Classified Practice **Advanced Theorems and Applications**

Q026 Written

Find the maximum and minimum of the source functions $y = -\frac{4}{3}\sin^2\theta + \cos\theta$ and $y = \cos 2\theta - 2\cos\theta$.

Q027 **Written**

Solve all source maximum/minimum problems in the final lesson, including absolute-value and double-angle forms, and state the corresponding angles where requested.

Q028 MCQ

The substitution for $\sin^2\theta + \cos\theta + 1$ **should be:**

Choose the correct option.

A $t = sin$ theta

B $t = cos$ theta

C $t = tan$ theta

D $t = theta$

Q029 **MCQ**

The maximum of $\sin^2\theta + \cos\theta + 1$ **is:**

Choose the correct option.

A 0

B 1

C 9/4

D 4

Classified Practice **Advanced Theorems and Applications**

Q030 MCQ

An extremal vertex is valid only if t lies in:

Choose the correct option.

A [-2,2]
B [-1,1]
C [0,1]
D all real numbers

Q031 **MCQ**

The identity for sin(a+b) is:

Choose the correct option.

A sin alpha cos beta + cos alpha sin beta

B sin alpha sin beta

C cos alpha cos beta

D tan alpha+tan beta

Classified Practice **Advanced Theorems and Applications**

Q032 **MCQ**

sin 75 degrees equals:

Choose the correct option.

A (sqrt6-sqrt2)/4
B (sqrt6+sqrt2)/4
C 1/2
D sqrt3/2

Q033 MCQ

A missing trig value is selected using:

Choose the correct option.

A the quadrant
B the denominator
C the period only
D a graph scale

Classified Practice **Advanced Theorems and Applications**

Q034 MCQ

The slope of y=mx+k is:

Choose the correct option.

A k
B m
C 1/m
D m+k

Q035 *MCQ*

The angle formula uses denominator:

Choose the correct option.

A 1+m1m2

B m1-m2

C m1+m2

D m1m2

Classified Practice **Advanced Theorems and Applications**

Q036 MCQ

For an acute angle, if tan theta1 is negative we use:

Choose the correct option.

A theta1

B pi-theta1

C 2theta1

D -theta1 only

Q037 MCQ

sin(2a) equals:

Choose the correct option.

A sin alpha

B 2 sin alpha cos alpha

C cos squared alpha

D tan alpha

Classified Practice **Advanced Theorems and Applications**

Q038 MCQ

cos(2a) can be written as:

Choose the correct option.

A cos squared alpha - sin squared alpha

B 2sin alpha

C tan alpha

D 1+sin alpha

Q039 **MCQ**

For theta in QIV, sin theta is:

Choose the correct option.

A positive

B negative

C zero always

D undefined

Classified Practice **Advanced Theorems and Applications**

Q040 MCQ

The half-angle formula for sin(a/2) uses:

Choose the correct option.

A (1-cos alpha)/2

B (1+cos alpha)/2

C tan alpha

D cos 2alpha

Q041 **MCQ**

The sign of a half-angle root is chosen from:

Choose the correct option.

A the original quadrant interval

B the coefficient only

C the denominator

D the period of tan

Classified Practice **Advanced Theorems and Applications**

Q042 MCQ

If cos a=-1/3 and 0<a<pi, cos(a/2) is:

Choose the correct option.

A negative

B positive

C zero

D undefined

Q043 MCQ

For double-angle equations, first rewrite in:

Choose the correct option.

A one trig function

B two unrelated angles

C degrees only

D a logarithm

Classified Practice **Advanced Theorems and Applications**

Q044 MCQ

The factorisation of sin(2 theta)+sin theta is:

Choose the correct option.

A sin theta(2cos theta+1)

B cos theta(2sin theta-1)

C sin theta+cos theta

D 2sin theta

Q045 **MCQ**

A tangent asymptote is always:

Choose the correct option.

A included

B excluded

C a maximum

D an endpoint

Classified Practice **Advanced Theorems and Applications**

Q046 MCQ

For a sin theta+b cos theta, r is:

Choose the correct option.

A a+b

B sqrt(a^2+b^2)

C a-b

D ab

Q047 **MCQ**

The harmonic target form is:

Choose the correct option.

A r sin(theta+alpha)

B r cos theta

C a tan theta

D sin theta only

Classified Practice **Advanced Theorems and Applications**

Q048 MCQ

The angle a is located from point:

Choose the correct option.

A P(a,b)

B P(r,theta)

C the origin only

D the x intercept

Q049 **MCQ**

After synthesis, t is usually:

Choose the correct option.

A theta+alpha

B $theta^2$

C tan theta

D cos theta

Classified Practice **Advanced Theorems and Applications**

Q050 MCQ

If max(sin theta+cos theta)=sqrt2, then sin theta+cos theta<3/2 is:

Choose the correct option.

A never true

B always true

C true only at pi

D undefined

Q051 *MCQ*

In a strict inequality, equality endpoints are:

Choose the correct option.

A included

B excluded

C doubled

D random

Classified Practice **Advanced Theorems and Applications**

Q052 MCQ

The range of 2sin(theta-pi/3) is:

Choose the correct option.

A [-1,1]
B [-2,2]
C [0,2]
D [-3,3]

Q053 MCQ

For a double-angle maximum problem, use:

Choose the correct option.

A the double-angle formula then a bounded variable

B a logarithm

C only a graph

D no substitution

Classified Practice **Advanced Theorems and Applications**

Q054 MCQ

A vertex outside [-1,1] is:

Choose the correct option.

A an allowed extremum
B rejected
C always the maximum
D always the minimum

Quiz MCQ 1 **Concept Quiz · MCQ**

The maximum of $2\sin(\theta - \pi/3)$ is:

Choose the correct option.

A -2

B -1

C 1

D 2

Classified Practice

Advanced Theorems and Applications

Quiz MCQ 2 **Concept Quiz · MCQ**

The addition-theorem value $\sin 75°$ is:

Choose the correct option.

A $(\sqrt{6}+\sqrt{2})/4$

B $(\sqrt{6}-\sqrt{2})/4$

C $1/2$

D 1

Quiz MCQ 3 **Concept Quiz · MCQ**

For $y = 2x^2 + 4x + 1$, the vertex x-value is:

Choose the correct option.

A -2

B -1

C 1

D 2

Classified Practice **Advanced Theorems and Applications**

Quiz MCQ 4 **Concept Quiz · MCQ**

The angle between slopes 2 and -3 is:

Choose the correct option.

A $\pi/6$
B $\pi/4$
C $\pi/3$
D $\pi/2$

Quiz WRITTEN 5 **Concept Quiz · Written**

Find the maximum and minimum of $y = \sin^2\theta + \cos\theta + 1$ **on** $0 \le \theta < 2\pi$.

Classified Practice **Advanced Theorems and Applications**

Quiz WRITTEN 6 **Concept Quiz · Written**

Find $\sin 2\alpha, \cos 2\alpha, \tan 2\alpha$ **when** $\sin\alpha = 1/3$ **and α is in QII.**

Quiz WRITTEN 7 **Concept Quiz · Written**

Rewrite $3\sin\theta - \sqrt{3}\cos\theta$ **as** $r\sin(\theta + \alpha)$.

Classified Practice **Advanced Theorems and Applications**

Quiz WRITTEN 8 **Concept Quiz · Written**

Solve $\sin\theta \leq \cos\theta$ **on** $0 \leq \theta < 2\pi$**.**

Answer key

Use the question number shown on each page.

Q001 $\max y = \frac{9}{4}$ at $\theta = \frac{\pi}{3}, \frac{5\pi}{3}$; $\min y = 0$ at $\theta = \pi$

Q002 $\max y = \frac{5}{4}$ at $\theta = \frac{7\pi}{6}, \frac{11\pi}{6}$; $\min y = -1$ at $\theta = \frac{\pi}{2}$

Q003 See the numbered solution: substitute $t = \sin\theta$ or $t = \cos\theta$, complete the square, then test $t = \pm 1$.

Q004 $\sin 75^\circ = \cos \pi/12 = \frac{\sqrt{6}+\sqrt{2}}{4}$

Q005 $\sin \alpha + \beta = \frac{4-3\sqrt{3}}{10}$; $\cos \alpha - \beta = \frac{3-4\sqrt{3}}{10}$

Q006 See the numbered solution; the three-angle challenge gives $\sin(\alpha+\beta+\gamma) = \frac{117}{125}$.

Q007 $\tan 105^\circ = -2+\sqrt{3}$; $\theta = \frac{\pi}{4}$

Q008 The first pair gives $\theta = \frac{\pi}{4}$; the second pair also gives $\theta = \frac{\pi}{4}$.

Q009 See the numbered solution; the two slopes are $m = 2/11, -2$, with triangle areas $49/165$ and $25/3$.

Q010 $\sin 2\alpha = -\frac{4\sqrt{2}}{9}$, $\cos 2\alpha = \frac{7}{9}$, $\tan 2\alpha = -\frac{4\sqrt{2}}{7}$

Q011 $\sin 2\alpha = \frac{4\sqrt{5}}{9}$, $\cos 2\alpha = \frac{1}{9}$, $\tan 2\alpha = 4\sqrt{5}$

Q012 For the challenge, $\cos 3\theta = -\frac{117}{125}$, $\tan 2\theta = \frac{24}{7}$, so the product is $-\frac{2808}{875}$.

Q013 $\sin\frac{\alpha}{2} = \frac{1}{\sqrt{5}}$, $\cos\frac{\alpha}{2} = \frac{2}{\sqrt{5}}$, $\tan\frac{\alpha}{2} = \frac{1}{2}$, $\cos\frac{5\pi}{8} = -\frac{\sqrt{2-\sqrt{2}}}{2}$

Q014 $\sin\frac{\alpha}{2} = \sqrt{\frac{2}{3}}$, $\cos\frac{\alpha}{2} = \frac{1}{\sqrt{3}}$, $\tan\frac{\alpha}{2} = \sqrt{2}$, $\sin\frac{3\pi}{8} = \frac{\sqrt{2+\sqrt{2}}}{2}$

Q015 Use the signed half-angle formulas; for the challenge,

$$\cos(A/2 - B/2) = \frac{\sqrt{3+\sqrt{5}} - \sqrt{15-5\sqrt{5}}}{6},$$

$$\sin(A/4) = \sqrt{\frac{1-\sqrt{5/6}}{2}}.$$

Q016 a $\theta = 0, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}$; b $\frac{\pi}{3} \le \theta \le \frac{5\pi}{3}$

Q017 $\theta = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{3\pi}{2}, \frac{11\pi}{6}$

Q018 See the numbered solution: replace double angles, solve in $s = \sin\theta$ or $c = \cos\theta$, and translate the accepted ranges.

Q019 $r = 2$, $\alpha = -\frac{\pi}{3}$, $\sin\theta - \sqrt{3}\cos\theta = 2\sin\theta - \frac{\pi}{3}$

Q020 $\sqrt{2}\sin\theta - \frac{\pi}{4}$, $\sqrt{2}\sin\theta + \frac{\pi}{4}$, $2\sqrt{3}\sin\theta - \frac{\pi}{6}$, $\sqrt{10}\sin\theta - \arctan\frac{1}{3}$

Q021 See the numbered solution; every result has $r = \sqrt{a^2+b^2} > 0$ and α chosen in $(-\pi, \pi)$.

Q022 a $\theta = \frac{\pi}{2}, \pi$; b $0 \le \theta < 2\pi$

Q023 $0 \le \theta \le \frac{\pi}{4}$ or $\frac{5\pi}{4} \le \theta < 2\pi$

Q024 See the numbered solution; synthesize first, then solve in the shifted variable over its full range.

Q025 First: $\max = 2, \min = -2$. Second: $\max = 7$ at $\theta = 0$; $\min = -2$ at $\theta = \frac{2\pi}{3}, \frac{4\pi}{3}$

Q026 For the first, $\max = 1, \min = -\frac{73}{48}$; for the second, use $t = \cos\theta$ and the endpoint/vertex test.

Q027 See the numbered solution: synthesize $a\sin\theta + b\cos\theta$ or substitute $t = \cos\theta$, then test the full allowed range.

Q028 B

Q029 C

Q030 B

Q031 A

Q032 B

Q033 A

Q034 B

Q035 A

Q036 B

Q037 B

Q038 A

Q039 B

Q040 A

Q041 A

Q042 B

Q043 A

Q044 A

Q045 B

Q046 B

Q047 A

Q048 A

Q049 A

Q050 B

Q051 B

Q052 B

Q053 A

Q054 B

Quiz MCQ 1 D

Quiz MCQ 2 A

Quiz MCQ 3 A

Quiz MCQ 4 B

Quiz WRITTEN 5 $\max = \frac{9}{4}, \min = 0$

Quiz WRITTEN 6 $-\frac{4\sqrt{2}}{9}, \frac{7}{9}, -\frac{4\sqrt{2}}{7}$

Quiz WRITTEN 7 $2\sqrt{3}\sin\theta - \frac{\pi}{6}$

Quiz WRITTEN 8 $[0, \frac{\pi}{4}] \cup [\frac{5\pi}{4}, 2\pi)$

Concept 1 | Laws of Exponents and Roots

English Student Edition • Focused formulas and guided practice

Formula opener

Product

$$a^m a^n = a^{m+n}$$

Power

$$(a^m)^n = a^{mn}$$

Root

$$a^{1/n} = \sqrt[n]{a}$$

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Classified Practice **Laws of Exponents and Roots**

Q001 **Written**

Find: **(a)** 10^0, **(b)** 2^{-3}, **(c)** $(0.2)^{-2}$.

Q002 **Written**

Let $a \neq 0, b \neq 0$. **Simplify: (a)** $a^{-5}a^{8}$, **(b)** $a^{5} \div a^{-3}$, **(c)** $(a^{-1}b)^{2}$, **(d)** $(2^{2})^{3} \div 2^{-2} \times 2^{-3}$.

Classified Practice Laws of Exponents and Roots

Q003 Written

Find: (a) 5^0, **(b)** 3^{-2}, **(c)** $(0.25)^{-2}$.

Classified Practice **Laws of Exponents and Roots**

Q004 **Written**

Simplify the eight source expressions: a^5a^4, $a^3 \div a^{-2}$, $(a^{-2}b^4)^{-1}$, $2^3 \times 2^{-7}$, $5^{-4} \div 5^{-6}$, $(a^3b^{-1})^3(a^2b^{-3})^{-2}$, $4^{-5}4^2 \div 4^{-2}$, **and** $2^4 \div (2^{-1})^3 \times 2^{-2}$.

Classified Practice **Laws of Exponents and Roots**

Q005 **Written**

Find the nine values: $3^0, 11^0, (2/3)^0, 4^{-3}, 5^{-3}, 6^{-2}, (1/3)^{-3}, (3/5)^{-2}, (0.04)^{-2}$.

Classified Practice **Laws of Exponents and Roots**

Q006 Written

Simplify the twelve same-base expressions in source Question 2: $a^2a^4, a^{-2}a^5, a^{-3}a^{-4}, a^3 \div a^5, a^8 \div a^{-3}, a^{-6} \div a^{-2}, (a^{-1})^3, (a^3b^4)^{-2}, (a^{-3}b^2)^{-3}, 3^5 3^{-3}, 4^4 4^{-3}, 5^5 5^{-5}$.

Classified Practice Laws of Exponents and Roots

Q007 Written

Simplify source Question 3: $5^4 \div 5^2$, $3^2 \div 3^{-2}$, $2^{-4} \div 2^{-5}$, $(a^{-2}b)^3(a^2b^{-1})^2$, $(a^{-3}b)^{-3}(a^4b^2)^{-2}$, $(2a^3b^{-1})^3(-3ab^{-2})^{-3}$, $3^2 3^{-1} \div 3^3$, $5^{-4}5^5 \div 5^{-2}$, $2^4 2^{-8} \div 2^{-6}$, $5^3(5^{-2})^2 \div 5^{-4}$, $3^5 \div (3^{-2})^{-3} \times 3$, **and** $(a/2)^3 a^{-4} \div (2a)^{-2}$.

Classified Practice **Laws of Exponents and Roots**

Q008 Written

Simplify completely with positive exponents: $\left(\frac{2a^2}{b^{-3}}\right)^{-2}(4a^{-3}b^2)^3 \div \left(\frac{a}{2b}\right)^{-4} \div (a^{-2}b^2)$.

Q009 Written

Find $\sqrt[3]{64}$ and $\sqrt[3]{1/27}$.

Classified Practice **Laws of Exponents and Roots**

Q010 **Written**

Perform: $\sqrt[3]{12}\sqrt[3]{18}$, $\sqrt[3]{250}/\sqrt[3]{2}$, $(\sqrt[6]{8})^2$, $\sqrt[3]{\sqrt{729}}$, **and** $\sqrt[3]{54} - \sqrt[3]{16} + \sqrt[3]{2}$.

Q011 **Written**

Find $\sqrt[4]{81}$ and $\sqrt[4]{1/16}$.

Classified Practice **Laws of Exponents and Roots**

Q012 Written

Perform: $\sqrt[4]{8}\sqrt[4]{32}$, $\sqrt[4]{48}/\sqrt[4]{3}$, $(\sqrt[6]{100})^3$, $\sqrt[3]{\sqrt[3]{64}}$, **and** $\sqrt[3]{81} - \sqrt[3]{3} + \sqrt[3]{24}$.

Q013 **Written**

Find: $\sqrt[3]{216}$, $\sqrt[4]{625}$, $\sqrt[5]{243}$, $\sqrt[3]{1/64}$, $\sqrt[3]{1/125}$, $\sqrt[4]{1/81}$, $\sqrt[3]{1}$, $\sqrt[5]{100000}$, $\sqrt[4]{0.0001}$.

Classified Practice

Laws of Exponents and Roots

Q014 Written

Perform the fifteen source calculations involving products, quotients, powers and sums of roots (including $\sqrt[3]{2}\sqrt[3]{4}$, $\sqrt[4]{27}\sqrt[4]{243}$, nested roots and the final three like-radical sums).

Classified Practice **Laws of Exponents and Roots**

Q015 **Written**

Perform $\sqrt[3]{\sqrt[4]{4096}} \div \sqrt{\sqrt[6]{4096}}$**.**

Q016 Written

Express in the form a^r: $\sqrt{5}$, $\sqrt[5]{2^4}$, and $1/\sqrt[5]{3^3}$.

Classified Practice **Laws of Exponents and Roots**

Q017 **Written**

Find: $27^{2/3}$, $64^{-3/2}$, $(25/4)^{-3/2}$, **and** $[(16/9)^{-3/4}]^{2/3}$.

Q018 **Written**

Express in the form a^r: $\sqrt{a}$, $\sqrt[4]{a^3}$, $\sqrt[6]{a^5}$, $\sqrt[3]{2^2}$, $1/\sqrt{a}$, $1/\sqrt[3]{a^2}$, **and** $1/\sqrt[4]{5^3}$.

Classified Practice **Laws of Exponents and Roots**

Q019 **Written**

Find $81^{3/4}$, $16^{-3/2}$, $(25/4)^{-3/2}$, **and** $[(27/125)^{1/2}]^{-4/3}$.

Q020 **Written**

Express the nine source expressions in the form a^r, including roots of a and numerical bases $2, 3, 5$.

Classified Practice Laws of Exponents and Roots

Q021 Written

Evaluate the twelve source expressions with rational powers: $25^{3/2}, 16^{3/4}, 64^{2/3}, 9^{-3/2}, 125^{-2/3}, 81^{-3/4}, (1/8)^{-1/3}, (125/8)^{-2/3}, (16/81)^{-3/4}, [(1/25$

Classified Practice **Laws of Exponents and Roots**

Q022 **Written**

Simplify $\left[\left((1000/27)^{-2/3}\right)^{7/15}\right]^{-15/14}$.

Q023 **Written**

Perform: (a) $2^{2/3}2^{1/2} \div 2^{7/6}$**, (b)** $6^{1/2}36^{1/4}$**, (c)** $\sqrt[3]{6}\sqrt{6}\sqrt[6]{6}$**, (d)** $\sqrt{2} \div \sqrt[6]{32} \times \sqrt[3]{16}$**.**

Classified Practice **Laws of Exponents and Roots**

Q024 **Written**

Perform: $5^{1/2}5^{1/3} \div 5^{-1/6}$, $4^{5/6}2^{4/3}$, $\sqrt{7}\sqrt[3]{7}\sqrt[6]{7}$, **and** $\sqrt[4]{27}\sqrt{27} \div \sqrt[4]{3}$.

Q025 **Written**

For $a > 0, b > 0$, perform: $\sqrt{a^3} \div \sqrt[3]{a} \div \sqrt[6]{a^5}$ and $\sqrt[3]{a}\sqrt{ab} \div \sqrt[6]{a^2/b^3}$.

Classified Practice **Laws of Exponents and Roots**

Q026 **Written**

Perform the twelve numerical calculations in source Question 1, including rational powers of 2, 3 and 5 and mixed radical products/quotients.

Q027 Written

For $a > 0, b > 0$, simplify the six source expressions built from $\sqrt{a}, \sqrt[3]{a}, \sqrt[4]{a}, \sqrt[6]{a}$ and mixed a,b radicals.

Classified Practice **Laws of Exponents and Roots**

Q028 **Written**

Simplify the ministry expression $\left(\frac{(243/32)^{-2/5}(9/4)^{-1}}{(125/64)^{1/3}(0.004)^{-1/2}}\right)^{-3/2}$.

Q029 **Written**

Assess the statement $\sqrt{a^2} = a$ for all real a, using positive and negative examples.

Classified Practice **Laws of Exponents and Roots**

Q030 MCQ

The value of 2^{-3} is:

Choose the correct option.

A 8
B 1/8
C -8
D 1/6

Q031 **MCQ**

When multiplying equal bases, we:

Choose the correct option.

A subtract exponents

B add exponents

C multiply bases only

D take reciprocals

Classified Practice **Laws of Exponents and Roots**

Q032 MCQ

The expression $a^5 \div a^{-3}$ simplifies to:

Choose the correct option.

A a^2

B a^8

C a^{-8}

D a^{15}

Q033 **MCQ**

A negative exponent means:

Choose the correct option.

A multiply by -1
B take the reciprocal power
C set the base to zero
D add one

Classified Practice **Laws of Exponents and Roots**

Q034 MCQ

The value of $(0.04)^{-2}$ is:

Choose the correct option.

A 25

B 400

C 625

D 1600

Q035 MCQ

The exponent of a in $(a^{-2}b)^3(a^2b^{-1})^2$ is:

Choose the correct option.

A -2

B -1

C 2

D 6

Classified Practice **Laws of Exponents and Roots**

Q036 MCQ

The positive-exponent form of the 6-1 challenge is:

Choose the correct option.

A a^7b^6
B $1/(a^7b^6)$
C $a^{-7}b^{-6}$
D $1/(a^6b^7)$

Q037 **MCQ**

The value of $3^5 \div (3^{-2})^{-3} \times 3$ is:

Choose the correct option.

A 1/3
B 1
C 3
D 9

Classified Practice

Laws of Exponents and Roots

Q038 MCQ

$\sqrt[3]{64}$ equals:

Choose the correct option.

A 2

B 4

C 8

D 16

Q039 **MCQ**

$\sqrt[3]{12}\sqrt[3]{18}$ **equals:**

Choose the correct option.

A 4

B 5

C 6

D 9

Classified Practice **Laws of Exponents and Roots**

Q040 MCQ

The fourth root of $1/16$ is:

Choose the correct option.

A $1/2$

B $1/4$

C 2

D 4

Q041 **MCQ**

The result of $\sqrt[3]{54} - \sqrt[3]{16} + \sqrt[3]{2}$ is:

Choose the correct option.

A $\sqrt[3]{2}$

B $2\sqrt[3]{2}$

C $3\sqrt[3]{2}$

D 0

Classified Practice **Laws of Exponents and Roots**

Q042 MCQ

$\sqrt[4]{625}$ **equals:**

Choose the correct option.

A 4

B 5

C 5^2

D 25

Q043 MCQ

When combining roots with the same index, multiply:

Choose the correct option.

A the indices

B the radicands

C the exponents only

D the signs only

Classified Practice **Laws of Exponents and Roots**

Q044 MCQ

The roots challenge evaluates to:

Choose the correct option.

A 0
B 1
C 2
D 4

Q045 **MCQ**

$\sqrt[5]{2^4}$ in rational-index form is:

Choose the correct option.

A $2^{4/5}$

B $2^{5/4}$

C 2^9

D $4^{1/5}$

Classified Practice **Laws of Exponents and Roots**

Q046 MCQ

The value of $27^{2/3}$ is:

Choose the correct option.

A 3

B 6

C 9

D 18

Q047 **MCQ**

A reciprocal square root is written as:

Choose the correct option.

A $a^{1/2}$

B $a^{-1/2}$

C a^{-2}

D a^{2}

Classified Practice **Laws of Exponents and Roots**

Q048 MCQ

The value of $81^{-3/4}$ is:

Choose the correct option.

A 27
B 1/9
C 1/27
D 1/81

Q049 **MCQ**

The exponent of a in $\sqrt[4]{a^3}$ is:

Choose the correct option.

A 3/4

B 4/3

C $-3/4$

D 3

Classified Practice **Laws of Exponents and Roots**

Q050 MCQ

The value of $[(64/27)^{-4/3}]^{-1/2}$ **is:**

Choose the correct option.

A 4/9

B 9/4

C 16/9

D 27/64

Q051 *MCQ*

The 6-3 challenge simplifies to:

Choose the correct option.

A 3/10

B 10/3

C 100/9

D 9/100

Classified Practice **Laws of Exponents and Roots**

Q052 **MCQ**

$2^{2/3}2^{1/2} \div 2^{7/6}$ **equals:**

Choose the correct option.

A $1/2$

B 1

C 2

D 4

Q053 **MCQ**

$4^{5/6}2^{4/3}$ **equals:**

Choose the correct option.

A 4

B 6

C 8

D 16

Classified Practice **Laws of Exponents and Roots**

Q054 MCQ

$\sqrt{a^3} \div \sqrt[3]{a} \div \sqrt[6]{a^5}$ **equals:**

Choose the correct option.

A $a^{1/3}$

B $a^{2/3}$

C $a^{5/3}$

D a

Q055 **MCQ**

The numerical result of $\sqrt[3]{6}\sqrt{6}\sqrt[6]{6}$ is:

Choose the correct option.

A 2

B 3

C 6

D 12

Classified Practice **Laws of Exponents and Roots**

Q056 MCQ

The source expression $\sqrt{a^3} \div \sqrt[3]{a} \div \sqrt[6]{a^5}$ tests:

Choose the correct option.

A integer powers
B combining rational exponents
C logarithms
D trigonometry

Q057 **MCQ**

The printed decimal in the 6-4 challenge is:

Choose the correct option.

A 0.04

B 0.004

C 0.4

D 4

Classified Practice **Laws of Exponents and Roots**

Q058 MCQ

For real a, $\sqrt{a^2}$ equals:

Choose the correct option.

A a always

B $-a$ always

C $|a|$

D 0

Quiz MCQ 1 **Concept Quiz · MCQ**

Which law gives $a^m a^n = a^{m+n}$?

Choose the correct option.

A same-base multiplication

B power of a power

C reciprocal law

D root product

Classified Practice **Laws of Exponents and Roots**

Quiz MCQ 2 **Concept Quiz · MCQ**

The value of $\sqrt[3]{250}/\sqrt[3]{2}$ is:

Choose the correct option.

A 2

B 5

C 10

D 125

Quiz MCQ 3 **Concept Quiz · MCQ**

The value of $64^{-3/2}$ is:

Choose the correct option.

A 512
B 1/64
C 1/512
D 64

Classified Practice **Laws of Exponents and Roots**

Quiz MCQ 4 **Concept Quiz · MCQ**

For a negative real a, $\sqrt{a^2}$ is:

Choose the correct option.

A a
B -a
C 0
D a^2

Quiz WRITTEN 5 **Concept Quiz · Written**

Simplify $a^{-3}a^{7} \div a^{2}$.

Classified Practice **Laws of Exponents and Roots**

Quiz WRITTEN 6 **Concept Quiz · Written**

Simplify $\sqrt[3]{54} - \sqrt[3]{16} + \sqrt[3]{2}$.

Quiz WRITTEN 7 **Concept Quiz · Written**

Evaluate $125^{-2/3}$.

Classified Practice **Laws of Exponents and Roots**

Quiz WRITTEN 8 **Concept Quiz · Written**

Explain why $\sqrt{a^2} = a$ is not true for every real a.

Answer key

Use the question number shown on each page.

Q001 $1, \frac{1}{8}, 25$

Q002 $a^3,\ a^8,\ \frac{b^2}{a^2},\ 32$

Q003 $1, \frac{1}{9}, 16$

Q004 $a^9,\ a^5,\ a^2b^{-4},\ \frac{1}{16},\ 25,\ a^5b^3,\ \frac{1}{4},\ 32$

Q005 $1, 1, 1, \frac{1}{64}, \frac{1}{125}, \frac{1}{36}, 27, \frac{25}{9}, 625$

Q006 $a^6, a^3, a^{-7}, a^{-2}, a^{11}, a^{-4}, a^{-3}, a^{-6}b^{-8}, a^9b^{-6}, 9, 4, 1$

Q007 $25, 81, 2, \frac{b}{a^2}, \frac{a}{b^7}, -\frac{8a^6b^3}{27}, \frac{1}{9}, 125, 4, 125, 1, \frac{a}{2}$

Q008 $\frac{1}{a^7b^6}$

Q009 $4, \frac{1}{3}$

Q010 $6, 5, 2, 3, 2\sqrt[3]{2}$

Q011 $3, \frac{1}{2}$

Q012 $4, 2, 10, \sqrt[3]{4}, 4\sqrt[3]{3}$

Q013 $6, 5, 3, \frac{1}{4}, \frac{1}{5}, \frac{1}{3}, 1, 10, 0.1$

Q014 $2, 4, 9, \sqrt[4]{27}, \frac{1}{4}, \frac{1}{5}, 7, 3, 4, 9, 4, \sqrt[3]{9}, \sqrt[3]{3}, 0, 0$

Q015 1

Q016 $5^{1/2},\ 2^{4/5},\ 3^{-3/5}$

Q017 $9, \frac{1}{512}, \frac{8}{125}, \frac{3}{4}$

Q018 $a^{1/2}, a^{3/4}, a^{5/6}, 2^{2/3}, a^{-1/2}, a^{-2/3}, 5^{-3/4}$

Q019 $27, \frac{1}{64}, \frac{8}{125}, \frac{25}{9}$

Q020 $2^{1/2}, 6^{1/2}, 7^{1/2}, a^{3/4}, a^{5/6}, 2^{2/3}, a^{-1/2}, a^{-2/3}, 5^{-3/4}$

Q021 $125, 8, 16, \frac{1}{27}, \frac{1}{25}, \frac{1}{27}, 2, \frac{4}{25}, \frac{27}{8}, 5, \frac{10}{9}, \frac{16}{9}$

Q022 $\frac{10}{3}$

Q023 $1, 6, 6, 2$

Q024 $5, 8, 7, 9$

Q025 $a^{1/3},\ b\sqrt{a}$

Q026 $1, 9, 5, 9, \sqrt{5}, 4, 5, 3, 2, 1, 8, \sqrt[3]{3}$

Q027 $a^{1/3},\ a^{7/6},\ a^4,\ a^{2/3}b^{5/6},\ a^{1/2}b^3,\ a^{13/6}b^{-5/6}$

Q028 $\left(\frac{2025\sqrt{10}}{64}\right)^{3/2}$

Q029 Sometimes: $\sqrt{a^2} = |a|$; it equals a only when $a \geq 0$.

Q030 B

Q031 B

Q032 B

Q033 B

Q034 C

Q035 A

Q036 B

Q037 B

Q038 B

Q039 C

Q040 A

Q041 B

Q042 B

Q043 B

Q044 B

Q045 A

Q046 C

Q047 B

Q048 C

Q049 A

Q050 C

Q051 B

Q052 B

Q053 C

Q054 A

Q055 C

Q056 B

Q057 B

Q058 C

Quiz MCQ 1 A

Quiz MCQ 2 B

Quiz MCQ 3 C

Quiz MCQ 4 B

Quiz WRITTEN 5 a^2

Quiz WRITTEN 6 $2\sqrt[3]{2}$

Quiz WRITTEN 7 1/25

Quiz WRITTEN 8 $\sqrt{a^2} = |a|$; e.g. $a = -3$ gives $3 \neq -3$

Concept 2 | Exponential Functions and Their Applications

English Student Edition • Focused formulas and guided practice

Formula opener

Exponential

$$y = ab^x$$

Growth

$$A = A_0(1 + r)^t$$

Decay

$$A = A_0(1 - r)^t$$

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Classified Practice **Exponential Functions and Their Applications**

Q001 **Written**

Draw the graphs of $y = 3^x$ and $y = (1/3)^x$, using the points $x = -1, 0, 1$.

Q002 **Written**

Draw $y = 2^x$ **and** $y = (1/2)^x$.

Classified Practice Exponential Functions and Their Applications

Q003 Written

Draw the six source graphs $y = 4^x,\ y = 10^x,\ y = (2/3)^x,\ y = (1/4)^x,\ y = (1/10)^x,\ y = (3/2)^x$.

Q004 Written

Draw $y = \pi^x$ for $-1 \leq x \leq 2$, marking the key points.

Classified Practice **Exponential Functions and Their Applications**

Q005 **Written**

Order the source sets $0.7^{-3}, 0.7^{-1}, 0.7^{2}$ **and** $5/8,\ 5/64,\ 6/16$ **using inequality signs.**

Q006 **Written**

Order the three displayed Try sets: $5^0, 5^{-4}, 5^{3/2}$; $(1/3)^7, (1/3)^2, (1/3)^{-5}$; **and** $3^0, 3^2, 3^{-2}$.

Classified Practice **Exponential Functions and Their Applications**

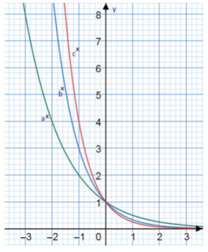
Q007 **Written**

Order the source sets with bases 3, 2, $3/2$, 7, 0.9, and 0.1, including the radical/fraction forms shown in the lesson.

Classified Practice **Exponential Functions and Their Applications**

Q008 **Written**

Order $(\sqrt{2})^5,\ (\sqrt[3]{4})^3,\ (\sqrt[4]{8})^2$, then use the opposite graph to compare the labelled values a, b, c.



Classified Practice **Exponential Functions and Their Applications**

Q009 **Written**

Solve the six Warm Up items: $4^{x-1} = 8^{x+3}$, $3^x = 81$, $7^x = 7$, $2^{2x-1} \geq 1/2$, $(1/27)^x < (1/3)^{x+4}$, **and** $8 \geq 2^{2x-1}$.

Q010 Written

Solve the eight Try equations/inequalities displayed in lesson 6-7, including $9^x = 243$, $3^{2x+1} = 125$, $2^{3x} \leq 9^{x+1}$, and the reciprocal-base inequalities.

Classified Practice **Exponential Functions and Their Applications**

Q011 **Written**

Solve all 24 source equations and inequalities in Exercise (a)–(x), including the $2, 3, 4, 5, 8, 9, 25, 27, 32, 64, 81, 125$ **common-base forms.**

Q012 **Written**

A food-safety model requires $4^{x^2-x} > 16$. Find when the batch becomes unsafe.

Classified Practice **Exponential Functions and Their Applications**

Q013 **Written**

Solve $4^x - 3 \cdot 2^x - 16 = 0$ and $8 \cdot 3^{-2x+1} - 3^{-x} \leq 0$ by substitution.

Q014 **Written**

Solve the four Try items in lesson 6-8: two quadratic equations and two inequalities in powers of 2 and 3.

Classified Practice **Exponential Functions and Their Applications**

Q015 **Written**

Solve the twelve source items (a)–(l), each reducible to a quadratic in 2^x or 3^x.

Q016 **Written**

Solve completely for real x: $(1/4)^{x^2-2x} - 5(1/2)^{x^2-2x} + 4 \leq 0.$

Classified Practice Exponential Functions and Their Applications

Q017 Written

For $-1 \le x \le 3$, find the maximum and minimum of $y = 2^{2x} - 4 \cdot 2^{x}$, with the corresponding x-values.

Q018 **Written**

For $-1 \le x \le 2$, find the maximum and minimum of $y = -9 + 2 \cdot 3^x + 3^{2x}$, with the corresponding x-values.

Classified Practice Exponential Functions and Their Applications

Q019 Written

Solve the three source extrema models: $-3 \leq x \leq 2,\ y = 2 \cdot 9^x - 12 \cdot 3^x + 7$; $-1 \leq x \leq 3,\ y = -4 + 2^{2x} - 3^{x+2}$; **and the third displayed interval/model.**

Q020 **Written**

For $-1 \leq x \leq 2$, optimise $y = 2 \cdot 4^{x} - 2^{x+2} + 11$.

Classified Practice **Exponential Functions and Their Applications**

Q021 **Written**

For $f(x) = a^x,\ g(x) = (1/a)^x$ with $a > 1$, analyse $h(x) = f(x) + g(x)$.

Q022 **MCQ**

Which base gives a strictly decreasing exponential graph?

Choose the correct option.

A 2

B 3

C 1/2

D π

Classified Practice **Exponential Functions and Their Applications**

Q023 **MCQ**

The horizontal asymptote of $y = a^x$ is:

Choose the correct option.

A $x = 0$
B $y = 0$
C $y = 1$
D $x = 1$

Q024 MCQ

The range of $y = 5^x$ is:

Choose the correct option.

A $\mathbb{R}$

B $[0, \infty)$

C $(0, \infty)$

D $(-\infty, 0)$

Classified Practice **Exponential Functions and Their Applications**

Q025 MCQ

For $y = \pi^x$ on $[-1, 2]$, which point is on the graph?

Choose the correct option.

A $(0, \pi)$
B $(1, \pi)$
C $(2, \pi)$
D $(-1, \pi)$

Q026 MCQ

For $0 < a < 1$ and $r < s$, which is true?

Choose the correct option.

A $a^r < a^s$

B $a^r > a^s$

C $a^r = a^s$

D Cannot compare

Classified Practice **Exponential Functions and Their Applications**

Q027 **MCQ**

The first step when bases differ is to:

Choose the correct option.

A add exponents

B make the bases equal

C take logarithms immediately

D ignore the base

Q028 MCQ

Which is largest? $3^{-2}, 3^0, 3^2$

Choose the correct option.

A 3^{-2}

B 3^0

C 3^2

D They are equal

Classified Practice **Exponential Functions and Their Applications**

Q029 MCQ

The challenge term $(\sqrt[3]{4})^3$ equals:

Choose the correct option.

A $2^{3/2}$

B 2^2

C 2^3

D 4^3

Q030 MCQ

Solving $4^{x-1} = 8^{x+3}$ gives:

Choose the correct option.

A $x = -11$
B $x = -3$
C $x = 3$
D $x = 11$

Classified Practice **Exponential Functions and Their Applications**

Q031 MCQ

For $0 < a < 1$, an exponential inequality requires:

Choose the correct option.

A the same exponent order

B a reversed exponent order

C no base conversion

D squaring both sides

Q032 MCQ

The solution of $3^x = 81$ is:

Choose the correct option.

A 2

B 3

C 4

D 5

Classified Practice **Exponential Functions and Their Applications**

Q033 MCQ

The food-safety condition $4^{x^2-x} > 16$ is unsafe for physical time when:

Choose the correct option.

A $0 \leq x \leq 2$

B $x > 2$

C $x < -1$

D $x < 2$

Q034 MCQ

In substitution methods, $t = a^x$ must satisfy:

Choose the correct option.

A $t < 0$

B $t = 0$

C $t > 0$

D $t \in \mathbb{R}$

Classified Practice **Exponential Functions and Their Applications**

Q035 MCQ

The positive root of $t^2 - 3t - 16 = 0$ is:

Choose the correct option.

A $(3 - \sqrt{73})/2$
B $(3 + \sqrt{73})/2$
C 8
D 16

Q036 MCQ

The challenge quadratic in t is:

Choose the correct option.

A $t^2 + 5t + 4$

B $t^2 - 5t + 4$

C $t^2 - 4t + 5$

D $t^2 + 4t - 5$

Classified Practice **Exponential Functions and Their Applications**

Q037 MCQ

After solving an exponential quadratic, reject roots with:

Choose the correct option.

A $t > 1$

B $t = 1$

C $t \leq 0$

D $t < 4$

Q038 **MCQ**

For extrema, the first required range is the range of:

Choose the correct option.

A x

B $t = a^x$

C a

D y only

Classified Practice **Exponential Functions and Their Applications**

Q039 MCQ

The minimum of $y = t^2 - 4t$ at $t = 2$ is:

Choose the correct option.

A -8

B -4

C 0

D 4

Q040 MCQ

The C06-K02 challenge has minimum value:

Choose the correct option.

A 9 at $x = 0$

B 9 at $x = 2$

C 27 at $x = 0$

D 27 at $x = 2$

Classified Practice **Exponential Functions and Their Applications**

Q041 **MCQ**

The global minimum of $a^x + a^{-x}$ for $a > 1$ is:

Choose the correct option.

A 0

B 1

C 2

D It has no minimum

Q042 MCQ

At which value of x does $a^x + a^{-x}$ attain its minimum?

Choose the correct option.

A $x = -1$

B $x = 0$

C $x = 1$

D It has no stationary point

Classified Practice **Exponential Functions and Their Applications**

Quiz MCQ 1 Concept Quiz · MCQ

Which statement about $y = a^x$ is always true for $a > 0,\ a \neq 1$?

Choose the correct option.

A range $(0, \infty)$
B range all reals
C vertical asymptote
D decreasing for every a

Quiz MCQ 2 **Concept Quiz · MCQ**

For base 0.5, increasing the exponent makes the value:

Choose the correct option.

A larger
B smaller
C unchanged
D zero

Classified Practice **Exponential Functions and Their Applications**

Quiz MCQ 3 **Concept Quiz · MCQ**

The positive-substitution condition is:

Choose the correct option.

A t<0

B $t = 0$

C t>0

D t can be any real

Quiz MCQ 4 **Concept Quiz · MCQ**

For $y = 2(t-1)^2 + 9$, the minimum is:

Choose the correct option.

A 0

B 1

C 9

D 11

Classified Practice **Exponential Functions and Their Applications**

Quiz WRITTEN 5 Concept Quiz · Written

Sketch $y = (1/4)^x$ and state its domain, range, and asymptote.

Quiz WRITTEN 6 **Concept Quiz · Written**

Order $2^{-3}, 2^{0}, 2^{5/2}$**.**

Classified Practice **Exponential Functions and Their Applications**

Quiz WRITTEN 7 **Concept Quiz · Written**

Solve $4^x - 5 \cdot 2^x + 4 = 0$.

Quiz WRITTEN 8 **Concept Quiz · Written**

Find the minimum of $a^x + a^{-x}$ for $a > 1$.

Answer key

Use the question number shown on each page.

Q001 Increasing 3^x; decreasing $(1/3)^x$; both pass through $(0, 1)$ and have $y = 0$ as asymptote.

Q002 2^x increases and $(1/2)^x$ decreases; domain $\mathbb{R}$, range $(0, \infty)$, horizontal asymptote $y = 0$.

Q003 Bases $4, 10, 3/2$: strictly increasing. Bases $2/3, 1/4, 1/10$: strictly decreasing. Every graph has $D = \mathbb{R}, R = (0, \infty)$, and asymptote $y = 0$.

Q004 Key points $(-1, 1/\pi), (0, 1), (1, \pi), (2, \pi^2)$; the restricted graph is increasing.

Q005 $0.7^2 < 0.7^{-1} < 0.7^{-3}$; and $5/64 < 5/8 < 6/16$.

Q006 $5^{-4} < 5^0 < 5^{3/2}; (1/3)^7 < (1/3)^2 < (1/3)^{-5}; 3^{-2} < 3^0 < 3^2$.

Q007 Each set is ordered by converting all terms to a common base, then comparing exponents; reverse the exponent order for $0 < a < 1$.

Q008 Rewrite every term as a power of 2: $2^{5/2}, 2^2, 2^{3/2}$, so $(\sqrt[4]{8})^2 < (\sqrt[3]{4})^3 < (\sqrt{2})^5$. The graph comparison follows the same increasing/decreasing rule.

Q009 $x = -11, 4, 1, x \geq 1/4, x > 2, x \leq 2$, respectively.

Q010 Convert each side to one base and compare exponents; the exact solution is the resulting linear equality or interval in x.

Q011 Every item reduces to a linear condition on x; equality items have one exact value and inequality items have the interval obtained after base/sign control.

Q012 $x^2 - x > 2 \Rightarrow (x - 2)(x + 1) > 0$, so $x < -1$ or $x > 2$. For time $x \geq 0$, the unsafe condition is $x > 2$ hours.

Q013 First: let $t = 2^x > 0$, $t^2 - 3t - 16 = 0$, so $t = (3 + \sqrt{73})/2$ and $x = \log_2((3 + \sqrt{73})/2)$. Second: $x \leq 1$.

Q014 Use $t = 2^x > 0$ or $t = 3^x > 0$; solve the resulting quadratic and retain only roots satisfying $t > 0$.

Q015 For each item, $t = a^x > 0$ gives a quadratic equation/inequality; convert every admissible t back to an exact logarithmic value or interval for x.

Q016 Let $t = (1/2)^{x^2 - 2x} > 0$. Then $t^2 - 5t + 4 \leq 0$, so $1 \leq t \leq 4$, which gives $0 \leq x^2 - 2x \leq 2$. Hence $1 - \sqrt{3} \leq x \leq 0$ or $2 \leq x \leq 1 + \sqrt{3}$.

Q017 Let $t = 2^x \in [1/2, 8]$. Then $y = t^2 - 4t = (t - 2)^2 - 4$: minimum -4 at $t = 2, x = 1$, maximum 32 at $t = 8, x = 3$.

Q018 Let $t = 3^x \in [1/3, 9]$. Then $y = t^2 + 2t - 9$; the minimum is at $t = 1/3$, and the maximum is at $t = 9$.

Q019 In each case set $t = a^x$, write the exact t-range, and compare the quadratic vertex and endpoints; report both extreme values and their x-locations.

Q020 Let $t = 2^x \in [1/2, 4]$. Then $y = 2t^2 - 4t + 11 = 2(t - 1)^2 + 9$: minimum 9 at $x = 0$, maximum 27 at $x = 2$.

Q021 h is non-monotonic on $\mathbb{R}$ and has global minimum $h(0) = 2$.

Q022 C

Q023 B

Q024 C

Q025 B

Q026 B

Q027 B

Q028 C

Q029 B

Q030 A

Q031 B

Q032 C

Q033 B

Q034 C

Q035 B

Q036 B

Q037 C

Q038 B

Q039 B

Q040 A

Q041 C

Q042 B

Quiz MCQ 1 A

Quiz MCQ 2 B

Quiz MCQ 3 C

Quiz MCQ 4 C

Quiz WRITTEN 5 Decreasing; $D = \mathbb{R}, R = (0, \infty), y =$

Quiz WRITTEN 6 $2^{-3} < 2^0 < 2^{5/2}$

Quiz WRITTEN 7 $x = 0 or x = 2$

Quiz WRITTEN 8 $2atx = 0$

Concept 3 | Introduction to Logarithms and Their Properties

English Student Edition • Focused formulas and guided practice

Formula opener

Definition

$$\log_b a = c \iff b^c = a$$

Product

$$\log_b(xy) = \log_b x + \log_b y$$

Power

$$\log_b(x^k) = k \log_b x$$

Eng Eslam Ahmed | 01120009622

Classified Practice **Introduction to Logarithms and Their Properties**

Q001 **Written**

Convert the source forms $2^5 = 32$ and $\log_3 9 = 2$ to the opposite notation.

Q002 **Written**

Convert $2^6 = 64$, $4^{-2} = 1/16$, and the two remaining Try expressions into logarithmic or exponential form.

Classified Practice Introduction to Logarithms and Their Properties

Q003 Written

Convert all Exercise items (a)+(l): the first six exponential statements to logarithms and the last six logarithmic statements to exponentials.

Q004 **Written**

If $\log_5 25 = x$, evaluate $\log_{1/2} x$ without a calculator.

Classified Practice Introduction to Logarithms and Their Properties

Q005 Written

Find the six direct logarithmic values shown in Warm Up, including $\log_{10} 1, \log_5 5, \log_2 8, \log_2 16, \log_5(1/5), \log_7(49^{3/2})$.

Q006 **Written**

Find the six Try values $\log_5 1, \log_3 3, \log_4 4, \log_3 27, \log_2(1/4), \log_5(1/125)$.

Classified Practice Introduction to Logarithms and Their Properties

Q007 Written

Find all 18 Exercise values (a)+(r), including reciprocal arguments and fractional powers.

Q008 **Written**

Put the printed logarithmic product $\log_{a^3}(a^2)$ into simplest form.

Classified Practice **Introduction to Logarithms and Their Properties**

Q009 **Written**

Perform $\log_6 2 + \log_6 3$ **and** $\log_2 30 - \log_2 3 - \log_2 5$.

Q010 **Written**

Perform the six Try calculations using product, quotient, and power laws, including $4\log_5 3 - 2\log_5 15 - \log_5 45$.

Classified Practice **Introduction to Logarithms and Their Properties**

Q011 Written

Perform all 18 source expressions (a)+(r), combining same-base logarithms exactly.

Q012 **Written**

Simplify the fractional expression $\log_5 75 + 3\log_5 \sqrt{2} - 2\log_5 \sqrt{6}$ **to one reduced fraction.**

Classified Practice **Introduction to Logarithms and Their Properties**

Q013 **Written**

Given $\log_{10} 2 = a$ **and** $\log_{10} 3 = b$, **express** $\log_{10} 24, \log_{10}(27/4), \log_{10} 45, \log_{10}(24/75)$ **in terms of** a, b.

Q014 **Written**

Given $\log_{10} 2 = a, \log_{10} 3 = b$**, express the four Try arguments** $72, 3^8/10, 15, 12$ **in terms of** a, b**.**

Classified Practice **Introduction to Logarithms and Their Properties**

Q015 **Written**

Express all 12 Exercise logarithms in terms of $a = \log_{10} 2$ and $b = \log_{10} 3$, including fractions and reciprocal arguments.

Q016 **Written**

If $\log_c 3 = x, \log_c 2 = y$, **write** $\log_c 12, \log_c(3/8), \log_c \sqrt{6}, \log_2 3, \log_{3^b}(2^b c)$ **in terms of** x, y.

Classified Practice Introduction to Logarithms and Their Properties

Q017 Written

Find $\log_{16} 64$ **and** $\log_5(1/125)$ **by changing to bases 2 and 5.**

Q018 **Written**

Evaluate the five Try logarithms using a suitable common base.

Classified Practice

Introduction to Logarithms and Their Properties

Q019 **Written**

Find all 15 Exercise values using change of base, including the mixed-base and fractional-argument items.

Q020 **Written**

Find $\log_9(\sqrt{3}/4\sqrt{2})$ **exactly.**

Classified Practice Introduction to Logarithms and Their Properties

Q021 Written

Solve the three warm-up expressions using a common base, including $\log_4 16 \cdot \log_9 3$.

Q022 **Written**

Solve the four Try products of logarithms after making the bases the same.

Classified Practice **Introduction to Logarithms and Their Properties**

Q023 **Written**

Simplify all 15 Exercise products and sums of logarithms by common-base conversion.

Q024 **Written**

Simplify $\log_x(\sqrt{y})\ \log_{y^2}(z^3)\ \log_{\sqrt{z}}(x^4)$ **for** $x > 1$.

Classified Practice **Introduction to Logarithms and Their Properties**

Q025 **Written**

Compare $f(x) = \log_4(x^2)$ and $g(x) = 2\log_4 x$ across the real line.

Q026 Written

Draw $y = \log_4 x$, state its positional relationship with $y = 4^x$, and draw $y = \log_{1/4} x$.

Classified Practice **Introduction to Logarithms and Their Properties**

Q027 Written

State graph relationships and draw the three Try logarithmic functions with bases 2, 5, and $1/3$.

Q028 **Written**

Complete both Exercise groups: state positional relationships and draw the six requested logarithmic graphs.

Classified Practice Introduction to Logarithms and Their Properties

Q029 **Written**

For $f(x) = \log_5(x-2)+1$ and $g(x) = 5^{x-1}+2$, state their exact geometric relationship and line of symmetry.

Q030 **Written**

Order the displayed logarithmic sets, including $\log_2 3, \log_2 5, 2\log_2(1/2), \log_{1/2} 3, \log_{1/2} 5$.

Classified Practice **Introduction to Logarithms and Their Properties**

Q031 **Written**

Order the four Try sets of logarithms using monotonicity and change of base.

Q032 Written

Order all 12 Exercise sets (a)+(l), including negative coefficients and reciprocal bases.

Classified Practice **Introduction to Logarithms and Their Properties**

Q033 **Written**

If $\log_c(3^2/2) = x$ **and** $\log_c(9) = y$**, write the relation between x and y for** $0 < c < 1$ **and** $c > 1$**.**

Q034 MCQ

$\log_2 32$ **equals:**

Choose the correct option.

A 2

B 4

C 5

D 32

Classified Practice **Introduction to Logarithms and Their Properties**

Q035 MCQ

$\log_{1/2} 2$ **equals:**

Choose the correct option.

A -2

B -1

C 1

D 2

Q036 MCQ

$\log_5(1/125)$ **equals:**

Choose the correct option.

A -3

B -2

C 2

D 3

Classified Practice **Introduction to Logarithms and Their Properties**

Q037 **MCQ**

$\log_7 49^{3/2}$ **equals:**

Choose the correct option.

A 1

B 2

C 3/2

D 7/2

Q038 MCQ

$\log_a M + \log_a N$ **equals:**

Choose the correct option.

A $\log_a(M + N)$

B $\log_a(MN)$

C $\log_a(M/N)$

D $\log_a(M - N)$

Classified Practice **Introduction to Logarithms and Their Properties**

Q039 **MCQ**

The coefficient $3\log_5 2$ becomes:

Choose the correct option.

A $\log_5(2+3)$

B $\log_5 2^3$

C $\log_5(3^2)$

D $3^2 \log_5 2$

Q040 MCQ

If $a = \log_{10} 2$ and $b = \log_{10} 3$, $\log_{10} 24$ is:

Choose the correct option.

A $a + b$

B $2a + 3b$

C $3a + b$

D $3a + 2b$

Classified Practice **Introduction to Logarithms and Their Properties**

Q041 **MCQ**

$\log_c \sqrt{6}$ **in terms of** $x = \log_c 3, y = \log_c 2$ **is:**

Choose the correct option.

A $x + y$

B $(x + y)/2$

C $2x + y$

D $x + 2y$

Q042 MCQ

$\log_{16} 64$ **equals:**

Choose the correct option.

A 1/2

B 3/2

C 2

D 4

Classified Practice **Introduction to Logarithms and Their Properties**

Q043 MCQ

Change of base gives $\log_a b =$:

Choose the correct option.

A $\log_c a / \log_c b$

B $\log_c b / \log_c a$

C $\log_a b / \log_c b$

D $\log_c a \log_c b$

Q044 **MCQ**

$\log_4 16 \cdot \log_9 3$ **equals:**

Choose the correct option.

A 1/2

B 1

C 2

D 4

Classified Practice **Introduction to Logarithms and Their Properties**

Q045 **MCQ**

The challenge product of three logarithms equals:

Choose the correct option.

A 6
B 8
C 12
D 24

Q046 **MCQ**

The domain of $y = \log_a x$ is:

Choose the correct option.

A all real x

B $x \geq 0$

C $x > 0$

D $x \neq 0$

Classified Practice **Introduction to Logarithms and Their Properties**

Q047 MCQ

The graph of $y = \log_a x$ is symmetric with $y = a^x$ about:

Choose the correct option.

A $x = 0$

B $y = 0$

C $y = x$

D $y = -x$

Q048 MCQ

For $a > 1$ and $M < N$, which is true?

Choose the correct option.

A $\log_a M > \log_a N$

B $\log_a M < \log_a N$

C They are equal

D Cannot compare

Classified Practice **Introduction to Logarithms and Their Properties**

Q049 MCQ

For $0 < a < 1$ and $M < N$, which is true?

Choose the correct option.

A $\log_a M > \log_a N$
B $\log_a M < \log_a N$
C They are equal
D Both are positive

Q050 MCQ

The vertical asymptote of $y = \log_a x$ is:

Choose the correct option.

A $x = 0$
B $y = 0$
C $x = 1$
D $y = 1$

Classified Practice **Introduction to Logarithms and Their Properties**

Q051 MCQ

A negative coefficient $-2\log_a M$ can be written as:

Choose the correct option.

A $\log_a M^{-2}$

B $\log_a(-2M)$

C $\log_a(M-2)$

D $2\log_a M$

Q052 **MCQ**

The base of $\log_a M$ is:

Choose the correct option.

A M

B a

C p

D $1/a$

Classified Practice **Introduction to Logarithms and Their Properties**

Q053 MCQ

The argument of $\log_a M$ is:

Choose the correct option.

A a
B M
C p
D 0

Q054 MCQ

$\log_2(1/8)$ **equals:**

Choose the correct option.

A -3

B -2

C 2

D 3

Classified Practice Introduction to Logarithms and Their Properties

Q055 MCQ

$\log_4 4$ **equals:**

Choose the correct option.

A 0

B 1

C 2

D 4

Q056 MCQ

$\log_3 1$ **equals:**

Choose the correct option.

A -1

B 0

C 1

D 3

Classified Practice **Introduction to Logarithms and Their Properties**

Q057 **MCQ**

$\log_a M - \log_a N$ **equals:**

Choose the correct option.

A $\log_a(MN)$

B $\log_a(M/N)$

C $\log_a(M-N)$

D $\log_a(N/M)$

Q058 **MCQ**

$2\log_3 5$ **equals:**

Choose the correct option.

A $\log_3 10$

B $\log_3 25$

C $\log_3 7$

D $\log_3(5/2)$

Classified Practice **Introduction to Logarithms and Their Properties**

Q059 MCQ

If $a = \log_{10} 2, b = \log_{10} 3, \log_{10} 12$ **is:**

Choose the correct option.

A $a + b$

B $2a + b$

C $a + 2b$

D $2a + 2b$

Q060 **MCQ**

$\log_{10} 5$ **in terms of** $a = \log_{10} 2$ **is:**

Choose the correct option.

A a

B $1 - a$

C $1 + a$

D $-a$

Classified Practice **Introduction to Logarithms and Their Properties**

Q061 *MCQ*

$\log_8 32$ **equals:**

Choose the correct option.

A 2/3

B 3/2

C 5/3

D 3

Q062 **MCQ**

$\log_{25} 125$ **equals:**

Choose the correct option.

A 1/2

B 3/2

C 2

D 5/2

Classified Practice **Introduction to Logarithms and Their Properties**

Q063 **MCQ**

$\log_a b \cdot \log_b a$ **equals:**

Choose the correct option.

A 0

B 1

C 2

D $\log_a b$

Q064 **MCQ**

$\log_2 8 \cdot \log_8 2$ **equals:**

Choose the correct option.

A 1/3

B 1

C 3

D 8

Classified Practice **Introduction to Logarithms and Their Properties**

Q065 MCQ

The best common base for 16 and 64 is:

Choose the correct option.

A 2

B 4

C 8

D 10

Q066 MCQ

The range of $y = \log_a x$ is:

Choose the correct option.

A $x > 0$

B $\mathbb{R}$

C $[0, \infty)$

D $y > 0$

Classified Practice Introduction to Logarithms and Their Properties

Quiz MCQ 1 Concept Quiz · MCQ

The logarithmic equivalent of $a^p = M$ is:

Choose the correct option.

A $\log_a M = p$
B $\log_p M = a$
C $\log_M a = p$
D $\log_a p = M$

Quiz MCQ 2 **Concept Quiz · MCQ**

$\log_2 8 + \log_2 4$ **equals:**

Choose the correct option.

A 2

B 3

C 5

D 12

Classified Practice **Introduction to Logarithms and Their Properties**

Quiz MCQ 3 **Concept Quiz · MCQ**

$\log_{16} 64$ **equals:**

Choose the correct option.

A 1/2

B 3/2

C 2

D 4

Quiz MCQ 4 Concept Quiz · MCQ

For $a > 1, M < N$ implies:

Choose the correct option.

A $\log_a M > \log_a N$

B $\log_a M < \log_a N$

C equality

D no relation

Classified Practice **Introduction to Logarithms and Their Properties**

Quiz WRITTEN 5 **Concept Quiz · Written**

Find $\log_3(1/27)$.

Quiz WRITTEN 6 **Concept Quiz · Written**

Simplify $\log_5 25 + \log_5 5$.

Classified Practice **Introduction to Logarithms and Their Properties**

Quiz WRITTEN 7 **Concept Quiz · Written**

Find $\log_{16} 64$ using change of base.

Quiz WRITTEN 8 **Concept Quiz · Written**

State the domain and range of $y = \log_a x$.

Answer key

Use the question number shown on each page.

Q001 $\log_2 32 = 5;\ 3^2 = 9$

Q002 $\log_2 64 = 6; \log_4(1/16) = -2$, with the same swap rule for the remaining items.

Q003 Each item follows $a^p = M \iff \log_a M = p$, preserving the printed bases and signs.

Q004 $x = 2$, so $\log_{1/2} 2 = -1$.

Q005 Values: $0, 1, 3, 4, -1, 3/2$.

Q006 $0, 1, 1, 3, -2, -3$.

Q007 Convert each argument to a power of its base; the answers are the corresponding integers or rational exponents.

Q008 $2/3$.

Q009 1 and 1.

Q010 Each expression reduces to one logarithm and then an integer; the displayed worked example gives -3.

Q011 Every item is reduced by $\log_a M + \log_a N = \log_a(MN)$, $\log_a M - \log_a N = \log_a(M/N)$, and $k \log_a M = \log_a(M^k)$.

Q012 1.

Q013 $3a + b,\ 2 - 2a + 3b,\ 2a + 1 - a,\ b + 2 - 2a$.

Q014 Use prime factorisation and the same linear combinations of a, b produced by the logarithm laws.

Q015 Each result is a linear combination of $a, b, 1$ obtained from prime factors.

Q016 $x + 2y,\ x - 3y,\ (x + y)/2,\ x/y,\ (1 + by)/(bx)$.

Q017 $3/2$ and -3.

Q018 Rewrite powers of 2, 3, or 5 and apply the change-of-base quotient exactly.

Q019 All values are exact rational numbers obtained from $\log_a b = \log_c b / \log_c a$.

Q020 $\dfrac{\log_2 3 - 5/2}{2 \log_2 3}$.

Q021 Use reciprocal and power identities; the worked values are exact constants.

Q022 Each product collapses to a rational number after change of base and cancellation.

Q023 Use $\log_a b \log_b a = 1$ and the power law; every printed item reduces exactly.

Q024 12.

Q025 They agree only for $x > 0$; f is defined for $x \neq 0$, while g requires $x > 0$.

Q026 Both logarithmic graphs have domain $x > 0$, range $\mathbb{R}$, asymptote $x = 0$; $y = \log_4 x$ increases and is the reflection of $y = 4^x$ in $y = x$, while $\log_{1/4} x$ decreases.

Q027 Base > 1 gives increasing curves; base between 0 and 1 gives decreasing curves; each is inverse-symmetric to its exponential.

Q028 Use inverse symmetry, domain $x > 0$, range $\mathbb{R}$, and increasing/decreasing classification from the base.

Q029 They are inverse-reflected curves with symmetry line $y = x + 1$.

Q030 Make bases equal, turn coefficients into powers, and reverse order for base $0 < a < 1$.

Q031 Each chain follows argument order for base > 1 and reverses it for base $0 < a < 1$.

Q032 Convert every term to one base-log form and apply monotonicity; negative coefficients become reciprocal arguments.

Q033 Use the same argument comparison; the relation reverses when the base crosses 1.

Q034 C

Q035 B

Q036 A

Q037 C

Q038 B

Q039 B

Q040 C

Q041 B

Q042 B

Q043 B

Q044 A

Q045 C

Q046 C

Q047 C

Q048 B

Q049 A

Q050 A

Q051 A

Q052 B

Q053 B

Q054 A

Q055 B

Q056 B

Q057 B

Q058 B

Q059 B

Q060 B

Q061 B

Q062 B

Q063 B

Q064 B

Q065 A

Q066 B

Quiz MCQ 1 A

Quiz MCQ 2 C

Quiz MCQ 3 B

Quiz MCQ 4 B

Quiz WRITTEN 5 -3

Quiz WRITTEN 6 3

Quiz WRITTEN 7 3/2

Quiz WRITTEN 8 Domain $x > 0$; range all real numbers

Concept 4 | Applications of Logarithms

English Student Edition • Focused formulas and guided practice

Formula opener

Change base

$$\log_b a = \frac{\ln a}{\ln b}$$

Domain

$$x > 0$$

Inverse

$$b^{\log_b x} = x$$

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Classified Practice **Applications of Logarithms**

Q001 **Written**

Solve $\log_2 x = 4$.

Q002 **Written**

Solve $\log_2(2x-1) = \log_2 3$.

Classified Practice **Applications of Logarithms**

Q003 **Written**

Solve all 12 source logarithmic equations, including same-base, product, quotient, and shifted-argument forms.

Q004 **Written**

Solve $\log_3(x+2) - \log_9((x-2)^2) = 1$ **completely.**

Classified Practice Applications of Logarithms

Q005 **Written**

Solve $\log_3 x < \log_3 81$ **and** $\log_{1/2}(x+2) + \log_{1/2}(x-5) > -3$.

Q006 **Written**

Solve the six source logarithmic inequalities using base monotonicity and argument conditions.

Classified Practice **Applications of Logarithms**

Q007 **Written**

Solve all 24 source logarithmic inequalities, including shifted arguments, coefficients, and rational arguments.

Q008 **Written**

Solve the source challenge inequality involving a reciprocal-base logarithmic sum and state the complete interval.

Classified Practice Applications of Logarithms

Q009 Written

Solve $5(\log_2 x)^2 - 16\log_2 x + 3 < 0.$

Q010 **Written**

Solve the three source substitution problems: two quadratic logarithmic inequalities/equations and one exponential-log value.

Classified Practice **Applications of Logarithms**

Q011 **Written**

Solve the source groups: three equations, three inequalities, and three exact logarithmic values.

Q012 **Written**

Solve $(\log_3 x)^2 - \log_3 x - 6 < 0$ **for the positive wave-intensity multiplier** x.

Classified Practice **Applications of Logarithms**

Q013 **Written**

For $y = -(\log_3 x)^2 + 4\log_3 x + 3$, $1 \le x \le 27$, find maximum/minimum values and the corresponding x.

Q014 **Written**

Find the maximum and minimum of the source logarithmic quadratic on its stated interval, with the corresponding values of x.

Classified Practice **Applications of Logarithms**

Q015 **Written**

Solve the three source extrema problems, each with a different base and interval.

Q016 **Written**

For $y = -(\log_2 x)^2 + 2\log_2 x + 4$, $1 \le x \le 8$, find the maximum and minimum health ratings and their concentrations.

Classified Practice **Applications of Logarithms**

Q017 **Written**

Using $\log_{10} 2 = 0.3010$ and $\log_{10} 3 = 0.4771$, determine the digits of 6^{30} and the first non-zero decimal place of $(2/3)^{50}$.

Q018 **Written**

Complete the source digit-count and first-non-zero-decimal-place sets for the displayed powers and fractions.

Classified Practice

Applications of Logarithms

Q019 **Written**

Using the printed common logs, determine the number of digits and the leading digit of 12^{50}.

Q020 **Written**

Solve $\log_x(x+2) \le \log_x(4-x)$ **for all real** x**.**

Classified Practice **Applications of Logarithms**

Q021 *MCQ*

The argument condition for $\log_3(x-5)$ is:

Choose the correct option.

A x<5

B $x \geq 5$

C x>5

D all real x

Q022 **MCQ**

If $\log_2(2x-1)=\log_2 3$**, then** $x=$**:**

Choose the correct option.

A 1

B 2

C 3

D 4

Classified Practice **Applications of Logarithms**

Q023 MCQ

The valid solution of the challenge equation is:

Choose the correct option.

A 1

B 2

C 4

D 8

Q024 MCQ

For same-base logarithms, equality of logs gives:

Choose the correct option.

A arguments equal
B bases equal
C arguments add
D no result

Classified Practice **Applications of Logarithms**

Q025 **MCQ**

For base $a > 1$, $\log_a M < \log_a N$ means:

Choose the correct option.

A M>N

B $M<N$

C $M = N$

D $MN<0$

Q026 **MCQ**

For base $0 < a < 1$, the inequality direction is:

Choose the correct option.

A preserved

B reversed

C removed

D squared

Classified Practice **Applications of Logarithms**

Q027 **MCQ**

The domain of $\log_{1/2}(x-5)$ **requires:**

Choose the correct option.

A x<5

B $x \leq 5$

C x>5

D $x \neq 5$

Q028 MCQ

The warm-up intersection is:

Choose the correct option.

A $0<x<5$

B $5<x<6$

C $x>6$

D $-3<x<6$

Classified Practice **Applications of Logarithms**

Q029 MCQ

The best substitution for a quadratic logarithmic equation is:

Choose the correct option.

A $t = x^2$

B $t = \log_a x$

C $t = a^x$

D $t = 1/x$

Q030 MCQ

The inequality $5t^2 - 16t + 3 < 0$ gives:

Choose the correct option.

A t<1/5

B 1/5<t<3

C t>3

D 0<t<1

Classified Practice **Applications of Logarithms**

Q031 MCQ

The substitution variable must satisfy:

Choose the correct option.

A t>0

B $t \in \mathbb{R}$

C t<0

D $t = 1$

Q032 MCQ

The challenge interval for $(\log_3 x)^2 - \log_3 x - 6 < 0$ is:

Choose the correct option.

A x<1/9

B 1/9<x<27

C x>27

D 0<x<3

Classified Practice **Applications of Logarithms**

Q033 MCQ

For $y = -(t-2)^2 + 7$, the maximum occurs at:

Choose the correct option.

A $t = 0$
B $t = 1$
C $t = 2$
D $t = 3$

Q034 MCQ

The correct first step for extrema is:

Choose the correct option.

A differentiate in x

B set $t = \log_a x$

C ignore the interval

D take $x = 0$

Classified Practice **Applications of Logarithms**

Q035 MCQ

For $1 \leq x \leq 27$, $t = \log_3 x$ lies in:

Choose the correct option.

A [-1,3]

B [0,3]

C [1,27]

D all real

Q036 MCQ

The water-tank challenge maximum health rating is:

Choose the correct option.

A 3

B 4

C 5

D 8

Classified Practice **Applications of Logarithms**

Q037 MCQ

The number of digits of N is determined by:

Choose the correct option.

A $\log_{10} N$

B $\lfloor \log_{10} N \rfloor + 1$

C N+1

D 10^{N}

Q038 MCQ

If $-9 < \log_{10} N < -8$, the first non-zero digit is at:

Choose the correct option.

A 8th

B 9th

C 10th

D 11th

Classified Practice **Applications of Logarithms**

Q039 **MCQ**

The common logarithm base is:

Choose the correct option.

A 2

B e

C 10

D any base

Q040 MCQ

The solution set of $\log_x(x+2) \leq \log_x(4-x)$ **is:**

Choose the correct option.

A (0,1)

B (1,4)

C (0,4)

D empty

Classified Practice **Applications of Logarithms**

Quiz MCQ 1 **Concept Quiz · MCQ**

Solve $\log_5(x-1)=2$.

Choose the correct option.

A $x=26$

B 0

C 1

D cannot be determined

Quiz MCQ 2 **Concept Quiz · MCQ**

For $0 < a < 1$ and $M < N$, compare $\log_a M$ and $\log_a N$.

Choose the correct option.

A $\log_a M > \log_a N$

B 0

C 1

D cannot be determined

Classified Practice **Applications of Logarithms**

Quiz MCQ 3 **Concept Quiz · MCQ**

Solve $4(\log_2 x)^2 - 5\log_2 x + 1 = 0$.

Choose the correct option.

A $x = 2, \sqrt{2}$

B 0

C 1

D cannot be determined

Quiz MCQ 4 **Concept Quiz · MCQ**

Find the maximum of $y = -(\log_3 x)^2 + 2\log_3 x + 4$ **on** $1 \leq x \leq 27$**.**

Choose the correct option.

A 5 at $x = 3$

B 0

C 1

D cannot be determined

Classified Practice **Applications of Logarithms**

Quiz WRITTEN 5 **Concept Quiz · Written**

Solve $\log_3(x+1) = 2$.

Quiz WRITTEN 6 **Concept Quiz · Written**

Solve $\log_2 x \geq 3$.

Classified Practice **Applications of Logarithms**

Quiz WRITTEN 7 **Concept Quiz · Written**

Solve $(\log_5 x)^2 - 4 < 0$.

Quiz WRITTEN 8 **Concept Quiz · Written**

How many digits does 6^{20} **have using** $\log_{10} 2 = 0.3010, \log_{10} 3 = 0.4771$**?**

Answer key

Use the question number shown on each page.

Q001 16

Q002 $x = 2$

Q003 Apply the argument condition first; then equate same-base arguments and reject any extraneous roots.

Q004 $x = 4$

Q005 $0 < x < 81$; $5 < x < 6$

Q006 Each solution is the intersection of the argument domain with the monotonicity interval.

Q007 Use argument conditions, convert coefficients by the power law, then solve and intersect the intervals.

Q008 Reverse the comparison after converting to a base below one, then intersect with both argument conditions.

Q009 $2^{1/5} < x < 8$

Q010 Substitute $t = \log_a x$, solve in t, enforce $x > 0$, and back-substitute.

Q011 All groups follow the same substitution, domain, solve, and back-substitution sequence.

Q012 $1/9 < x < 27$

Q013 Maximum 7 at $x = 9$; minimum 3 at $x = 1$.

Q014 Use the vertex in $t = \log_a x$ and compare it with both endpoints.

Q015 For every item, map the domain to t, locate the vertex, and check endpoints.

Q016 Maximum 5 at $x = 2$; minimum 4 at $x = 1$.

Q017 24 digits; 9th decimal place.

Q018 Use $n - 1 < \log_{10} N < n$ for digits and $-n < \log_{10} N < -n + 1$ for the first non-zero place.

Q019 55 digits; leading digit 1.

Q020 No real solution.

Q021 C

Q022 B

Q023 C

Q024 A

Q025 B

Q026 B

Q027 C

Q028 B

Q029 B

Q030 B

Q031 B

Q032 B

Q033 C

Q034 B

Q035 B

Q036 C

Q037 B

Q038 B

Q039 C

Q040 D

Quiz MCQ 1 A

Quiz MCQ 2 B

Quiz MCQ 3 C

Quiz MCQ 4 D

Quiz WRITTEN 5 $x = 8$

Quiz WRITTEN 6 $x \geq 8$

Quiz WRITTEN 7 $1/25 < x < 25$

Quiz WRITTEN 8 16 digits

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

English Student Edition • Focused formulas and guided practice

Formula opener

Derivative

$$\frac{d}{dx}x^n = nx^{n-1}$$

Product

$$(uv)' = u'v + uv'$$

Chain

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$$

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Classified Practice **Fundamentals of Calculus (Limits and Derivatives)**

Q001 **Written**

For $f(x) = x^2 + 2x$, find the average rate of change from -1 to 3, and from 3 to $3 + h$.

Q002 **Written**

Find the average rate of change for the three source Try functions on their stated intervals.

Classified Practice **Fundamentals of Calculus (Limits and Derivatives)**

Q003 **Written**

Find the average rate of change for all nine source Exercise functions, including fixed and h-form intervals.

Q004 **Written**

Find the average rate of change of a circle's area as its diameter changes from 10 cm to 18 cm.

Classified Practice **Fundamentals of Calculus (Limits and Derivatives)**

Q005 **Written**

Find the source limits by direct substitution and by simplifying the indeterminate fraction before substitution.

Q006 **Written**

Find the three source Try limits, including polynomial and h-quotient forms.

Classified Practice **Fundamentals of Calculus (Limits and Derivatives)**

Q007 **Written**

Find all nine source Exercise limits, including x-limits and h-limits.

Q008 **Written**

If $\lim_{x \to 3}(x^2 - 5x + k)/(x - 3) = L$ **is finite, find** Lk**.**

Classified Practice **Fundamentals of Calculus (Limits and Derivatives)**

Q009 **Written**

Find the differential coefficient of $f(x) = 3x^2$ at $x = -2$ using the definition.

Q010 **Written**

Find the differential coefficients at the given values for the two source Try functions using the definition.

Classified Practice **Fundamentals of Calculus (Limits and Derivatives)**

Q011 **Written**

Find the differential coefficients for all six source Exercise functions at their stated values.

Q012 **Written**

A dropped stone has displacement $s(t) = 4.9t^2$ metres. Find its instantaneous velocity at $t = 10$ seconds using the definition.

Classified Practice **Fundamentals of Calculus (Limits and Derivatives)**

Q013 **Written**

Differentiate $f(x) = x^2 + 3x$ using the definition of the derivative.

Q014 **Written**

Differentiate the three source Try functions using the definition.

Classified Practice **Fundamentals of Calculus (Limits and Derivatives)**

Q015 **Written**

Differentiate all nine source Exercise functions using the definition.

Q016 **Written**

A regular hexagon has side length x. Write its area and differentiate it using the definition.

Classified Practice **Fundamentals of Calculus (Limits and Derivatives)**

Q017 **Written**

If $f(x) = 5$ for every x except that it is undefined at x=2, does $\lim_{x\to 2} f(x)$ exist?

Q018 MCQ

The average rate of change from a to b is:

Choose the correct option.

A [f(a)+f(b)]/(b-a)

B [f(b)-f(a)]/(b-a)

C f'(a)

D b-a

Classified Practice **Fundamentals of Calculus (Limits and Derivatives)**

Q019 MCQ

For $f(x) = x^2 + 2x$ from -1 to 3, the average rate is:

Choose the correct option.

A 2

B 3

C 4

D 8

Q020 MCQ

The average area-rate for diameter 10 to 18 is:

Choose the correct option.

A 7π

B 14π

C 28π

D 56π

Classified Practice   **Fundamentals of Calculus (Limits and Derivatives)**

Q021 MCQ

The h-form average rate is found by:

Choose the correct option.

A setting $h = 1$
B cancelling the common h
C differentiating first
D squaring h

Q022 MCQ

A direct-substitution limit is valid when:

Choose the correct option.

A the expression is defined
B the denominator is zero
C the numerator is infinite
D h is negative

Classified Practice **Fundamentals of Calculus (Limits and Derivatives)**

Q023 **MCQ**

The 0/0 form requires:

Choose the correct option.

A factor/cancel first
B differentiate twice
C ignore the denominator
D change the base

Q024 MCQ

In the finite-limit challenge, k equals:

Choose the correct option.

A 3

B 5

C 6

D 9

Classified Practice **Fundamentals of Calculus (Limits and Derivatives)**

Q025 MCQ

For a polynomial, the limit equals:

Choose the correct option.

A its value at the point
B zero
C infinity
D the slope

Q026 MCQ

The definition of the differential coefficient at a is:

Choose the correct option.

A lim [f(a+h)-f(a)]/h

B f(a+h)+f(a)

C [f(a)-f(0)]/a

D h/f(a)

Classified Practice **Fundamentals of Calculus (Limits and Derivatives)**

Q027 MCQ

For $f(x) = 3x^2$, $f'(-2)$ is:

Choose the correct option.

A -12

B -6

C 6

D 12

Q028 MCQ

The stone's velocity at 10 s is:

Choose the correct option.

A 49

B 98

C 196

D 4.9

Classified Practice **Fundamentals of Calculus (Limits and Derivatives)**

Q029 **MCQ**

After cancelling h, take the limit as:

Choose the correct option.

A $h \to 0$

B $h \to 1$

C $x \to 0$

D $a \to \infty$

Q030 MCQ

The derivative function is:

Choose the correct option.

A the coefficient at one point
B the differential coefficient for every x
C the average rate only
D the original function

Classified Practice **Fundamentals of Calculus (Limits and Derivatives)**

Q031 *MCQ*

The derivative of $x^2 + 3x$ from the definition is:

Choose the correct option.

A x+3

B 2x+3

C 2x

D x^2

Q032 MCQ

The area of a regular hexagon of side x is:

Choose the correct option.

A $3\sqrt{3}x$

B $(3\sqrt{3}/2)x^2$

C $6x^2$

D $\sqrt{3}x$

Classified Practice **Fundamentals of Calculus (Limits and Derivatives)**

Q033 MCQ

A function need not be defined at a for its limit to:

Choose the correct option.

A exist
B be infinite only
C equal zero only
D fail

Q034 MCQ

The derivative of a constant function is:

Choose the correct option.

A 0

B 1

C x

D the constant

Classified Practice **Fundamentals of Calculus (Limits and Derivatives)**

Quiz MCQ 1 **Concept Quiz · MCQ**

Find the average rate of change of x^2 from 1 to 3.

Choose the correct option.

A 4
B 1
C 2
D 3

Quiz MCQ 2 **Concept Quiz · MCQ**

Find $\lim_{x \to 2}(x^2 + 1)$**.**

Choose the correct option.

A 0

B 5

C 2

D 3

Classified Practice **Fundamentals of Calculus (Limits and Derivatives)**

Quiz MCQ 3 **Concept Quiz · MCQ**

Find $f'(x)$ for $f(x) = x^2$ using the definition.

Choose the correct option.

A 0
B 1
C 2x
D 3

Quiz MCQ 4 **Concept Quiz · MCQ**

If $s(t) = 4.9t^2$, find the velocity at t=5.

Choose the correct option.

A 0

B 1

C 2

D 49

Classified Practice **Fundamentals of Calculus (Limits and Derivatives)**

Quiz WRITTEN 5 Concept Quiz · Written

Find the average rate of change of $3x + 1$ from 2 to 7.

Quiz WRITTEN 6 **Concept Quiz · Written**

Evaluate $\lim_{h\to 0}(6 + 2h)$**.**

Classified Practice **Fundamentals of Calculus (Limits and Derivatives)**

Quiz WRITTEN 7 Concept Quiz · Written

Find the differential coefficient of $2x^2$ at x=3 from the definition.

Quiz WRITTEN 8 **Concept Quiz · Written**

State $\lim_{x\to 2} f(x)$ when f(x)=5 for all x except x=2 where it is undefined.

Answer key

Use the question number shown on each page.

Q001 4; $8+h$

Q002 2; $-5/3$; $h-6$

Q003 Apply $[f(b)-f(a)]/(b-a)$ to each printed item exactly.

Q004 $7\pi\ \mathrm{cm}^2/\mathrm{cm}$

Q005 -2; 6

Q006 Substitute after removing any common factor; polynomial limits are obtained directly.

Q007 Each limit is obtained by direct substitution or cancellation of the factor causing 0/0.

Q008 6

Q009 $f'(-2)=-12$

Q010 Substitute a into $[f(a+h)-f(a)]/h$, cancel h, and let h tend to zero.

Q011 Use the definition at each specified point and simplify exactly.

Q012 $98\ \mathrm{m\,s^{-1}}$

Q013 $f'(x)=2x+3$

Q014 Expand each $f(x+h)-f(x)$, cancel h, and take the limit.

Q015 Apply $f'(x)=\lim_{h\to 0}[f(x+h)-f(x)]/h$ to every printed polynomial.

Q016 $A=\frac{3\sqrt{3}}{2}x^2$, $A'(x)=3\sqrt{3}\,x$

Q017 Yes; the limit is 5.

Q018 B

Q019 C

Q020 A

Q021 B

Q022 A

Q023 A

Q024 C

Q025 A

Q026 A

Q027 A

Q028 B

Q029 A

Q030 B

Q031 B

Q032 B

Q033 A

Q034 A

Quiz MCQ 1 A

Quiz MCQ 2 B

Quiz MCQ 3 C

Quiz MCQ 4 D

Quiz WRITTEN 5 3

Quiz WRITTEN 6 6

Quiz WRITTEN 7 12

Quiz WRITTEN 8 5

Concept 2 | Differentiation and Tangent Lines

English Student Edition • Focused formulas and guided practice

Formula opener

Gradient

$$m = f'(x_1)$$

Tangent

$$y - y_1 = m(x - x_1)$$

Normal

$$m_n m = -1$$

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Classified Practice **Differentiation and Tangent Lines**

Q001 **Written**

Differentiate the four Warm Up demands: $y = x^2 - 3x + 2$, $y = (2x + 1)^3$, $V = \pi r^2 h$ with respect to (r), and find $f'(-1)$ for $f(x) = x^3 - 3x + 1$.

Q002 **Written**

Differentiate the six printed Try functions, the two variable-labelled functions $S = \pi r^2$ and $h = vt - \frac{1}{2}gt^2$, and evaluate $f'(2), f'(-3)$ for $f(x) = 4x^3 - 3x^2 + 2$.

Classified Practice **Differentiation and Tangent Lines**

Q003 Written

Differentiate every source Exercise function $18 polynomial/product forms$**, the six functions with variables indicated in brackets, and evaluate the two values in each of Exercises 3–5.**

Classified Practice **Differentiation and Tangent Lines**

Q004 **Written**

For $f(x) = \sqrt[3]{x^5} + \frac{1}{x^2}$**, find** $f'(1) + f'(-1)$.

Q005 **Written**

Find the quadratic $f(x) = ax^2 + bx + c$ **satisfying** $f'(0) = -2, f'(2) = 2, f(1) = 2.$

Classified Practice **Differentiation and Tangent Lines**

Q006 **Written**

Find the quadratic satisfying $f'(0) = 2, f'(1) = 4, f(2) = 6.$

Q007 **Written**

Find the four quadratics satisfying the four source triples of derivative and function conditions.

Classified Practice **Differentiation and Tangent Lines**

Q008 Written

Find the quadratic (f(x)) satisfying $f'(2) + f'(4) = 0, f(3) = 0, f(1) = 12.$

Q009 **Written**

Find the tangent line to $y = x^2 - 4x + 1$ **at** $(3, -2)$.

Classified Practice **Differentiation and Tangent Lines**

Q010 **Written**

Find the tangent line for the three source functions at their stated points: $y = x^2 + 3$ **at** $(-2, 7)$, $y = x^2 - 3x + 2$ **at ((1,0)), and** $y = -x^3 + 2x + 4$ **at** $(-2, 8)$.

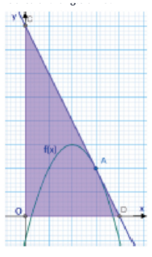
Q011 Written

Find the tangent line for all nine source Exercise functions at their given points.

Classified Practice **Differentiation and Tangent Lines**

Q012 Written

For $f(x) = -x^2 + 4x - 1$, the tangent at x-coordinate 3 meets the axes at (D,C). Find the area of triangle (ODC).



Classified Practice **Differentiation and Tangent Lines**

Q013 **Written**

Solve the two source Warm Up demands: tangents from (C(0,2)) to $y = x^2 - 3x + 6$, and tangents to $y = x^3 + 4x^2 - 3$ with gradient 3.

Q014 **Written**

Solve the two source Try demands: tangents from $C(1,-1)$ to $y=x^2-x+3$, and tangents to $y=x^3+2x-3$ with gradient 5.

Classified Practice Differentiation and Tangent Lines

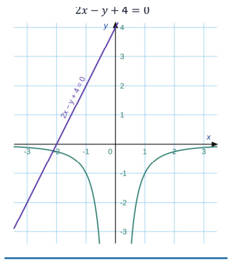
Q015 Written

Solve all six source Exercise demands involving external points or prescribed gradients.

Classified Practice **Differentiation and Tangent Lines**

Q016 **Written**

For $f(x) = -1/x^2$, find the tangent line parallel to $2x - y + 4 = 0$.



Classified Practice **Differentiation and Tangent Lines**

Q017 MCQ

The derivative of x^n is:

Choose the correct option.

A x^{n-1}

B nx^{n-1}

C $n+x$

D x^n

Q018 MCQ

The derivative of $V = \pi r^2 h$ with respect to r is:

Choose the correct option.

A πh

B $2\pi r h$

C $2\pi r$

D πr^2

Classified Practice **Differentiation and Tangent Lines**

Q019 MCQ

For $f(x) = x^3 - 3x + 1$, $f'(-1)$ is:

Choose the correct option.

A -6

B -3

C 0

D 3

Q020 MCQ

The challenge function can be written as:

Choose the correct option.

A $x^{5/3} + x^{-2}$

B $x^5 + x^2$

C $x^{3/5} + x^2$

D $x^{5/3} + x^2$

Classified Practice **Differentiation and Tangent Lines**

Q021 MCQ

For $f(x) = ax^2 + bx + c$, (f'(x)) is:

Choose the correct option.

A $ax + b$

B $2ax + b$

C $2a + b + c$

D $ax^2 + b$

Q022 **MCQ**

The condition $f'(0) = -2$ gives:

Choose the correct option.

A $a = -2$

B $b = -2$

C $c = -2$

D $2a = -2$

Classified Practice **Differentiation and Tangent Lines**

Q023 MCQ

The warm-up quadratic is:

Choose the correct option.

A $x^2 - 2x + 3$

B $x^2 + 2x + 3$

C $2x^2 - x + 3$

D $x^2 - 2x - 3$

Q024 MCQ

The challenge coefficient a is:

Choose the correct option.

A 1

B 2

C 3

D 6

Classified Practice **Differentiation and Tangent Lines**

Q025 *MCQ*

The tangent gradient at x=a is:

Choose the correct option.

A (f(a))

B (a)

C (f'(a))

D $1/f'(a)$

Q026 MCQ

The tangent to $y = x^2 - 4x + 1$ at $(3, -2)$ is:

Choose the correct option.

A $y = x - 5$
B $y = 2x - 8$
C $y = -2x + 4$
D $y = 3x - 11$

Classified Practice **Differentiation and Tangent Lines**

Q027 MCQ

In the triangle challenge, the x-intercept of the tangent is:

Choose the correct option.

A 2

B 3

C 4

D 8

Q028 MCQ

The area of triangle (ODC) in the challenge is:

Choose the correct option.

A 8
B 12
C 16
D 32

Classified Practice **Differentiation and Tangent Lines**

Q029 MCQ

When the contact point is not given, set it to:

Choose the correct option.

A ((0,0))
B ((a,f(a)))
C ((f(a),a))
D ((1,f'(a)))

Q030 MCQ

The gradient of $2x - y + 4 = 0$ is:

Choose the correct option.

A -2

B -1

C 1

D 2

Classified Practice **Differentiation and Tangent Lines**

Q031 MCQ

For $f(x) = -1/x^2$, (f'(x)) is:

Choose the correct option.

A $-2/x^3$

B $2/x^3$

C $1/x^2$

D $-1/x^3$

Q032 **MCQ**

The parallel tangent in the challenge is:

Choose the correct option.

A $y = 2x + 3$
B $y = 2x - 3$
C $y = -2x - 3$
D $y = x - 2$

Classified Practice **Differentiation and Tangent Lines**

Quiz WRITTEN 1 Concept Quiz · Written

Differentiate $3x^3 - 2x$.

Quiz WRITTEN 2 **Concept Quiz · Written**

Find the quadratic with $f'(0) = 1, f'(1) = 3, f(1) = 2.$

Classified Practice **Differentiation and Tangent Lines**

Quiz WRITTEN 3 Concept Quiz · Written

Find the tangent to $y = x^2$ at ((2,4)).

Quiz WRITTEN 4 **Concept Quiz · Written**

For $f(x) = -1/x$, find the tangent parallel to $y = x + 2$.

Classified Practice **Differentiation and Tangent Lines**

Quiz MCQ 5 Concept Quiz · MCQ

The derivative of a constant is:

Choose the correct option.

A 0

B 1

C x

D the constant

Quiz MCQ 6 **Concept Quiz · MCQ**

For $f(x) = ax^2 + bx + c$, the coefficient equations come from:

Choose the correct option.

A three function values

B two derivatives and one function value

C one derivative only

D the tangent gradient only

Classified Practice **Differentiation and Tangent Lines**

Quiz MCQ 7 **Concept Quiz · MCQ**

The tangent equation in point-slope form is:

Choose the correct option.

A $y - y_1 = m(x - x_1)$
B $y = mx + c$
C $y + y_1 = m(x + x_1)$
D $x - x_1 = m(y - y_1)$

Quiz MCQ 8 **Concept Quiz · MCQ**

A tangent parallel to $y = 3x - 1$ has gradient:

Choose the correct option.

A -3

B 1/3

C 3

D 1

Answer key

Use the question number shown on each page.

Q001 $2x - 3; 6(2x+1)^2; 2\pi rh; 0$

Q002 $3, -6x+8, 2x+3, 6x^2-8x+6, 6(2x-3)^2, 4x^2+x-1; 2\pi r, v-gt; 36, 12$

Q003 Apply the power rule term-by-term; for a product or power, expand first. The value sets are: Exercise 3 (33,65); Exercise 4 $-24, -3$; Exercise 5 (3,28).

Q004 $\frac{10}{3}$

Q005 $f(x) = x^2 - 2x + 3$

Q006 $f(x) = x^2 + 2x$

Q007 $x^2 - 4x + 8; -2x^2 + 3x + 3; -5x^2 + 26x - 14; -\frac{3}{2}x^2 + 2x + \frac{3}{2}$

Q008 $f(x) = 3x^2 - 18x + 27$

Q009 $y = 2x - 8$

Q010 $y = -4x - 1; y = -x + 1; y = -10x - 12$

Q011 $y = 4x - 5; y = 4x - 1; y = x - 4; y = x - 7; y = -4x + 3; y = 7x - 3; y = 9x - 16; y = -x - 2; y = x + 1$

Q012 16 square units

Q013 $y = -7x + 2, y = x + 2; \quad y = 3x + 15, y = 3x - \frac{95}{27}$

Q014 $y = 5x - 6, y = -3x + 2; \quad y = 5x - 5, y = 5x - 1$

Q015 $y = 6x - 8, y = -2x; y = x + 1, y = -7x + 25; y = -5x + 6, y = 3x - 2; y = 9x - 11, y = 9x + 21; y = -x; y = -2x + 16, y = -2x - 8$

Q016 $y = 2x - 3$

Q017 B

Q018 B

Q019 C

Q020 A

Q021 B

Q022 B

Q023 A

Q024 C

Q025 C

Q026 B

Q027 C

Q028 C

Q029 B

Q030 D

Q031 B

Q032 B

Quiz WRITTEN 1 $9x^2 - 2$

Quiz WRITTEN 2 $x^2 + x$

Quiz WRITTEN 3 $y = 4x - 4$

Quiz WRITTEN 4 $y = x - 4$

Quiz MCQ 5 A

Quiz MCQ 6 B

Quiz MCQ 7 A

Quiz MCQ 8 C

Concept 3 | Applications of Differentiation: Polynomial Functions

English Student Edition • Focused formulas and guided practice

Formula opener

Rate

$$\frac{dy}{dx}$$

Stationary

$$f'(x) = 0$$

Second derivative

$$f''(x) > 0 \Rightarrow \text{minimum}$$

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Classified Practice **Applications of Differentiation: Polynomial Functions**

Q001 Written

Find the extreme values and sketch the graphs of the three cubic functions printed in the Warm Up.

Q002 **Written**

Find the extreme values and sketch the three source Try cubics.

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q003 **Written**

Find the extreme values and draw the graphs for all nine source cubic functions in Exercise.

Q004 **Written**

For $P(x) = x^3 - 3ax^2 + (3a^2 - 3)x + 4$, $a > 0$, find the flow rates giving the maximum and minimum peak efficiency.

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q005 **Written**

On $[-3, 4]$, find the maximum and minimum values of $y = -x^3 + 12x + 5$.

Q006 **Written**

Find the maximum and minimum values of the two source Try cubics on their stated closed intervals.

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q007 **Written**

Find the maximum and minimum values of all six source cubics on their given closed intervals.

Q008 **Written**

For $f(x) = x^3 - 6x$ on $[-3, 3]$, find the absolute deepest dive and highest ascent.

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q009 **Written**

If $f(x) = -x^3 + ax + b$ has a local maximum of 9 at $x = 2$, find a, b, and the local minimum.

Q010 Written

Solve the two source Try problems: determine the coefficients from a local minimum/maximum and another stationary point.

Classified Practice

Applications of Differentiation: Polynomial Functions

Q011 **Written**

Solve all six source Exercise problems asking for cubic coefficients from extrema.

Q012 **Written**

For $f(x) = x^3 + ax^2 + bx + c$, the graph has a local peak 178 at $x = -2$ and a local trough at $x = 4$. Find a, b, c.

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q013 **Written**

If $f(x) = x^3 - 3kx^2 + kx + 2$ has no extreme values, find the range of k.

Q014 **Written**

Determine the parameter ranges in the source Try problems for a cubic to have no extreme values or to have both extremes.

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q015 **Written**

Determine the parameter ranges for all six source Exercise cubics according to whether they have extreme values.

Q016 **Written**

For $f(x) = x^3 + 3kx^2 + 12kx + 10$, **find the exact k-range that guarantees both a local maximum and minimum.**

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q017 **Written**

If $f(x) = x^3 - 3kx^2 + (k^2 + 5)x + 3$ is always monotonically increasing, find the range of k.

Q018 **Written**

Find the parameter ranges in the two source Try problems for the cubic to be monotonically increasing.

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q019 **Written**

Find the parameter ranges in all six source Exercise problems for monotonic increase.

Q020 **Written**

For $f(x) = \frac{1}{3}x^3 - kx^2 + (k+6)x + 10$, find the k-range for monotonic increase on $x \in [1, \infty)$.

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q021 **Written**

Find the number of distinct real roots of $x^3 - 6x^2 + 9x - 3 = 0$.

Q022 **Written**

Find the number of distinct real roots for the three source Try equations.

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q023 **Written**

Find the number of distinct real roots for all nine source cubic equations.

Q024 **Written**

Find the number of distinct real roots of $-h^3 + 12h^2 - 36h + 40 = 0$.

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q025 **Written**

If $-x^3 + 3x^2 + 9x + a = 0$ has three distinct real roots, find the range of a.

Q026 **Written**

Solve the two source Try parameter problems for three or two distinct real roots.

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q027 Written

Solve all four source Exercise parameter problems for the required number of distinct roots.

Q028 Written

For $f(x) = x^3 - 6x^2 + 9x - 2a^2 + 7a$, find all real a such that $f(x) = 2$ has exactly three distinct real roots.

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q029 **Written**

Prove the source inequalities using an auxiliary cubic and a derivative sign table.

Q030 **Written**

Prove all three source cubic inequalities on their stated domains.

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q031 **Written**

Find the largest k such that $x^3 - 3x^2 - 3x + k + 3 \leq 0$ for every $x \in [-1, 3]$.

Q032 **Written**

Find the extreme values and sketch the two source quartic functions.

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q033 **Written**

Find the extreme values and graphs for all six source quartic functions.

Q034 **Written**

For $f(x) = x^4 - 4x^3 + k$, find the exact k-range for exactly two distinct real roots.

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q035 MCQ

At a local maximum, the sign of f' changes:

Choose the correct option.

A $- \to +$

B $+ \to -$

C $+ \to +$

D $- \to -$

Q036 MCQ

For $P'(x) = 3[(x - a)^2 - 1]$, the local maximum occurs at:

Choose the correct option.

A $a - 1$

B a

C $a + 1$

D $1 - a$

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q037 **MCQ**

Absolute extrema on a closed interval require comparing:

Choose the correct option.

A stationary points only
B endpoints only
C stationary points and endpoints
D the y-intercept

Q038 MCQ

The absolute maximum of $x^3 - 6x$ on $[-3, 3]$ is:

Choose the correct option.

A $4\sqrt{2}$

B 9

C 6

D 0

Classified Practice Applications of Differentiation: Polynomial Functions

Q039 MCQ

If $f(x) = -x^3 + ax + b$ has a stationary point at 2, then a is:

Choose the correct option.

A 2

B 6

C 12

D -12

Q040 MCQ

The derivative roots in the challenge are:

Choose the correct option.

A $-4, 2$

B $-2, 4$

C $2, 4$

D $-3, 3$

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q041 *MCQ*

No two distinct extrema for a cubic means the derivative discriminant is:

Choose the correct option.

A $D > 0$

B $D = 1$

C $D \leq 0$

D $D < 0$ only

Q042 MCQ

The challenge range for both extrema is:

Choose the correct option.

A $0 < k < 4$

B $k < 0$ or $k > 4$

C $-4 < k < 0$

D $k \leq 4$

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q043 MCQ

For a positive-leading cubic to be monotone increasing, its derivative must be:

Choose the correct option.

A negative everywhere

B non-negative everywhere

C zero everywhere

D linear

Q044 MCQ

The challenge domain in 7-13 is:

Choose the correct option.

A $(-\infty, \infty)$

B $[1, \infty)$

C $[-1, 1]$

D $(0, 1)$

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q045 MCQ

Three distinct roots occur when the x-axis lies:

Choose the correct option.

A above both extrema
B between the local maximum and minimum
C at an endpoint only
D at the local maximum only

Q046 MCQ

The 7-14 challenge has how many distinct real roots?

Choose the correct option.

A 0

B 1

C 2

D 3

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q047 **MCQ**

Three intersections with $y = a$ require the level to lie:

Choose the correct option.

A strictly between the extrema

B above the maximum

C below the minimum

D at an extremum

Q048 MCQ

The 7-15 warm-up range is:

Choose the correct option.

A $-27 < a < 5$

B $-5 < a < 27$

C $a \leq -27$

D $a \geq 5$

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q049 **MCQ**

To prove LHS $\geq$ RHS, define:

Choose the correct option.

A $f = \text{LHS} + \text{RHS}$

B $f = \text{LHS} - \text{RHS}$

C $f = \text{RHS} - 1$

D $f = f'$

Q050 MCQ

The largest k in the challenge is:

Choose the correct option.

A $2 - 4\sqrt{2}$
B $4\sqrt{2} - 2$
C $2 + 4\sqrt{2}$
D $-2 - 4\sqrt{2}$

Classified Practice

Applications of Differentiation: Polynomial Functions

Q051 **MCQ**

A repeated root of a derivative may be:

Choose the correct option.

A always a local maximum
B always a local minimum
C stationary but not an extreme
D impossible

Q052 **MCQ**

Exactly two distinct roots for $x^4 - 4x^3 + k = 0$ require:

Choose the correct option.

A $k = 27$

B $k < 27$

C $k > 27$

D $k \leq 0$ only

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q053 MCQ

A local minimum is identified by the change:

Choose the correct option.

A $+ \to -$

B $- \to +$

C $+ \to +$

D $- \to -$

Q054 MCQ

For $y = x^3 - 6x^2 + 16$, the critical x-values are:

Choose the correct option.

A $0, 4$

B $1, 3$

C $-2, 2$

D $2, 4$

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q055 MCQ

The derivative sign table is used to determine:

Choose the correct option.

A only the y-intercept

B increase/decrease and extrema

C the constant term

D the domain

Q056 MCQ

On a closed interval, an endpoint can be:

Choose the correct option.

A ignored

B an absolute extreme

C a derivative root only

D never a minimum

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q057 MCQ

For $y = -x^3 + 12x + 5$, the stationary x-values are:

Choose the correct option.

A $-2, 2$

B $-1, 1$

C $0, 4$

D $2, 4$

Q058 MCQ

The absolute minimum of $x^3 - 6x$ on $[-3, 3]$ is at:

Choose the correct option.

A $x = -3$
B $x = -\sqrt{2}$
C $x = \sqrt{2}$
D $x = 3$

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q059 MCQ

To determine a cubic from an extreme value, use both $f(a) = M$ and:

Choose the correct option.

A $f(a) = 0$
B $f'(a) = 0$
C $f''(a) = 0$
D $a = 0$

Q060 **MCQ**

In the warm-up $f(x) = -x^3 + 12x - 7$, the local minimum occurs at:

Choose the correct option.

A $x = -2$

B $x = 0$

C $x = 2$

D $x = 7$

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q061 MCQ

For $f(x) = x^3 - 3kx^2 + kx + 2$, no two distinct stationary roots requires:

Choose the correct option.

A $0 \leq k \leq 1/3$

B $k > 1/3$

C $k < 0$

D $k \geq 1/3$

Q062 MCQ

Two distinct stationary points occur when the derivative discriminant is:

Choose the correct option.

A $D < 0$

B $D = 0$

C $D > 0$

D $D = 1$

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q063 MCQ

For a positive-leading quadratic derivative, $D \leq 0$ implies it is:

Choose the correct option.

A non-negative for all x

B negative for all x

C linear

D undefined

Q064 MCQ

In the 7-13 challenge, monotonicity is required on:

Choose the correct option.

A $[1, \infty)$

B $[-1, 1]$

C $(-infty, 0)$

D $[0, 1]$

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q065 MCQ

A cubic with both stationary values above the x-axis has:

Choose the correct option.

A three roots

B one distinct real root

C no real roots

D two repeated roots

Q066 MCQ

To count cubic roots, compare the extrema with:

Choose the correct option.

A the y-axis

B the x-axis

C the derivative axis

D the domain only

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q067 MCQ

At an extremal parameter level, a cubic and a horizontal line have:

Choose the correct option.

A one distinct intersection
B two distinct intersections
C three distinct intersections
D no intersection

Q068 MCQ

For $x^4 - 4x^3 + k$, the global minimum occurs at:

Choose the correct option.

A $x = 0$

B $x = 1$

C $x = 3$

D $x = 4$

Classified Practice **Applications of Differentiation: Polynomial Functions**

Quiz WRITTEN 1 Concept Quiz · Written

Find the local maximum and minimum of $x^3 - 3x^2 + 1$.

Quiz WRITTEN 2 **Concept Quiz · Written**

Find a, b if $f(x) = -x^3 + ax + b$ has a local maximum 4 at x=1.

Classified Practice **Applications of Differentiation: Polynomial Functions**

Quiz WRITTEN 3 **Concept Quiz · Written**

Find the k-range for $x^3 - 3kx^2 + kx$ to have no two distinct stationary points.

Quiz WRITTEN 4 **Concept Quiz · Written**

Find the largest k such that $x^3 - 3x^2 - 3x + k + 3 \leq 0$ on $[-1, 3]$.

Classified Practice **Applications of Differentiation: Polynomial Functions**

Quiz MCQ 5 Concept Quiz · MCQ

At a local minimum, f' changes:

Choose the correct option.

A $+ \to -$

B $- \to +$

C $+ \to +$

D $- \to -$

Quiz MCQ 6 **Concept Quiz · MCQ**

A cubic with a positive-leading derivative and $D < 0$ is:

Choose the correct option.

A always increasing

B always decreasing

C constant

D undefined

Classified Practice **Applications of Differentiation: Polynomial Functions**

Quiz MCQ 7 **Concept Quiz · MCQ**

The 7-15 challenge requires the level to satisfy:

Choose the correct option.

A $t < 0$

B $0 < t < 4$

C $t > 4$

D $t = 4$

Quiz MCQ 8 **Concept Quiz · MCQ**

The global minimum of $x^4 - 4x^3 + k$ is:

Choose the correct option.

A $k - 27$

B $k + 27$

C $k - 3$

D $27 - k$

Answer key

Use the question number shown on each page.

Q001 Use the derivative sign table: for x^3-6x^2+16, local maximum 16 at $x=0$ and local minimum -16 at $x=4$; classify the other two from their stationary points.

Q002 Apply the same derivative sign-table test; a stationary point where the sign of f' does not change is not an extreme value.

Q003 Differentiate each polynomial, solve the quadratic derivative, use the sign table, and evaluate f at every genuine stationary point.

Q004 Local maximum at $x=a-1$; local minimum at $x=a+1$.

Q005 Maximum 21 at $x=2$; minimum -11 at $x=-2$.

Q006 Evaluate both endpoints and every stationary point inside the interval; the greatest value is the absolute maximum and the least is the absolute minimum.

Q007 For each function, compare the endpoint values with the values at all critical x-values lying in the interval.

Q008 Absolute minimum -9 at $x=-3$; absolute maximum 9 at $x=3$.

Q009 $a=12, b=-7$; local minimum -23 at $x=-2$.

Q010 Use $f'(a)=0, f(a)=M$, and the second stationary condition to form three simultaneous equations in the coefficients.

Q011 Each set is determined by two derivative roots/conditions and one extreme-value condition; solve the resulting simultaneous equations.

Q012 $a=-3,\ b=-24,\ c=150.$

Q013 $0\le k\le \frac{1}{3}$.

Q014 Use $D\le 0$ for no extremes and $D>0$ for two distinct stationary points.

Q015 Apply the derivative-discriminant criterion separately to each printed cubic.

Q016 $k<0 or k>4.$

Q017 $-\frac{\sqrt{10}}{2}\le k\le \frac{\sqrt{10}}{2}.$

Q018 Set the derivative discriminant $D\le 0$, with the derivative leading coefficient positive.

Q019 Differentiate each cubic and impose that its quadratic derivative never becomes negative.

Q020 $k\le 3.$

Q021 Three distinct real roots.

Q022 Use the cubic graph: count x-axis intersections from the stationary points and end behavior.

Q023 Each answer is obtained from the number of x-axis intersections of its corresponding cubic graph.

Q024 One distinct real root.

Q025 $-27<a<5.$

Q026 Rewrite each equation as $g(x)=a$, then compare the parameter line with the local maximum and minimum of g.

Q027 Use the local extrema of the parameter-free cubic and count intersections with the horizontal parameter line.

Q028 $\frac{7-\sqrt{65}}{4}<a<\frac{7-\sqrt{33}}{4} or \frac{7+\sqrt{33}}{4}<a<\frac{7+\sqrt{65}}{4}.$

Q029 Set $f(x)=\mathrm{LHS}-\mathrm{RHS}$, prove $f(x)\ge 0$ on the stated domain, and identify its minimum.

Q030 Move the right-hand side to the left, then show the resulting auxiliary cubic has non-negative minimum on the domain.

Q031 $k\le 2-4\sqrt{2}$; the largest value is $2-4\sqrt{2}$.

Q032 For $3x^4+4x^3-12x^2$: local maximum 0 at $x=0$, local minima -32 at $x=-2$ and -5 at $x=1$. For $-x^4-4x^3-8$: maximum 19 at $x=-3$; $x=0$ is stationary but not an extreme.

Q033 Differentiate each quartic, construct the derivative sign table, and evaluate each genuine turning point.

Q034 $k<27.$

Q035 B

Q036 A

Q037 C

Q038 B

Q039 C

Q040 B

Q041 C

Q042 B

Q043 B

Q044 B

Q045 B

Q046 B

Q047 A

Q048 A

Q049 B

Q050 A

Q051 C

Q052 B

Q053 B

Q054 A

Q055 B

Q056 B

Q057 A

Q058 A

Q059 B

Q060 A

Q061 A

Q062 C

Q063 A

Q064 A

Q065 B

Q066 B

Q067 B

Q068 C

Quiz WRITTEN 1 Maximum 1 at $x = 0$; minimum -3 at $x = 2$

Quiz WRITTEN 2 $a = 3,\ b = 2$

Quiz WRITTEN 3 $-\frac{\sqrt{10}}{2} \leq k \leq \frac{\sqrt{10}}{2}$

Quiz WRITTEN 4 $2 - 4\sqrt{2}$

Quiz MCQ 5 B

Quiz MCQ 6 A

Quiz MCQ 7 B

Quiz MCQ 8 A

Concept 4 | Integration: Definite Integrals and Applications

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Formula opener

Antiderivative

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Definite

$$\int_a^b f(x)dx = F(b) - F(a)$$

Area

$$A = \int_a^b |f(x)|dx$$

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Classified Practice

Integration: Definite Integrals and Applications

Q001 **Written**

Find the two indefinite integrals in the Warm Up.

Q002 **Written**

Find the six indefinite integrals in the source Try section, expanding a squared bracket first when needed.

Classified Practice **Integration: Definite Integrals and Applications**

Q003 **Written**

Solve the first ten source indefinite-integral exercises (constants, linear and quadratic polynomials).

Q004 **Written**

Solve the remaining eight source integrals, including products and squared brackets.

Classified Practice **Integration: Definite Integrals and Applications**

Q005 **Written**

Find $F(x)$ when $F'(x) = 3x^2 + 8x - 1$ and $F(1) = 7$.

Q006 Written

Find the two source Try functions from their derivatives and one given value of F.

Classified Practice **Integration: Definite Integrals and Applications**

Q007 **Written**

Find $F(x)$ for all six source Exercise conditions.

Q008 **Written**

Find the curve $y = f(x)$ through $(3, 2)$ whose tangent gradient is $3x^2 - 2x + 1$.

Classified Practice **Integration: Definite Integrals and Applications**

Q009 **Written**

Solve the source Try tangent-gradient problem through $(1, 1)$**.**

Q010 **Written**

Solve the three source tangent-gradient exercises, finding each function from its point and gradient.

Classified Practice **Integration: Definite Integrals and Applications**

Q011 **Written**

Evaluate the three definite integrals in the Warm Up using the Fundamental Theorem of Calculus.

Q012 **Written**

Evaluate the six source Try definite integrals, simplifying the integrand before applying the limits.

Classified Practice **Integration: Definite Integrals and Applications**

Q013 **Written**

Evaluate the first twelve source Exercise definite integrals, including reversed and equal limits.

Q014 **Written**

Evaluate the remaining twelve source Exercise integrals, including sums, differences and products.

Classified Practice **Integration: Definite Integrals and Applications**

Q015 **Written**

Use the four properties of definite integrals to simplify the three Warm Up expressions.

Q016 **Written**

Solve the four source Try definite-integral transformations.

Classified Practice **Integration: Definite Integrals and Applications**

Q017 **Written**

Solve the first six source Exercise questions using interval splitting and reversal.

Q018 **Written**

Solve the final six source Exercise questions using the factorized definite-integral identity.

Classified Practice **Integration: Definite Integrals and Applications**

Q019 **Written**

Find $f(x)$ from the source equation $f(x) = x^2 + \int_0^1 f(t)\,dt$.

Q020 **Written**

Solve the two source Try integral equations for $f(x)$.

Classified Practice **Integration: Definite Integrals and Applications**

Q021 **Written**

Find $f(x)$ for all five source integral equations.

Q022 **Written**

Differentiate $\int_3^x (2t^2 + 3t - 5)\, dt$ with respect to x.

Classified Practice **Integration: Definite Integrals and Applications**

Q023 **Written**

Solve the two source Try problems involving differentiation of a variable-limit integral and an unknown lower limit.

Q024 **Written**

Differentiate the three source expressions with respect to x.

Classified Practice **Integration: Definite Integrals and Applications**

Q025 **Written**

Find $f(x)$ and a for the three source equations with a variable upper limit.

Q026 MCQ

The constant of integration is required because:

Choose the correct option.

A every primitive is unique

B primitives differ by a constant

C the integrand is constant

D the limits are equal

Classified Practice **Integration: Definite Integrals and Applications**

Q027 MCQ

The integral of x^n for $n \neq -1$ is:

Choose the correct option.

A $nx^{n-1} + C$

B $x^{n+1}/(n+1) + C$

C $x^n/n + C$

D $x^{n-1}/(n-1) + C$

Q028 MCQ

Before integrating $(x-1)^2$, first:

Choose the correct option.

A differentiate it

B expand it

C set $x = 0$

D add limits

Classified Practice **Integration: Definite Integrals and Applications**

Q029 **MCQ**

The integral of $3x^2 - 2x + 1$ is:

Choose the correct option.

A $x^3 - x^2 + x + C$

B $3x^3 - 2x^2 + x + C$

C $x^3 - 2x + 1 + C$

D $6x - 2 + C$

Q030 MCQ

To find $F(x)$ from $F'(x)$, you should first:

Choose the correct option.

A differentiate again

B integrate $F'(x)$

C set $F = 0$

D ignore the initial value

Classified Practice **Integration: Definite Integrals and Applications**

Q031 MCQ

The condition $F(1) = 7$ determines:

Choose the correct option.

A the derivative

B the domain

C the constant C

D the leading coefficient

Q032 MCQ

If $F'(x) = 6x - 2$ and $F(0) = 1$, then $F(x)$ is:

Choose the correct option.

A $3x^2 - 2x + 1$

B $6x^2 - 2x + 1$

C $3x^2 - 2x$

D $6x - 2 + 1$

Classified Practice **Integration: Definite Integrals and Applications**

Q033 MCQ

The gradient of the tangent at (x, y) is:

Choose the correct option.

A $f(x)$

B $f'(x)$

C $f''(x)$

D $x + y$

Q034 MCQ

A point (x_0, y_0) on $y = f(x)$ gives:

Choose the correct option.

A $f'(x_0) = y_0$

B $f(x_0) = y_0$

C $f(0) = x_0$

D $C = 0$

Classified Practice **Integration: Definite Integrals and Applications**

Q035 MCQ

If $f'(x) = 3x^2 - 2x + 1$, an antiderivative is:

Choose the correct option.

A $3x^3 - 2x^2 + x$

B $x^3 - x^2 + x + C$

C $6x - 2$

D $x^3 - 2x + 1$

Q036 **MCQ**

The Fundamental Theorem gives $\int_a^b f(x)dx =$:

Choose the correct option.

A $F(a) - F(b)$

B $F(b) - F(a)$

C $F(a) + F(b)$

D 0 always

Classified Practice **Integration: Definite Integrals and Applications**

Q037 MCQ

When the limits are equal, $\int_a^a f(x)dx$ equals:

Choose the correct option.

A 1

B a

C 0

D $f(a)$

Q038 MCQ

A reversed definite integral satisfies:

Choose the correct option.

A $\int_a^b f = \int_b^a f$

B $\int_a^b f = -\int_b^a f$

C both are zero

D only if $f = 0$

Classified Practice **Integration: Definite Integrals and Applications**

Q039 MCQ

To evaluate a definite integral of a product, first:

Choose the correct option.

A integrate each factor separately
B expand the product
C differentiate the product
D change the limits

Q040 **MCQ**

Adjacent integrals with the same integrand can be:

Choose the correct option.

A multiplied

B combined over the joined interval

C set equal to zero

D reversed only

Classified Practice **Integration: Definite Integrals and Applications**

Q041 **MCQ**

To use $\int_{\alpha}^{\beta}(x-\alpha)(x-\beta)dx$, the factors must have:

Choose the correct option.

A different roots

B the displayed roots (a,b)

C zero coefficients

D equal limits

Q042 MCQ

The identity for the factored integral is:

Choose the correct option.

A $(\beta - \alpha)^2/6$

B $-(\beta - \alpha)^3/6$

C $(\beta + \alpha)^3/6$

D 0

Classified Practice **Integration: Definite Integrals and Applications**

Q043 MCQ

A minus sign before an integral can be removed by:

Choose the correct option.

A squaring the integrand

B reversing its limits

C adding C

D differentiating

Q044 **MCQ**

In $f(x) = p(x) + \int_a^b f(t)dt$, the definite integral is:

Choose the correct option.

A a function of x

B a constant

C always zero

D a derivative

Classified Practice **Integration: Definite Integrals and Applications**

Q045 **MCQ**

The best first step for an integral equation is to let:

Choose the correct option.

A $A = f(x)$

B $A = \int_a^b f(t)dt$

C $A = x$

D $A = C$ only

Q046 MCQ

For the warm-up $f(x) = x^2 + \int_0^1 f(t)dt$, the constant integral is:

Choose the correct option.

A $1/3$

B $-1/3$

C 0

D 1

Classified Practice **Integration: Definite Integrals and Applications**

Q047 **MCQ**

$\frac{d}{dx}\int_a^x f(t)dt$ **equals:**

Choose the correct option.

A $f(a)$

B $f(x)$

C $F(x)$

D 0

Q048 MCQ

When differentiating an integral, the dummy variable t is replaced by:

Choose the correct option.

A the lower limit

B the upper-limit variable

C C

D zero

Classified Practice **Integration: Definite Integrals and Applications**

Q049 **MCQ**

If $\int_a^a f(t)dt = 0$, this can determine:

Choose the correct option.

A the derivative
B the unknown lower limit a
C the integrand degree
D the upper limit

Q050 MCQ

For $\int_3^x (2t^2 + 3t - 5)dt$, the derivative is:

Choose the correct option.

A $2x^2 + 3x - 5$
B $2x^3 + 3x^2 - 5x$
C $3x^2 + 3x - 5$
D 0

Classified Practice **Integration: Definite Integrals and Applications**

Quiz WRITTEN 1 **Concept Quiz · Written**

Find $int(4x^3 - 6x + 2)dx$.

Quiz WRITTEN 2 **Concept Quiz · Written**

Find $F(x)$ if $F'(x) = 2x - 5$ and $F(2) = 1$.

Classified Practice **Integration: Definite Integrals and Applications**

Quiz WRITTEN 3 **Concept Quiz · Written**

Evaluate $\int_{-1}^{2}(3x^2 - 4x + 1)dx$.

Quiz WRITTEN 4 **Concept Quiz · Written**

Find $f(x)$ **if** $f(x) = x^2 + \int_0^1 f(t)dt$.

Classified Practice **Integration: Definite Integrals and Applications**

Quiz MCQ 5 **Concept Quiz · MCQ**

The derivative of $\int_0^x (3t^2 + 1)dt$ is:

Choose the correct option.

A $3x^2 + 1$

B $x^3 + x$

C $6x$

D 0

Quiz MCQ 6 **Concept Quiz · MCQ**

A primitive of $5x^4$ is:

Choose the correct option.

A $x^5 + C$

B $5x^5 + C$

C $20x^3 + C$

D $x^4/5 + C$

Classified Practice **Integration: Definite Integrals and Applications**

Quiz MCQ 7 **Concept Quiz · MCQ**

For $F'(x) = 4x + 2$ and $F(0) = 3$, $F(x)$ is:

Choose the correct option.

A $2x^2 + 2x + 3$

B $4x^2 + 2x + 3$

C $2x^2 + 2$

D $4x + 5$

Quiz MCQ 8 **Concept Quiz · MCQ**

$\int_1^3 f(x)dx + \int_3^5 f(x)dx$ **equals:**

Choose the correct option.

A $\int_1^5 f(x)dx$
B $\int_1^3 f(x)dx$
C $\int_3^5 f(x)dx$
D 0

Answer key

Use the question number shown on each page.

Q001 Integrate term by term and add C to each primitive.

Q002 Expand products or powers, collect like powers, then integrate every term.

Q003 Apply linearity and the power rule; retain the constant of integration.

Q004 Expand each product or square, simplify, and integrate term by term.

Q005 $F(x) = x^3 + 4x^2 - x + 3$.

Q006 Integrate the derivative and use the given point to determine C.

Q007 For each item, integrate $F'(x)$, then impose the stated value of F.

Q008 $f(x) = x^3 - x^2 + x - 19$.

Q009 Integrate the given gradient and use the point on the curve to determine the constant.

Q010 Integrate the polynomial gradient and apply the given point condition in each case.

Q011 Find an antiderivative F and calculate $F(b) - F(a)$; equal limits give zero.

Q012 Expand products, integrate, and evaluate the antiderivative at the upper and lower limits.

Q013 Use the Fundamental Theorem; a reversed interval changes the sign and equal limits give 0.

Q014 Use linearity, expand products, then apply the endpoint-substitution rule.

Q015 Scale constants, reverse limits, split intervals, or use the factored-interval identity as appropriate.

Q016 Swap limits when necessary, join adjacent intervals, and normalize a linear factor before using the identity.

Q017 Rewrite each expression until adjacent intervals have matching integrands and compatible limits.

Q018 Scale to $\int_a^b (x-a)(x-b)\,dx = -\frac{(b-a)^3}{6}$.

Q019 Let $A = \int_0{}^1 f(t)\,dt$; then $A = frac13 + A$, so $A = -frac13$ and $f(x) = x^2 - frac13$.

Q020 Introduce a constant for the definite integral, evaluate it using the proposed $f(t)$, and solve the linear equation.

Q021 Let the definite integral be A, substitute the expression for $f(t)$, solve for A, then recover $f(x)$.

Q022 $2x^2 + 3x - 5$.

Q023 Replace t by x when differentiating, then use equal limits or the given condition to determine a.

Q024 Apply the variable-limit Fundamental Theorem directly to each integrand.

Q025 Differentiate to get $f(x)$, then set the integral to zero at $x = a$ to solve for a.

Q026 B

Q027 B

Q028 B

Q029 A

Q030 B

Q031 C

Q032 A

Q033 B

Q034 B

Q035 B

Q036 B

Q037 C

Q038 B

Q039 B

Q040 B

Q041 B

Q042 B

Q043 B

Q044 B

Q045 B

Q046 B

Q047 B

Q048 B

Q049 B

Q050 A

Quiz WRITTEN 1 $x^4 - 3x^2 + 2x + C$

Quiz WRITTEN 2 $F(x) = x^2 - 5x + 7$

Quiz WRITTEN 3 6

Quiz WRITTEN 4 $f(x) = x^2 - 1/3$

Quiz MCQ 5 A

Quiz MCQ 6 A

Quiz MCQ 7 A

Quiz MCQ 8 A

Concept 5 | Applications of Integration to Geometry

English Student Edition • Focused formulas and guided practice

Formula opener

Area

$$A = \int_a^b (y_{top} - y_{bottom}) dx$$

Volume

$$V = \pi \int_a^b y^2 dx$$

Bounds

$$a \le x \le b$$

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Classified Practice Applications of Integration to Geometry

Q001 Written

Find the area bounded by $y = 2x^2 + 1$, the x-axis, $x = -2$, and $x = 1$.

Q002 Written

Find the area bounded by $y = x^2 + x - 6$, the x-axis, $x = -2$, and $x = 1$.

Classified Practice Applications of Integration to Geometry

Q003 Written

Find the area bounded by $y = -x^2 + 4$ and the x-axis.

Q004 **Written**

Find the area under $y = -x^2 - 2x + 3$ and above the x-axis.

Classified Practice Applications of Integration to Geometry

Q005 Written

Find the area between $y = x^2 - 2$ and $y = -2x + 1$.

Q006 Written

Find the area between $y = x^2 + 2x - 4$ and $y = -x^2 + 2x + 4$.

Classified Practice Applications of Integration to Geometry

Q007 **Written**

Find the area between $y = -x^2 + 4x$ and $y = x$.

Q008 Written

Find the area between $y = -x^2 + 8$ and $y = 2x$ for $-2 \leq x \leq 3$.

Classified Practice Applications of Integration to Geometry

Q009 Written

Find the area bounded by $y = x^3 - 5x^2 + 6x$ and the x-axis.

Q010 Written

Find the area bounded by $y = -x^3 + 4x$ and the x-axis.

Classified Practice **Applications of Integration to Geometry**

Q011 **Written**

Solve the source cubic area exercises by finding every x-intercept and adding the absolute signed pieces.

Q012 **Written**

Evaluate $\int_{-2}^{1} |x+1|\,dx$.

Classified Practice **Applications of Integration to Geometry**

Q013 **Written**

Evaluate $\int_0^3 |x^2 - 4|\, dx$.

Q014 **Written**

Evaluate $\int_0^2 |x^2 + 3x - 4| \, dx$.

Classified Practice Applications of Integration to Geometry

Q015 Written

Find the area bounded by $y = x^2 - x$ and its tangents at $(0, 0)$ and $(2, 2)$.

Q016 **Written**

Find the area bounded by $y = x^3 + x^2 - 3x + 6$ **and its tangent at** $(1, 5)$.

Classified Practice **Applications of Integration to Geometry**

Q017 **Written**

For $y = x^2$, tangents through $(3, 8)$ touch at $x = 2, 4$. Find the enclosed area.

Q018 **Written**

If $y = kx$ bisects the area under $y = 2x - x^2$ and above the x-axis, find k.

Classified Practice Applications of Integration to Geometry

Q019 Written

Solve the source bisection exercises for the given parabolas and lines, using the required area ratio.

Q020 **MCQ**

When the graph is below the x-axis, area is:

Choose the correct option.

A the signed integral

B the negative of the integral

C always zero

D undefined

Classified Practice

Applications of Integration to Geometry

Q021 MCQ

A bounded area under a curve is found by first checking:

Choose the correct option.

A the graph and sign

B only the y-intercept

C the font size

D the derivative only

Q022 MCQ

The area under $y = -x^2 + 4$ between its roots is:

Choose the correct option.

A 8/3

B 16/3

C 32/3

D 64/3

Classified Practice **Applications of Integration to Geometry**

Q023 MCQ

Between two curves, the integrand is:

Choose the correct option.

A lower-upper

B upper-lower

C product

D derivative

Q024 **MCQ**

If curves cross inside the interval, area must be:

Choose the correct option.

A split at the crossing

B ignored

C doubled always

D negative

Classified Practice **Applications of Integration to Geometry**

Q025 **MCQ**

The area between $y = x^2 - 2$ and $y = -2x + 1$ is:

Choose the correct option.

A 16/3

B 32/3

C 64/3

D 9/2

Q026 **MCQ**

For cubic area, the sign changes at:

Choose the correct option.

A every turning point

B each x-intercept

C the y-intercept only

D the midpoint

Classified Practice **Applications of Integration to Geometry**

Q027 **MCQ**

The area for $y = -x^3 + 4x$ is:

Choose the correct option.

A 4

B 6

C 8

D 12

Q028 MCQ

A negative lobe contributes:

Choose the correct option.

A negative area

B positive absolute area

C no area

D a derivative

Classified Practice Applications of Integration to Geometry

Q029 MCQ

For an absolute-value integral, first locate:

Choose the correct option.

A the roots
B the vertex only
C the y-axis
D the tangent

Q030 MCQ

The value of $\int_0^2 |x^2+3x-4|\,dx$ is:

Choose the correct option.

A 3

B 4

C 5

D 6

Classified Practice **Applications of Integration to Geometry**

Q031 MCQ

If f changes sign at c, split the integral:

Choose the correct option.

A at c

B at 0 only

C never

D after doubling

Q032 **MCQ**

A tangent line at x=a has slope:

Choose the correct option.

A f(a)
B f'(a)
C a^2
D the area

Classified Practice **Applications of Integration to Geometry**

Q033 **MCQ**

For two tangent lines, split at their:

Choose the correct option.

A intersection x-coordinate

B y-intercept only

C vertex only

D origin

Q034 MCQ

The area in the source parabola-tangent Warm Up is:

Choose the correct option.

A 1/2

B 1

C 3/2

D 3

Classified Practice **Applications of Integration to Geometry**

Q035 **MCQ**

Bisection means each part has:

Choose the correct option.

A the same area

B the same slope

C the same root

D zero area

Q036 MCQ

For $y=2x-x^2$, the total area above the x-axis is:

Choose the correct option.

A 1

B 2

C 3

D 4

Classified Practice

Applications of Integration to Geometry

Q037 MCQ

The line y=kx must satisfy the stated:

Choose the correct option.

A area ratio

B font ratio

C root count

D period

Q038 MCQ

In a bisection problem, the cut-off area is:

Choose the correct option.

A the total area
B half the total area
C twice the total area
D zero

Classified Practice Applications of Integration to Geometry

Quiz MCQ 1 Concept Quiz · MCQ

The area under $y = -x^2 + 4$ and above the x-axis is:

Choose the correct option.

A 16/3

B 8

C 32/3

D 64/3

Quiz MCQ 2 **Concept Quiz · MCQ**

The area between $y = x^2 - 2$ and $y = -2x + 1$ is:

Choose the correct option.

A 16/3
B 32/3
C 64/3
D 9/2

Classified Practice **Applications of Integration to Geometry**

Quiz MCQ 3 **Concept Quiz · MCQ**

Evaluate $\int_{-2}^{1} |x+1|\, dx$:

Choose the correct option.

A 2

B 5/2

C 3

D 7/2

Quiz MCQ 4 **Concept Quiz · MCQ**

The area enclosed by $y = x^2$ and tangents through $(3, 8)$ is:

Choose the correct option.

A 1/3

B 1/2

C 2/3

D 4/3

Classified Practice **Applications of Integration to Geometry**

Quiz WRITTEN 5 **Concept Quiz · Written**

Find the area bounded by $y = x^3 - 5x^2 + 6x$ and the x-axis.

Quiz WRITTEN 6 **Concept Quiz · Written**

Evaluate $\int_0^2 |x^2 + 3x - 4|\, dx$.

Classified Practice **Applications of Integration to Geometry**

Quiz WRITTEN 7 Concept Quiz · Written

Find the area bounded by $y = x^2 - x$ and its tangents at $(0, 0)$ and $(2, 2)$.

Quiz WRITTEN 8 **Concept Quiz · Written**

If $y = kx$ bisects the area under $y = 2x - x^2$, find k.

Answer key

Use the question number shown on each page.

Q001 9

Q002 $\frac{33}{2}$

Q003 $\frac{32}{3}$

Q004 $\frac{32}{3}$

Q005 $\frac{32}{3}$

Q006 $\frac{64}{3}$

Q007 $\frac{9}{2}$

Q008 30

Q009 $\frac{37}{12}$

Q010 8

Q011 Use the numbered solution for each source cubic.

Q012 $\frac{5}{2}$

Q013 $\frac{23}{3}$

Q014 5

Q015 $\frac{3}{2}$

Q016 $\frac{64}{3}$

Q017 $\frac{2}{3}$

Q018 $k = 2 - \sqrt[3]{\frac{3}{4}}$

Q019 Use the numbered solution for each source ratio.

Q020 B

Q021 A

Q022 C

Q023 B

Q024 A

Q025 B

Q026 B

Q027 C

Q028 B

Q029 A

Q030 C

Q031 A

Q032 B

Q033 A

Q034 C

Q035 A

Q036 C

Q037 A

Q038 B

Quiz MCQ 1 C

Quiz MCQ 2 B

Quiz MCQ 3 B

Quiz MCQ 4 C

Quiz WRITTEN 5 $\frac{37}{12}$

Quiz WRITTEN 6 5

Quiz WRITTEN 7 $\frac{3}{2}$

Quiz WRITTEN 8 $2 - \sqrt[3]{\frac{3}{4}}$

Concept 1 | Fundamentals of Random Variables

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Formula opener

Expectation

$$E(X) = \sum xp(x)$$

Variance

$$\mathrm{Var}(X) = E(X^2) - [E(X)]^2$$

Probability

$$\sum p(x) = 1$$

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Classified Practice **Fundamentals of Random Variables**

Q001 Written

For the sum of two dice, construct the probability distribution and find P$5 <= X <= 8$.

Q002 Written

Find the probability distributions for the absolute difference of two dice and the number of white balls drawn from the stated bag; evaluate the requested interval probabilities.

Classified Practice **Fundamentals of Random Variables**

Q003 **Written**

Construct both source probability tables (four coins; three balls from 4 red and 2 blue) and evaluate the given ranges.

Q004 **Written**

Find EX for tails in four coins and for red balls in a draw of three from a bag with 4 red and 5 white.

Classified Practice **Fundamentals of Random Variables**

Q005 **Written**

Find expected values for tails in three coins, the absolute difference of two dice, and white balls in the stated bag.

Q006 **Written**

Find EX for all six source random-variable experiments (cards, dice, balls, tickets, coins and repeated die rolls).

Classified Practice **Fundamentals of Random Variables**

Q007 Written

For tails in four fair coins, find E(3X+2), E(-4X+1), and EX^2.

Q008 **Written**

For white balls drawn from 4 red and 3 white, find the four requested transformed expected values.

Classified Practice **Fundamentals of Random Variables**

Q009 **Written**

Solve both source Exercise groups of transformed expectations for coins and for balls.

Q010 **Written**

For black balls drawn from the stated box, find EX and VX.

Classified Practice **Fundamentals of Random Variables**

Q011 **Written**

Find the mean and variance for the two source Try random variables (a numbered card and winning tickets).

Q012 **Written**

Find EX and VX for all five source Exercise random variables.

Classified Practice **Fundamentals of Random Variables**

Q013 Written

Find the standard deviation of tails when two biased coins have P(head)=4/7 and P(tail)=3/7.

Q014 **Written**

Find the standard deviation from each source probability table and for white balls drawn from 4 red and 3 white.

Classified Practice **Fundamentals of Random Variables**

Q015 **Written**

Find the standard deviation for both source probability-table exercises.

Q016 **Written**

For the number of winning tickets in the source draw, find VY and sigma$Y$ for Y=2X+1 and Y=-3X-2.

Classified Practice **Fundamentals of Random Variables**

Q017 **Written**

For red balls drawn from the stated bag, find variance and standard deviation for the three linear transformations of X.

Q018 **Written**

For a single die, find VY and sigma$Y$ for Y=X+2, Y=3X+1 and Y=-2X+4.

Classified Practice **Fundamentals of Random Variables**

Q019 **Written**

Solve the two source transformation problems: a net prize after an entry fee, and finding a,b from E(aX+b)=7/3 and sigma(aX+b)=3.

Q020 Written

Solve the two source Try problems involving net winnings and determining a,b for a die transformation.

Classified Practice **Fundamentals of Random Variables**

Q021 **Written**

Solve all four source transformation exercises, including the number-line coin walk, games, and the calibrated die model.

Q022 **MCQ**

A probability distribution must have probabilities that sum to:

Choose the correct option.

A 0

B 1

C the number of outcomes

D the mean

Classified Practice **Fundamentals of Random Variables**

Q023 MCQ

For two fair dice, P$X = 2$ is:

Choose the correct option.

A 1/6
B 1/36
C 2/36
D 0

Q024 MCQ

The number of heads in three coins can be:

Choose the correct option.

A 1,2,3

B 0,1,2,3

C 0,3 only

D any integer

Classified Practice **Fundamentals of Random Variables**

Q025 MCQ

The expected value is calculated as:

Choose the correct option.

A sum P$X = x$

B sum xP$X = x$

C largest x

D variance squared

Q026 MCQ

The mean number of tails in four fair coins is:

Choose the correct option.

A 1

B 2

C 3

D 4

Classified Practice **Fundamentals of Random Variables**

Q027 **MCQ**

For a fair die, EX is:

Choose the correct option.

A 3

B 3.5

C 4

D 6

Q028 MCQ

E(aX+b) equals:

Choose the correct option.

A aEX

B aEX+b

C EX+b only

D a+b

Classified Practice **Fundamentals of Random Variables**

Q029 **MCQ**

In general, $\mathrm{E}X^2$ is:

Choose the correct option.

A $E[X]^2$

B equal to $\mathrm{E}X$

C a weighted square sum

D always zero

Q030 MCQ

If EX=2, then E(3X+2) is:

Choose the correct option.

A 6

B 8

C 10

D 12

Classified Practice **Fundamentals of Random Variables**

Q031 MCQ

The variance identity is:

Choose the correct option.

A $E[X] - E[X^2]$

B $E[X^2] - E[X]^2$

C $E[X]^2$

D sqrt V

Q032 MCQ

Variance can never be:

Choose the correct option.

A positive
B zero
C negative
D fractional

Classified Practice **Fundamentals of Random Variables**

Q033 MCQ

The first step in a variance problem is to:

Choose the correct option.

A draw the distribution

B take a square root

C ignore probabilities

D add the values

Q034 MCQ

Standard deviation is:

Choose the correct option.

A VX

B sqrt$V(X)$

C VX squared

D EX

Classified Practice **Fundamentals of Random Variables**

Q035 **MCQ**

Standard deviation has units:

Choose the correct option.

A squared units
B the same units as X
C no units
D units cubed

Q036 MCQ

V(aX+b) equals:

Choose the correct option.

A aVX+b

B a squared times VX

C |a|VX

D VX+b

Classified Practice **Fundamentals of Random Variables**

Q037 **MCQ**

sigma(aX+b) equals:

Choose the correct option.

A a sigma$X$

B |a| sigma$X$

C a squared sigma$X$

D sigma$X$+b

Q038 MCQ

For Y=X+3, the variance is:

Choose the correct option.

A $\mathrm{V}X+3$

B $\mathrm{V}X$

C $3\mathrm{V}X$

D $9\mathrm{V}X$

Classified Practice **Fundamentals of Random Variables**

Q039 MCQ

A net prize after a fixed fee is modeled as:

Choose the correct option.

A $Y = X + b$

B $Y = aX + b$

C $Y = X^2$

D $Y = a + b$ only

Q040 MCQ

If a>0 and sigma(aX+b)=3, then a is:

Choose the correct option.

A 3 sigma$X$

B 3/sigma$X$

C sigma$X$/3

D 3+b

Classified Practice **Fundamentals of Random Variables**

Q041 **MCQ**

Changing b in aX+b changes:

Choose the correct option.

A variance only
B standard deviation only
C the mean only
D nothing

Q042 MCQ

If EX=2 and EX^2=7, then VX is:

Choose the correct option.

A 3

B 5

C 7

D 9

Classified Practice **Fundamentals of Random Variables**

Quiz WRITTEN 1 Concept Quiz · Written

Find EX and VX for the number of heads in three fair coins.

Quiz WRITTEN 2 **Concept Quiz · Written**

For Y=4X-1 with VX=2, find VY and sigma$Y$.

Classified Practice **Fundamentals of Random Variables**

Quiz WRITTEN 3 Concept Quiz · Written

For a fair die, find E(2X+5).

Quiz WRITTEN 4 **Concept Quiz · Written**

If sigma$X$=2 and EX=3, find sigma(-5X+7).

Classified Practice **Fundamentals of Random Variables**

Quiz MCQ 5 **Concept Quiz · MCQ**

If VX=5, V(3X+1) is:

Choose the correct option.

A 5
B 8
C 15
D 45

Quiz MCQ 6 **Concept Quiz · MCQ**

For a fair die $\mathrm{E}X$ is:

Choose the correct option.

A 2.5
B 3
C 3.5
D 4

Classified Practice **Fundamentals of Random Variables**

Quiz MCQ 7 **Concept Quiz · MCQ**

A probability table total must be:

Choose the correct option.

A 0
B 1
C 2
D the mean

Quiz MCQ 8 **Concept Quiz · MCQ**

If sigma$X$=4, sigma(X-9) is:

Choose the correct option.

A 0

B 4

C 13

D 16

Answer key

Use the question number shown on each page.

Q001 Values 2 to 12 have frequencies 1,2,3,4,5,6,5,4,3,2,1 over 36; P5 <= X <= 8=20/36=5/9.

Q002 Use the sample space or combinations to list every possible X and sum the relevant probabilities.

Q003 Use binomial or hypergeometric counts, then add probabilities for the requested values.

Q004 For four fair coins EX=2; for the hypergeometric draw EX=4/3.

Q005 Use EX=sum xP$X = x$ after constructing each distribution.

Q006 Compute each mean from its probability distribution; symmetry gives quick checks for fair cards and dice.

Q007 Use $E(aX + b) = aEX$+b; calculate EX^2 directly from the table.

Q008 Apply linearity for aX+b and use the weighted square sum for X^2.

Q009 For every expression use $E(aX + b) = aEX$+b; evaluate powers from the distribution table.

Q010 $EX = 6/7; VX = 20/49$.

Q011 Construct each probability table, then use VX=EX^2-EX^2.

Q012 Use the distribution table for each experiment and the variance identity.

Q013 The table gives VX=24/49, so sigma$X$=2 sqrt6/7.

Q014 Use sigma$X$=sqrt$V(X)$ after evaluating the variance from the table.

Q015 Compute EX, EX^2, then sigma$X$=sqrt$E(X^2) - E(X)^2$.

Q016 Use $V(aX + b) = a^2VX$ and sigma(aX+b)=|a|sigma$X$.

Q017 Apply the squared multiplier to VX and the absolute multiplier to sigma$X$.

Q018 Use the variance and standard-deviation transformation formulas for each Y.

Q019 Use $E(aX + b) = aEX$+b and sigma(aX+b)=|a|sigma$X$, with a>0.

Q020 Model the payout as $Y = aX + b$, then apply the mean and standard-deviation rules.

Q021 Express every quantity as $Y = aX + b$ and use the appropriate mean, variance or standard-deviation rule.

Q022 B

Q023 B

Q024 B

Q025 B

Q026 B

Q027 B

Q028 B

Q029 C

Q030 C

Q031 B

Q032 C

Q033 A

Q034 B

Q035 B

Q036 B

Q037 B

Q038 B

Q039 B

Q040 B

Q041 C

Q042 A

Quiz WRITTEN 1 EX=3/2 and VX=3/4

Quiz WRITTEN 2 VY=32 and sigma$Y$=4 sqrt2

Quiz WRITTEN 3 12

Quiz WRITTEN 4 10

Quiz MCQ 5 D

Quiz MCQ 6 C

Quiz MCQ 7 B

Quiz MCQ 8 B

Concept 2 | Expected Value and Variance of Multiple Variables

English Student Edition • Focused formulas and guided practice

Formula opener

Linearity

$$E(aX+b)=aE(X)+b$$

Variance

$$\operatorname{Var}(aX+b)=a^2\operatorname{Var}(X)$$

Independent

$$\operatorname{Var}(X+Y)=\operatorname{Var}X+\operatorname{Var}Y$$

Eng Eslam Ahmed | 01120009622

Classified Practice **Expected Value and Variance of Multiple Variables**

Q001 **Written**

Given the probability distributions of X and Y in the source tables, find E(X+Y).

Q002 **Written**

Three coins are tossed once; let X, Y and Z be the head indicators. Find E(X+Y+Z).

Classified Practice **Expected Value and Variance of Multiple Variables**

Q003 Written

For the two source tables with X=1,2,3 and Y=4,5,6, find E(X+Y).

Q004 **Written**

For the two bags in the source exercise, find the expected total number of red balls drawn.

Classified Practice **Expected Value and Variance of Multiple Variables**

Q005 **Written**

Three independent subsystems each receive two signals with success probability 0.5. Find E(X+Y+Z).

Q006 **Written**

Two 500-point coins and three 100-point coins are tossed. Find the expected total score Z.

Classified Practice **Expected Value and Variance of Multiple Variables**

Q007 **Written**

Two 500-point coins and two 100-point coins are tossed. Find EZ.

Q008 **Written**

A die is thrown twice and the score is Z=3X+5Y. Find EZ.

Classified Practice **Expected Value and Variance of Multiple Variables**

Q009 Written

For three 100-point coins and one 50-point coin, find the expected score from heads.

Q010 **Written**

In two independent raffles, buy two tickets from a 2/10 winner raffle and two from a 1/6 winner raffle. Find the expected prizes.

Classified Practice **Expected Value and Variance of Multiple Variables**

Q011 **Written**

Three coins are tossed and a die is rolled; X is heads and Y is the die score. Find E(XY).

Q012 **Written**

For independent X with probabilities 2/5,2/5,1/5 at 1,2,3 and Y uniform on 2,4,6, find E(XY).

Classified Practice **Expected Value and Variance of Multiple Variables**

Q013 **Written**

Ahmed tosses three coins and Mohamed tosses two coins. Let X and Y be their head counts. Find E(XY).

Q014 Written

A card numbered 1 to 4 is replaced and a second card is drawn. Find E(XY).

Classified Practice **Expected Value and Variance of Multiple Variables**

Q015 Written

For independent X and Y with EX=1 and EY=2/7, find E(XY).

Q016 **Written**

A card X is drawn from 1,2,3 and three-coin heads Y is independent. Find V(X+Y) and sigma(X+Y).

Classified Practice **Expected Value and Variance of Multiple Variables**

Q017 **Written**

Two independent dice are thrown. Find the variance and standard deviation of their sum.

Q018 Written

An independent coin head count X and die score Y are combined. Find V(X+Y) and sigma(X+Y).

Classified Practice **Expected Value and Variance of Multiple Variables**

Q019 **Written**

Ahmed tosses two coins and Mohamed tosses three coins. Find the variance of the total head count.

Q020 Written

Independent X is uniform on 1 to 4 and Y on 1 to 6. Find V(X+Y) and sigma(X+Y).

Classified Practice **Expected Value and Variance of Multiple Variables**

Q021 **Written**

Three independent special dice take values 2,4,6 equally. Find E(XYZ) and V(X+Y+Z).

Q022 **Written**

Three independent dice take values 1,3,5 equally. Find E(XYZ) and V(X+Y+Z).

Classified Practice

Expected Value and Variance of Multiple Variables

Q023 **Written**

Three ordinary dice are thrown. Find the expected product and the variance of their sum.

Q024 **Written**

Three friends have 5, 4 and 3 independent rolls to score a six. Find E(XYZ) and V(X+Y+Z).

Classified Practice **Expected Value and Variance of Multiple Variables**

Q025 **Written**

If X, Y and Z are independent head counts from 4, 3 and 2 fair coins, find E(XYZ).

Q026 **MCQ**

E(X+Y) equals:

Choose the correct option.

A $\mathrm{E}X$-$\mathrm{E}Y$

B $\mathrm{E}X$+$\mathrm{E}Y$

C $\mathrm{E}X\mathrm{E}Y$

D $\mathrm{E}X/\mathrm{E}Y$

Classified Practice **Expected Value and Variance of Multiple Variables**

Q027 **MCQ**

For three variables, E(X+Y+Z) equals:

Choose the correct option.

A EXEYEZ

B EX+EY+EZ

C EX-EY-EZ

D 0

Q028 MCQ

The expected heads in one fair coin toss is:

Choose the correct option.

A 0

B 1/2

C 1

D 2

Classified Practice **Expected Value and Variance of Multiple Variables**

Q029 **MCQ**

For two signals with p=0.5, EX is:

Choose the correct option.

A 0.5

B 1

C 2

D 4

Q030 MCQ

If EX=2 and EY=5, E(X+Y) is:

Choose the correct option.

A 3

B 7

C 10

D 25

Classified Practice **Expected Value and Variance of Multiple Variables**

Q031 MCQ

E(aX+bY) equals:

Choose the correct option.

A aEX+bEY

B abE(XY)

C EX+EY

D a+b

Q032 MCQ

The expected score of one fair 100-point coin is:

Choose the correct option.

A 25

B 50

C 100

D 200

Classified Practice **Expected Value and Variance of Multiple Variables**

Q033 MCQ

For a fair die, E(3X+5Y) with two rolls is:

Choose the correct option.

A 14

B 21

C 28

D 35

Q034 **MCQ**

A fixed entry fee changes:

Choose the correct option.

A variance only

B the mean only

C neither mean nor variance

D standard deviation only

Classified Practice **Expected Value and Variance of Multiple Variables**

Q035 MCQ

The expected number of winners in 2 of 10 tickets with 2 winners is:

Choose the correct option.

A 1/5

B 2/5

C 1/2

D 4/5

Q036 MCQ

E(XY)=EXEY requires:

Choose the correct option.

A $X = Y$

B independence

C zero means

D equal variances

Classified Practice **Expected Value and Variance of Multiple Variables**

Q037 **MCQ**

For three fair coins, EX is:

Choose the correct option.

A 1/2

B 1

C 3/2

D 3

Q038 MCQ

For a fair die, EY is:

Choose the correct option.

A 2.5

B 3

C 3.5

D 6

Classified Practice **Expected Value and Variance of Multiple Variables**

Q039 MCQ

With replacement, two card draws are:

Choose the correct option.

A dependent

B independent

C identical outcomes

D impossible

Q040 MCQ

If EX=2 and EY=3 for independent variables, E(XY) is:

Choose the correct option.

A 5

B 6

C 1

D 9

Classified Practice **Expected Value and Variance of Multiple Variables**

Q041 MCQ

For independent X,Y, V(X+Y) equals:

Choose the correct option.

A $VX\text{-}VY$

B $VX+VY$

C $VXVY$

D 0

Q042 **MCQ**

The variance of a fair die is:

Choose the correct option.

A 7/2

B 35/12

C 35/6

D 1/4

Classified Practice **Expected Value and Variance of Multiple Variables**

Q043 **MCQ**

Sigma(X+Y) is:

Choose the correct option.

A V(X+Y)

B sqrt$V(X+Y)$

C VX+VY

D E(X+Y)

Q044 MCQ

The variance of the sum of two independent fair dice is:

Choose the correct option.

A 35/12

B 35/6

C 70/6

D 49/4

Classified Practice **Expected Value and Variance of Multiple Variables**

Q045 **MCQ**

For independent sums, a constant shift changes:

Choose the correct option.

A variance
B standard deviation
C neither spread measure
D both spread measures

Q046 MCQ

For independent X,Y,Z, E(XYZ) equals:

Choose the correct option.

A EX+EY+EZ

B EXEYEZ

C 0

D E(XY)+EZ

Classified Practice **Expected Value and Variance of Multiple Variables**

Q047 **MCQ**

For independent X,Y,Z, V(X+Y+Z) equals:

Choose the correct option.

A $VXVYVZ$

B $VX+VY+VZ$

C $VX\text{-}VY\text{-}VZ$

D 0

Q048 **MCQ**

For a binomial count with n=5 and p=1/6, EX is:

Choose the correct option.

A 1/6

B 5/6

C 5/36

D 6

Classified Practice **Expected Value and Variance of Multiple Variables**

Q049 **MCQ**

For three values 2,4,6 equally likely, EX is:

Choose the correct option.

A 2

B 3

C 4

D 6

Q050 MCQ

For three independent fair dice, V(sum) is:

Choose the correct option.

A 35/12

B 35/6

C 35/4

D 343/8

Classified Practice **Expected Value and Variance of Multiple Variables**

Quiz WRITTEN 1 **Concept Quiz · Written**

If EX=4 and EY=7, find E(3X-2Y).

Quiz WRITTEN 2 **Concept Quiz · Written**

For independent variables with EX=2 and EY=5, find E(XY).

Classified Practice **Expected Value and Variance of Multiple Variables**

Quiz WRITTEN 3 **Concept Quiz · Written**

Two independent fair dice are thrown. Find V(X+Y) and sigma(X+Y).

Quiz WRITTEN 4 **Concept Quiz · Written**

Three independent Binomial counts have (n,p)=(2,1/2),(3,1/2),(4,1/2). Find V(X+Y+Z).

Classified Practice **Expected Value and Variance of Multiple Variables**

Quiz MCQ 5 **Concept Quiz · MCQ**

If EX=3 and EY=4, E(2X+Y) is:

Choose the correct option.

A 7

B 10

C 14

D 24

Quiz MCQ 6 **Concept Quiz · MCQ**

For independent variables EX=2, EY=5, E(XY) is:

Choose the correct option.

A 7

B 10

C 25

D 1/2

Classified Practice **Expected Value and Variance of Multiple Variables**

Quiz MCQ 7 **Concept Quiz · MCQ**

For two independent die rolls, V(X+Y) is:

Choose the correct option.

A 35/12
B 35/6
C 35/4
D 7/2

Quiz MCQ 8 **Concept Quiz · MCQ**

For three independent variables, V(X+Y+Z) is:

Choose the correct option.

A the product of variances

B the sum of variances

C the sum of means

D always zero

Answer key

Use the question number shown on each page.

Q001 $EX = 35/3, EY = 9/2, so E(X+Y) = 97/6.$

Q002 $E(X+Y+Z) = 1/2 + 1/2 + 1/2 = 3/2.$

Q003 $E(X+Y) = EX + EY = 2 + 5 = 7.$

Q004 Bag A: 6/7; Bag B: 6/7; total $E = 12/7$.

Q005 Each mean is $2(0.5) = 1$, so $E(X+Y+Z) = 3$.

Q006 $Z = 500X + 100Y, so EZ = 5001 + 1003/2 = 650.$

Q007 $EZ = 5001 + 1001 = 600.$

Q008 $EZ = 37/2 + 57/2 = 28.$

Q009 $EZ = 350 + 25 = 175.$

Q010 $EZ = 22/10 + 21/6 = 11/15.$

Q011 X and Y are independent, so $E(XY) = EX$EY=3/27/2=21/4.

Q012 $EX = 8/5, EY = 4, so E(XY) = 32/5.$

Q013 $E(XY) = EXEY = 3/21 = 3/2.$

Q014 $EX = EY = 5/2, so E(XY) = 25/4.$

Q015 $E(XY) = EXEY = 2/7.$

Q016 VX=2/3, VY=3/4, so $V(X+Y) = 17/12$ and sigma=sqrt17/12.

Q017 $V(X+Y) = 35/12 + 35/12 = 35/6$; sigma=sqrt35/6.

Q018 $V = 1/4 + 35/12 = 19/6$; sigma=sqrt19/6.

Q019 $V = 21/4 + 31/4 = 5/4.$

Q020 $V = 5/4 + 35/12 = 25/6$; sigma=5/sqrt6.

Q021 EX=4, VX=8/3; $E(XYZ) = 64$ and $V(sum) = 8$.

Q022 EX=3, VX=8/3; $E(XYZ) = 27$ and $V(sum) = 8$.

Q023 $E(product)$ =7/2^3 $= 343/8$; $V(sum) = 335/12$=35/4.

Q024 Means are 5/6, 2/3, 1/2; $E(product) = 5/18$ and $V(sum) = 5/3$.

Q025 $E(XYZ) = 23/21$=3.

Q026 B

Q027 B

Q028 B

Q029 B

Q030 B

Q031 A

Q032 B

Q033 C

Q034 B

Q035 B

Q036 B

Q037 C

Q038 C

Q039 B

Q040 B

Q041 B

Q042 B

Q043 B

Q044 B

Q045 C

Q046 B

Q047 B

Q048 B

Q049 C

Q050 C

Quiz WRITTEN 1 -2

Quiz WRITTEN 2 10

Quiz WRITTEN 3 $V = 35/6$, sigma=sqrt35/6

Quiz WRITTEN 4 9/4

Quiz MCQ 5 B

Quiz MCQ 6 B

Quiz MCQ 7 B

Quiz MCQ 8 B